

US Interfacility Transport: Sourced Evidence Master v3.7

A complete, source-verified evidence base for the United States interfacility medical transport market: who moves, between which care settings, at what clinical acuity, paid by whom, at what price, served by which suppliers, and where the whitespace is measurable. v3 carries the entire verified v2.7 evidence base forward unchanged and adds the platform IFT database and the full government-connector layer. Built and verified 09-11 July 2026. Navigate with the Index tab: every tab, one click.

What this workbook is, in one paragraph

Every number in these 323 tabs comes from a named government dataset, a peer-reviewed study, or a published primary document, or is computed by a visible Excel formula from numbers that do. Nothing is modeled, blended, or assumed. Where a figure the market usually wants does not exist in any public source, this workbook says so and names exactly what would produce it. Figures that failed that rule are quarantined on Excluded_Not_Sourced. PUBLIC-WEB company material is carried with its label and dates, never silently mixed with government data.

How to read any number in three steps

Step 1: every tab tags its figures with Fact IDs (F01 to F598). Step 2: the Fact_Ledger resolves each ID to a source, tier, and locator down to the table or page. Step 3: the Source_Index identifies all 418 sources on one row each, with the full citation and URL on the Source_Register. If a figure is labeled DERIVED, the ledger names its inputs and the cell shows the arithmetic. New in v3: Pull_Manifest gives the exact endpoint, dataset UUID, filters, SHA-256 and timestamp of every live extraction, and V3_Change_Log records every edit made to carried-forward v2.7 content.

The colour and label conventions

Blue font: a value hardcoded from a source document or dataset extraction. Black font: an Excel formula. Green font: a link to another tab. Tier A: verified against the primary document or extracted directly from the primary dataset by the committed pipeline. Tier B: published but secondary, or a figure that moves over time. SOURCED: carried from an upstream dataset and not independently reopened. DERIVED: computed here, with inputs named. PUBLIC-WEB: company/press material, labeled, dated, never blended with government figures. FRAMEWORK: definitions and scaffolds only - never numbers.

The map: what each section answers		
Section	Tabs	The question it answers
Governance	Index, README .. Connector_Estate_Map, Engagement_Data_Map	Why every number is here, the findings, the complete audit trail, the v3 change log, the live-pull manifest, and the connector estate that feeds the data tabs.
Demand	Demand_Drivers .. Growth_Evidence_Registry, InDepth_Q01-Q10	How many patients move, from which settings, for which conditions, driven by which measured forces - with the verbatim evidence registry behind each claim.
Medicare claims core	Medicare_PSPS .. MA_Geo_Variation	The public claims spine: PSPS 2010-2024 including denial rates, the MUP utilization/price series 2013-2024, dialysis/ESRD denominators, and the Medicare Advantage utilization bounds.
Price and payment	Payment_Rules .. Commercial_Context_APCD	What Medicare pays, locality by locality (GPCI), a worked derived rate card, service-level economics, and published commercial context.
Supply	Supplier_Landscape .. Facility_Universe_State + the full registries (Hosp/SNF/Dialysis/IRF/LTCH/Hospice/HHA, PECOS, HSA)	Who provides transport and where: billing organizations, enrolled suppliers, workforce, certification vintages, ownership churn, and the facility universe that originates and receives transfers.
Market structure	State_Saturation_Raw .. Imbalance_Ledger + SP_Index and the 51 state profiles	State and metro screens, county-level supply density and whitespace bands, and demand-over-supply imbalances.
Company & competitive	MMT_NPI_Estate .. Contract_Benchmarks	The subject company NPI estate and dossier, named competitors, payment-integrity evidence, and contract benchmarks. PUBLIC-WEB material is labeled.
Client alignment & costs	Sizing_Playbook .. Cross_Source_Recon	How the team converges on a defensible number, production costs, and why four sources give four different counts.
Reference	IFT_Alt_Measures .. Dataset_Linkage_Map	Alternative counting routes, every code system that records a transfer, the KPI dictionary, and the regulatory register.

Integrity

Fault_Audit .. Excluded_Not_Sourced

Every known threat to validity, and the quarantine of everything that failed the sourcing rule.

What this workbook deliberately does not contain

A single asserted total addressable market figure. A credible TAM requires a commercial transport volume, a Medicare Advantage transport volume, and an acuity-weighted price, and none of the three exists in any public source. The three-scenario TAM MODEL with its assumption register lives on TAM_Model_National; v3 adds the published bounds for its weakest input (MA_Geo_Variation) but does not convert the model into a fact. The quarantine list on Excluded_Not_Sourced grew in v3: the platform database sweep added 63 more modeled figures that are named and excluded rather than blended.

Revision summary		
Revision	Date	Contents
1	09 July 2026	Initial evidence base: 20 tabs, facts F01 to F109, sources S01 to S38, findings 1 to 25, all verified against primary documents.
2	10 July 2026	The complete Medicare origin-to-destination dollar map rebuilt from 392,409 raw claim rows (reconciling to the cent), the air and dialysis channels, commercial rates, macro drivers, code crosswalks, and the dataset linkage map. Facts through F135; sources through S58.
3	10 July 2026	Per-beneficiary normalization, acuity by channel, the supplier landscape, and the payment add-on cliff. Facts through F147; sources through S61.
4	10 July 2026	The demand and supply stacks, the imbalance ledger with the whitespace register, an eleven-year supplier series, workforce, state saturation, metro landscape, alternative counting routes, and the consolidated fault audit. Facts through F163; sources through S72.
5	10 July 2026	Final consolidation: the official state-file cross-check (bounding the supplier-file suppression tail at 1.0%), corrections from independent external review, and full internal-consistency verification. Facts through F165; 73 sources; zero formula errors.
6 (v3.0)	10 July 2026	The platform integration: all v2.7 content carried forward unchanged (faithful-copy proof on Verification_Log Panel I) plus 209 new tabs built from the platform IFT database and 321 live government-API extractions - the MUP ambulance utilization/price series 2013-2024, the PSPS denial-rate series 2010-2024, CMS market saturation 2020-2025 with county whitespace bands, BLS QCEW EMS industry series 2014-2025, the PECOS supplier universe, the facility O/D universe, CY2025 GPCI localities with a derived rate card, certification-vintage supply series, the OMB county-to-CBSA crosswalk, and the sourced clinical/growth/company evidence registries. Facts through F430; 304 sources; 189 charts; ~6,970 printed pages (as shipped in v3.0 - this row records that revision. not the
7 (v3.1)	11 July 2026	The usability and closure pass: a hyperlinked Index tab covering every tab; findings 42-51 appended with LIVE formula references into the new evidence; the measured Census 65+/85+ state base 2020-2024 (State_Age_65plus - closes the state-grain half of P20); the OEWS May 2024 EMS occupation wage ladder (OEWS_EMS_Wages - closes P4); four gallery charts added to the Charts tab; state profiles gain 65+ joins; every source now carries a URL, a repo-file locator, or an explicit no-URL-captured statement; and two citation corrections applied at source (the PMID 25397857 attribution and a dead MedPAC link).
8 (v3.2)	11 July 2026	The raw-granularity doubling: the provider-grain Medicare ambulance registry (every billing NPI by name, 2019 and 2024, ~90,000 rows - the most granular public Medicare ambulance record); hospital-to-ZIP transfer corridors (top-10 origin ZIPs per hospital); the county 65+/85+ age base 2020 vs 2024; county chronic-disease prevalence (CKD/CHD/stroke/diabetes, CDC PLACES, methodology stated); the quarterly QCEW industry series 2014-2025; the HCRIS hospital capacity panel FY2020-2022 (17,974 hospital-years); and the OIG ambulance-exclusion registry. All Tier A dataset extractions on Pull_Manifest.

9 (v3.3)	11 July 2026	The presentation overhaul: every chart rebuilt through a single hardened house-style layer - bottom category axes with real year/state labels, explicit series colours and weights, legends only where they earn their space, subtle gridlines, zero chart-on-chart overlaps (19 collisions found and removed); the two dual-axis combo charts split into paired single-axis charts; the carried v2.7 charts re-parsed at full fidelity so every series regained its category labels and names; the Methodology tab rebuilt as a nine-section decision record covering the v3 pipeline, the verification gates and a sixty-second audit path (and repairing a v2.7 URL typo); and a workbook-wide format sweep - print setup normalized on every tab, truncated columns widened, header rows tightened.
10 (v3.4)	11 July 2026	The specificity-and-analysis pass (worker handoff + eight-hour extension order): the facility-pay evidence layer (Facility_Pay_Layer - GADCS revenue-source census, the DocGo outside-per-trip decomposition, USAspending V225, all verified against the primary documents); the subject company's measured Medicare book (MMT_Medicare_Book - consolidated and per-NPI volumes, acuity mix with volume-weighted RVU, mileage economics, three-vintage trajectory); the handoff 5.1-5.3 repairs (committed fact-tag scan - zero stale tags found - and data-quality rows on all seven v3.2 raw tabs); findings 52 onward with live references; plus the analysis and cohort layers this revision row is extended by as they land.
11 (v3.5)	12 July 2026	The completion pass: the market-wide 990 contractor sweep across the footprint nonprofit systems (Footprint_990_Sweep - the largest public window into facility-direct transport payments, with the top-five information floor stated); the Medicaid ambulance rate card extended toward the full ten-state footprint; and the retrieved parked pulls (SNF return-leg quality, Hospital-at-Home participants) landed where public data allowed and left as bordered PENDING with the named dataset where it did not. Findings continue past the v3.4 register; every new tab carries a read panel, a finding and a data-quality row.
12 (v3.6)	12 July 2026	Return-leg structure pass: joins the SNF Quality Reporting Program claims measures to the CMS Nursing Home provider-info file on the CCN (all reporting SNFs, 1:1) and cross-tabs the bounce-back (potentially preventable readmission) and discharge-to-community rates by ownership, certified-bed scale and 5-star rating (SNF_ReturnLeg_Structure) - the structural cut of where return-leg transport demand concentrates. This revision also restores the verification gate script (verify.py), which the v3.5 package had picked up in a stale form.
13 (v3.7)	12 July 2026	CIM-grade presentation pass (Run 3, Block U): a committed format gate (format_gate.py) and a presentation pass (cim_format.py) now enforce the deck-ready standard on every tab - gridlines off, a section tab colour, freeze panes, a bold A1 title, and an explicit number format on every numeric cell (no raw floats, no unformatted counts, no Python artifacts), including the carried tabs the builder never touched. Adds the Style_Standard reference tab stating the rules. No evidence changed; every tab is now screenshot-worthy at 100% zoom.

Pending register: named enhancements, none assumed

Carried from v2.7 with v3 status: P1 HCUPnet condition-level transfer rates - OPEN (point estimates only without licensed microdata). P2 Ambulance-desert county tables - PARTIALLY CLOSED (county provider-count bands on Market_Saturation_Ambulance measure thin-supply counties; a population-weighted desert definition still needs a licensed drive-time layer). P3 ZCTA/county-to-CBSA crosswalk - CLOSED (CBSA_Crosswalk_Reference, OMB Bulletin 23-01). P4 BLS OEWS state wage files - CLOSED in v3.1 (OEWS_EMS_Wages: May 2024 state file, EMT/paramedic/ambulance-driver occupations, beside the QCEW industry series on QCEW_EMS_Employment). P5 USRDS prevalent dialysis counts - OPEN (Enrollment_ESRD_State carries the Medicare ESRD denominator instead). P14 locality price adjustment - CLOSED (GPCI_Localities + Derived_Rate_Card). P20: state-grain 65+ base - CLOSED in v3.1 for the state grain (State_Age_65plus, Census Vintage 2024 measured estimates 2020-2024); the ACS API route stays OPEN for tract/county age detail (needs a free CENSUS_API_KEY). P21: NPPES monthly full-replacement file for the national supplier universe by taxonomy - OPEN (the live API caps at 1,200 rows per query; PECOS_Suppliers_State is the enrollment-based count today).

Methodology - the decision record

The question here is simple: why should anyone trust this workbook? This tab is the decision record. It walks through the inclusion rule, why each evidence tab exists, the definitional choices that move the numbers, exactly how the data was pulled so it can be rerun, the rules for comparing figures, the corrections made along the way, and what the workbook still cannot do. Read this first. Sections 5 and 9 cover the deterministic pull pipeline, the verification gates and the sixty-second audit path added in the v3 builds.

1. The question this workbook answers, and the one it refuses to answer

The question. How large is US interfacility transport, who moves, between which settings, at what acuity, paid by whom, and at what price. Every tab answers one slice of that and cites the document that produced it.

The refusal. This workbook does not size a total addressable market. A TAM requires a commercial-payer volume, a Medicare Advantage volume and an acuity-weighted price, none of which exists in any public source. The initial build carried three such assumptions (a 4.27x gross-up, a 2x to 4x commercial multiple, an acuity mix); all three were moved to Excluded_Not_Sourced and stayed there. revision 2 replaced two of them with published bounds rather than restoring the estimate.

The consequence you must accept. Some cells on this workbook say the number does not exist. That is the finding. An analyst who needs a TAM must buy data (Dataset_Linkage_Map lists which), not blend a public one.

2. The inclusion rule, applied without exception

Rule	Why	What it prevents	
A number enters only if a named publisher document produced it, or an Excel formula computes it from numbers that did.	Traceability must survive the analyst. Anyone opening a cell sees either a source-pinned constant or the arithmetic that made it.	Numbers with no parent. The single most common failure in diligence workbooks is a figure that no one can source six weeks later.	
Blue font means hardcoded from a document. Black means a formula. Green means a link to another tab.	The colour is the audit trail. A reader can see the provenance of a cell without clicking it.	Hardcoded values masquerading as calculations, and stale pastes of numbers that should have updated.	
Every fact carries a Fact ID (F01-F165 in v2.7; v3 extends the ledger - see Fact_Ledger), a Tier, a Source ID and a locator down to the table or page.	Tier separates what was verified against the primary document from what was asserted by an upstream workbook.	Laundering. An assumption quoted twice does not become evidence.	
Anything labeled ILLUSTRATIVE, FRAMEWORK or INDUSTRY ESTIMATE is quarantined on Excluded_Not_Sourced.	These labels are how modeled numbers enter a deck and never leave.	A framework assumption being read off a chart axis as though it were measured.	
Derived figures are labeled DERIVED and show their inputs.	A ratio of two published numbers is legitimate. A ratio of a published number and a guess is not.	Same-source, same-year discipline on every multiple (see section 6).	
The colour key, live:	Blue - hardcoded straight from the cited document (e.g. 8,911)	Black - a visible formula over sourced cells (e.g. =B12/C12 typed in the cell)	Green - a live link to another tab (e.g. =Fact_Ledger!D9)
Tier A	Verified against the primary document or dataset in this build; the locator points into that document.	Tier B	Published but secondary, or a figure that moves with revisions; carried with that caveat on the fact row.

3. Why each evidence tab exists

Tab	The question it settles	Why it had to be built	Primary sources
Acute_IFT_Series	How many acute-to-acute transfers happen, from the ED and from inpatient beds.	HCUP is the only national all-payer census of hospital discharges. Its two origin channels are mutually exclusive and must be added, not averaged.	S01, S04
Receiving_Side	Does the receiving side of the ledger agree with the sending side.	A transfer has two ends. NIS TRAN_IN counts arrivals independently of DISPUNIFORM departures. Agreement is the workbook internal validity test.	S04
EMS_Transports	How many ambulance legs exist, and what share are interfacility.	Hospital data counts patients moved; EMS data counts vehicles moving them. The two are different units and neither substitutes for the other.	S12, S25, S55
Medicare_PSPS	What does Medicare fee-for-service actually pay, by acuity and by origin-destination pair.	It is the only public file that carries dollars, acuity and endpoints on the same row.	S13, S39
Medicare_IFT_Series	Is the Medicare hospital-to-hospital book growing or shrinking.	No published table contains this series. It was built by pulling fifteen annual PSPS files and filtering the HH modifier.	S32
Medicare_OD_Matrix	Where does the entire Medicare ambulance dollar go, pair by pair.	revision 1 carried the HH pair only. Building the full matrix revealed that the SNF interface is 2.75x the hospital pair, which reframes the market definition.	S39

Utilization_Normalized	Is the Medicare decline demand or denominator.	FFS enrollment fell 15.5% over the same window. Without dividing by it, every trend on this workbook overstates decline by roughly a factor of two.	S39, S60
Acuity_by_Channel	Are the interfacility channels one clinical product or several.	Staffing, unit economics and payer strategy all follow acuity. SNF-to-hospital is 94% emergency; hospital-to-SNF is 99% non-emergency. One name, opposite products.	S39
Air_Ambulance_IFT	Where does high-acuity interfacility volume actually sit.	Rotary-wing is 65.6% hospital-to-hospital and grows per beneficiary while ground shrinks. Ground acuity mix cannot describe it.	S40, S44
Dialysis_ESRD_Channel	What happens to a scheduled-transport book when a payer decides the volume is discretionary.	The RSNAT rollout is a natural experiment with a date. The claims date-stamp the intervention, which makes it the template for Medicare Advantage behaviour elsewhere.	S39, S41, S42
Payment_Rules	What governs the Medicare price, and when does it change.	RVUs are frozen since 2002, geographic adjustments since 2008, and the statutory add-ons expire 1 January 2028. The price is legislative, not market.	S14, S16, S28, S29, S61
Payer_Rates_Commercial	What does everyone other than Medicare pay.	The commercial multiple was the largest excluded assumption in v1. FAIR Health supplies published bounds; GAPB supplies the balance-billing regime.	S43, S44, S45, S46, S47, S48
Supplier_Landscape	Who bills, and how concentrated is the market.	8,721 NPIs with the largest under 1% of national volume. Any consolidation thesis starts here, and every count is a floor because of small-cell suppression.	S59
Cost_and_Capacity	What does a transport cost to produce.	GADCS is the only national cost collection, and MedPAC says it cannot yet support a margin. That warning travels with every cost figure.	S22, S30, S31
Macro_Demand_Drivers	What moves demand from outside the hospital.	Census 65+, Medicare Advantage share, rural closures, coordination infrastructure. Each is a measured series, not a narrative.	S50, S51, S52, S53, S54, S55
Code_Crosswalks	How is one transfer written down in five different systems.	A transfer is an O-D modifier in claims, a discharge status on a UB-04, a DISP_ED value in HCUP, an eResponse.05 value in NEMSIS. Joining them requires knowing all four.	S56, S57, S58
Dataset_Linkage_Map	What would it take to answer what this workbook cannot.	It names the join keys, the access terms and the two purchases that close the largest gaps.	All
Cross_Source_Recon	Why do four sources give four different numbers.	They measure different units on different universes. The wedge is enumerated driver by driver so the spread is explained rather than averaged away.	S01, S04, S12, S13
v3 evidence layers (live public-API pulls; one row per family; every pull is on Pull_Manifest with endpoint and SHA-256):			
Enrollment_ESRD_State + Enroll_State_*	The fee-for-service denominator, nationally and per state, 2013-2025.	Every per-beneficiary rate in the workbook divides by these cells; the MA migration is the single biggest trend confounder.	CMS Medicare Monthly Enrollment
MUP_Ambulance_National/State, MUP_State_*, MUP_Providers_2013/19/24	Who bills ground ambulance, down to the individual NPI, and at what average submitted/allowed/paid amounts.	Supplier concentration and rate benchmarks need provider grain, not national sums.	CMS Medicare Physician & Other Practitioners
PSPS_Denial_Series + PSPS_Detail_*	Denial behaviour by code, modifier and level of service, 2010-2024.	Denial rates are the revenue-cycle risk signal; no published table carries this series.	CMS Physician/Supplier Procedure Summary
Market_Saturation_Ambulance + MS_County_*, MS_CtyWin_*	County-level supplier density and utilization.	Saturation and dark-market analysis is a county question.	CMS Market Saturation & Utilization
QCEW_EMS_Employment/State/County/Quarterly	The ambulance industry labor base: employers, jobs, wages, quarterly to 2025.	Crew labor is ~69% of ground ambulance cost; this is the cost-inflation instrument.	BLS QCEW NAICS 621910
Facility_Universe_State + Hosp/SNF/Dialysis/IRF/LTCH/Hospice/HHA registries, PECOS_Registry	The universe of transfer endpoints and enrolled ambulance suppliers, row per facility.	Origin-destination analysis needs the actual endpoint rolls, not counts.	CMS Provider Data Catalog; PECOS
HSA_Hospital_Catchment + HSA_Corridors	Which ZIPs each hospital actually draws from - the top-15 corridors per hospital.	Transfer-lane planning is a corridor question; this is the only public flow file.	CMS Hospital Service Area file
State_Age_65plus, County_Age_65plus, PLACES_County_Chronic	The demand base: 65+/85+ population and chronic-disease prevalence by county.	Transport demand is driven by age and morbidity where the patient lives.	Census Vintage 2024 estimates; CDC PLACES
HCRIS_Hospital_Panel	Hospital cost-report fundamentals, 17,974 hospital-years.	Occupancy and finance context for the sending side.	CMS HCRIS extract (vendored in repo)
LEIE_Ambulance_Exclusions	Ambulance-related program-integrity exclusions.	Counterparty screening; the OIG file is the reference list.	HHS OIG LEIE

Metro_ * families + SP_ * state profiles	The same evidence re-cut per target metro and per state, formula-driven off the registry tabs.	Market entry decisions are made metro by metro and state by state.	Derived - whole-column INDEX/MATCH/SUMIFS over the registry tabs
Company/lnDepth/Reference tab families	Connector-derived and repository evidence: unit economics, growth registries, question-by-question deep dives.	Carries the RCM_MC evidence base into the same sourcing regime.	Repository modules + documents on Source_Index

4. Definitional choices that change the answer, and the choice this workbook made		
Choice	Options	What was chosen and why
What counts as interfacility	(a) Hospital-to-hospital only. (b) Any facility-to-facility pair. (c) Any non-scene transport.	(b), reported by pair. Restricting to (a) hides 2.4M SNF-interface transports; (c) sweeps in dialysis and discharge-home legs that behave nothing alike. Every tab reports pairs separately so the reader can pick.
Ground base codes	Six codes (A0426-A0429, A0433, A0434) or seven including paramedic intercept A0432.	Six. A0432 is a rural intercept payment, not a transport (3,312 services in 2024). Stated on Acuity_by_Channel so counts tie.
Service basis	PSPS submitted services, or MUP-PHY final-action approved services.	PSPS for all volumes and dollars; MUP only for supplier counts. The two differ by 1.10x in 2024 and must never be mixed. Data_Quality_Register #34.
FFS denominator	Original Medicare any part (33.9M in 2024) or Part A and B both (27.7M).	A and B, the population that can generate a carrier ambulance claim. The choice moves every per-1,000 rate by about a fifth, so it is stated on the tab and in DQ #35.
Fractional service counts	Truncate to integers or sum as floats.	Sum as floats. PSPS carries decimal service counts on shared services. The revision 1 truncation cost 39 services out of 10.6M. Correction #7.
Rural closure count	Sheps 154 since 2010, or Chartis 182.	Sheps, with the Chartis number recorded beside it and a warning never to mix them. Different conventions on Rural Emergency Hospital conversions. DQ #31.
Air interfacility inference	Use O-D modifiers for both rotary and fixed-wing, or rotary only.	Rotary only. Fixed-wing is 94.6% II (airport to airport), so its modifiers identify runways. DQ #29.

5. The data pipeline, stated so it can be rerun

PSPS. GET [https://data.cms.gov/data-api/v1/dataset/{uuid}/data?filter\[HCPD_CD\]={code}](https://data.cms.gov/data-api/v1/dataset/{uuid}/data?filter[HCPD_CD]={code}), paginated at 5,000 rows, one call per HCPD per data year. Fifteen data years, twelve ambulance codes, 392,409 rows. Dataset UUIDs are printed on Verification_Log Panel F. Two schema traps: the PSPS_ field prefix begins with data year 2020 (not 2021), and service counts are decimal.

MUP-PHY. Same host, filter[HCPD_CD]={code}, 2019 and 2024 datasets, eight base codes, 44,614 NPI-by-code rows. Rows with 10 or fewer beneficiaries are suppressed at source, so counts are floors.

Enrollment. Same host, Medicare Monthly Enrollment, filter[BENE_GEO_LVL]=National and filter[MONTH]=Year, 2013 to 2025.

Everything else was opened by hand: PDFs downloaded and parsed, HTML read, PubMed queried by PMID, NPPES and the Medicare Coverage Database queried live. Dates of access are on the Source_Index.

Determinism. No sampling, no imputation, no model. Every figure is a sum, a ratio, or a value copied from a document. The workbook recalculates to zero formula errors across 55,295 formulas workbook-wide (verified in an independent LibreOffice recalculation; Verification_Log Panel K carries the stamped counts).

v3 pipeline. Every API-backed tab is produced by a deterministic pull-assemble-verify pipeline shipped in the repository at RCM_MC/scripts/ift_evidence_v3 (pull.py through pull7.py, assemble.py, verify.py). Each pull is cached to disk and recorded on Pull_Manifest with the exact endpoint, dataset version UUID, filters, pages fetched, rows kept, SHA-256 of the canonical payload and the UTC retrieval time. Re-running the chain reproduces this workbook byte-for-byte up to retrieval dates.

Dataset versions. CMS datasets are pinned per data year by parsing the version UUID out of the public DCAT catalog, so a later CMS re-release cannot silently change a number: the UUID on Pull_Manifest is the version that produced the cell.

Verification gates. V1: every carried v2.7 cell must recalculate to its original value (10,463 cells compared, zero differences allowed). V2: the whole workbook must recalculate with zero Excel error cells. V3: derived cells on pull-backed tabs are recomputed independently in python from the cached payloads. V7: scale gates on tabs, printed pages and file size. V8: fact and source IDs must be contiguous with no ghost references. The build runs twice so the verified numbers are stamped back into Verification_Log Panel K and re-verified.

6. Rules for comparing numbers, which is where diligence workbooks usually break

Same source, same year, or say so. The 5.0x rotary air multiple divides FAIR Health Medicare allowed by FAIR Health Medicare allowed, both 2020, both from one white paper. The ground multiple of roughly 1.6x divides FAIR 2020 commercial by a 2024 claims-derived Medicare figure and is therefore labeled CROSS-YEAR AND CROSS-BASIS on its own tab. It is bounding evidence, not a rate card.

Never add a patient count to a vehicle count. HCUP counts discharges, NEMSIS counts activations, PSPS counts billed services. Cross_Source_Recon enumerates the wedge drivers instead of averaging four incompatible numbers into a fifth wrong one.

Deflate before you interpret. Any Medicare fee-for-service series read as demand must first be divided by fee-for-service enrollment, which fell 15.5% over 2019 to 2024 (Utilization_Normalized).

A suppressed cell is a floor, not a zero. CMS masks cells under 11 services in PSPS and under 11 beneficiaries in MUP. Every count in this workbook derived from those files is a floor and says so.

A confidence interval that does not exist cannot be invented. HCUP weighted summary statistics carry no standard errors. HCUPnet does NOT supply them; they require licensed microdata and the AHRQ variance method (finding confirmed by independent external review, 10 July 2026). Until that route is taken, no HCUP trend in this workbook is called significant.

7. Corrections made to prior work, kept visible on purpose				
#	What was wrong	What is right	Where	
1	CY2024 Ambulance Inflation Factor carried as 3.0%	2.6%, per the CMS manual and MedPAC	Payment_Rules	
2	Medicare FFS ambulance spend as \$4.0B in 2023	\$3.9B in 2022 program spending	Payment_Rules	
3	GADCS for-profit cost per transport as \$1,778	\$1,788	Cost_and_Capacity	
4	GADCS mean of \$2,673 quoted without its median	Median is \$1,340. The mean is skewed and MedPAC says the data cannot support a margin	Cost_and_Capacity	
5	Conversion factor assumed at \$278.00	\$272.44, CY2024, published	Payment_Rules	
6	2007 NIS total omitted 16,542 records	Restated	Acute_IFT_Series	
7	PSPS service counts truncated to integers	Decimal counts summed as floats. 871,753 becomes 871,756 on the 2024 HH base set	Medicare_OD_Matrix	
8	PSPS_ field prefix said to begin in 2021	It begins with data year 2020	Verification_Log	
9	Two draft read-notes overstated their own findings	Corrected against computed values before release: HH per-capita decline is 8.0% (not flat) and the HH emergency-share shift is 2.4 points (not 5.0)	Verification_Log Panel G	
v3	Cell-level corrections made during the v3 builds (citation attribution, dead links, count inconsistencies)	Logged one per row with the old value, the new value and the reason	V3_Change_Log	

8. What this workbook still cannot do, and what would fix it				
Gap	Why it exists	The fix	Cost	
No confidence interval on any HCUP trend	Weighted summary statistics ship without standard errors	License NEDS or NIS microdata and apply AHRQ Methods 2015_09; HCUPnet is point estimates only	Purchase plus DUA	
No hospital-pair transfer flows	National HCUP files mask hospital identity	Purchase one state SID/SEDD with hospital IDs	Low four figures per state	
No corporate ownership roll-up	An NPI is not a parent company	Build the PECOS reassignment and ownership graph	Free, engineering time	
No real commercial rate card	FAIR Health publishes averages, not rates	Parse Transparency in Coverage machine-readable files for A042x by NPI	Free, heavy compute	
No Medicare Advantage transport volume	MA encounter data is not public	Encounter data via ResDAC DUA, or a licensed claims panel	Months, or paid	
No activation-level IFT flag nationally	NEMSIS public reports are aggregated	Request the NEMSIS research dataset	Free, paperwork	
No 2025 EMS data	The NEMSIS 2025 annual report is not published	Expected around August 2026	Wait	
No Medicare ambulance margin	MedPAC found GADCS reporting errors severe enough to disqualify it	MedPAC Recommendation 6 asks Congress to keep collecting, streamlined	Years	

9. How to verify any number in sixty seconds

Step 1 - read the colour. Blue: the value was typed from the cited document and nothing computed it. Black: a formula; every input is itself blue, black or green, so the arithmetic is fully inspectable. Green: a live link to another tab - follow it.

Step 2 - find the fact. Facts F01-F598 live on Fact_Ledger: one row per fact with its value, unit, year, tier, source ID, the locator inside the source (table, page or cell) and the tab the fact lives on.

Step 3 - follow the source. Sources S01-S418 live on Source_Index with publisher, document title, URL and access date. For API-backed cells, Pull_Manifest carries the exact endpoint, dataset version UUID, filters and the SHA-256 of the cached payload that produced the number.

Step 4 - recompute. Open the workbook and let it recalculate: 55,295 formulas resolve with zero errors, and every carried v2.7 cell reproduces its original value. The Index tab links to all 323 tabs; the repository pipeline re-runs the pulls end to end.

This tab was rebuilt in v3.3: same decision record, house-style layout. Carried sections 1-8 are v2.7 wording verbatim except three repairs - the data-api URL artifact ("revision 1" for "v1") was restored, the fact-ID range and the workbook formula count were updated to current truth - plus the v3 pipeline rows in section 5, the v3 family rows in section 3 and new section 9. Logged on V3_Change_Log.

Verification log

The question: was every source actually opened and every extraction actually tested? This log records each verification pass, panel by panel, including the reconciliations that matched to the cent and the working hypotheses that were tested and rejected. The rejected ones stay visible on purpose: an audit trail that only records successes is not an audit trail.

Panel A. Values carried in this workbook, checked against the document that published them.

Evidence layer	Values checked	Matched exactly	Corrections	How it was checked	Primary document opened	Outcome
CMS PSPS ambulance claims	60	60	-	Downloaded 110,568 raw claim-summary rows for 2022-2024 from the CMS data.cms.gov API and re-aggregated every origin-destination pair, every acuity HCPCS and every annual total from scratch.	data.cms.gov/data-api/v1/dataset/(usid)/data, Physician/Supplier Procedure Summary, 2022, 2023 and 2024	PERFECT: 12 origin-destination pairs across three years, the residual bucket, the annual totals, and all six hospital-to-hospital acuity codes across three years. Every cell matched to the unit.
HCUP NEDS, DISP_ED	68	67	1	Downloaded 20 AHRQ weighted summary statistics PDFs (2007-2023, plus the revised 2007 and 2011 editions) and parsed the DISP_ED frequency table from each.	hcup-us.ahrq.gov/db/nation/heds/stats/NEDS_(year)_(MaskedStats_Core_Weighted).PDF	67 of 68 exact. The workbook correctly uses the REVISED (revision 2) editions for 2007 and 2011, where AHRQ restated transfers by up to 1.24%. The single gap is data year 2023 destination-unknown, and it is a source-mixing issue, not an error. See Cross_Source_Recon.
HCUP NIS, DISPUNIFORM and TRAN_IN	74	73	1	Downloaded 17 AHRQ NIS weighted summary statistics PDFs and parsed DISPUNIFORM, TRAN_IN, TRAN_OUT and ED_BIRTH.	hcup-us.ahrq.gov/db/nation/nis/tools/stats/	73 of 74 exact. 2008 through 2023 is flawless. TRAN_OUT = 1 equals DISPUNIFORM = 2 in every single year, confirming it is a reliable and not independent evidence. The one gap (the 2007 NIS total omits 16,542 discharged alive, destination unknown) records. 2007 is excluded from every trend statistic.
AHA community hospital admissions	4	4	-	Downloaded the AHA Fast Facts PDFs and read the admissions lines directly.	aha.org, Fast Facts on US Hospitals, 2025 and 2026 editions	EXACT. 2023: total 34,436,650, community 32,345,846. 2024: total 35,658,563, community 33,053,725. The 2023 community figure is the denominator in the 2023 IFT ratio.
NEMSIS 2024 interfacility counts	7	7	-	Downloaded the NEMSIS 2024 National EMS Data Report and parsed the eResponse.05 table.	nemsis.org, NEMSIS End-of-Year Report 2024	EXACT. Total 60,298,684. Hospital-to-Hospital 5,510,664. Other Routine Medical Transport 4,467,204. Hospital to Non-Hospital 1,418,924. Non-Hospital to Non-Hospital 335,432. Non-Hospital to Hospital 247,636. Transports to a hospital ED 27,768,728.
Ambulance Inflation Factor, CY2020-CY2025	6	5	1	Read the full AIF history table in the CMS Claims Processing Manual.	CMS Pub. 100-04, Ch.15 Sec.20.4, Transmittal 13464 (14 Nov 2025)	CORRECTION. CY2024 is 2.6%, not the 3.0% the source market study carried. All other years match. CY2026 (2.0%) added as new.
Ambulance RVUs, seven ground levels	7	7	-	Read the RVU table in both the CMS public use file description and the MedPAC fact sheet.	CMS Ambulance Fee Schedule public use file; MedPAC Payment Basics (Ambulance), Oct 2024, Table 1	EXACT on all seven. BLS 1.00, BLS-emergency 1.60, ALS1 1.20, ALS1-emergency 1.90, paramedic intercept 1.75, ALS2 2.75, SCT 3.25. Upgraded from GOV-B to GOV-A.
GADCS cost per transport	3	2	1	Downloaded the 201-page RAND report and read Table 6.1.	RAND Corporation for CMS, GADCS Report: Year 1 and Year 2 Cohort Analysis, 19 Dec 2024	CORRECTION. Mean \$2.673 and government \$3.127 confirmed. Private for-profit is \$1,788, NOT the \$1,778 that circulates in the secondary coverage.
GADCS crew labor share of cost	1	1	-	Read Figure 5.1 and Table 5.1.	Same report	CONFIRMED at 69.4%: labor is \$18,869M of the \$27.2B aggregated ground ambulance cost base. An earlier draft of this workbook flagged this as an unverifiable conflation. That flag was wrong and has been withdrawn.
Medicare FFS ambulance spend	1	-	1	Read p.1 of the fact sheet, then cross-checked against the raw claims.	MedPAC, Payment Basics: Ambulance Services Payment System, Oct 2024	CORRECTION. Source workbook said \$4.08 for 2023. MedPAC reports \$3.98 of program spending in 2022, excluding beneficiary cost sharing. Reconciled against claims: carrier-billed ground NCH payments were \$3.728 in 2022; air and institutional claims close the gap.
NEDS 2023 dual-table discrepancy	2	2	-	Downloaded the 2023 NEDS Introduction PDF (published 14 Jan 2026) and located both tables.	hcup-us.ahrq.gov/db/nation/heds/NEDSIntroduction2023.pdf	BOTH FIGURES CONFIRMED AS PRINTED. Table A7 reports 21,049,219 ED admissions and 2,552,895 transfers. Table D3, in the SAME document, reports 20,973,122, matching the Weighted Summary Statistics. Table A7's deltas sum to exactly zero: a reallocation, not a revision.
CMS PSPS, 2010-2021 extension	n/a	n/a	n/a	Pulled every claim row carrying the HH origin-destination modifier for twelve additional data years, using the same API. Pre-2021 files drop the PSPS_ prefix on the amount fields.	data.cms.gov, Physician/Supplier Procedure Summary, 2010 through 2021	NEW SERIES. No values to check: this evidence did not exist in any source workbook. Produces the fifteen-year Medicare interfacility series on Medicare_IFT_Series. The 2022-2024 overlap reproduces the verified figures exactly.
MedPAC June 2026 mandated report	n/a	n/a	n/a	Downloaded the 325-page June 2026 Report to the Congress and read Chapter 6, including Tables 6-6, 6-10, 6-11 and Recommendation 6.	MedPAC, Report to the Congress, June 2026, Chapter 6, published 15 June 2026	NEW SOURCE, PUBLISHED AFTER THE SOURCE WORKBOOKS. Supersedes the RAND GADCS unweighted means for operator benchmarking, quantifies the scale effect on cost, and states that GADCS-based revenue-to-cost ratios are not reliable indicators of payment adequacy today.
CMS Timely and Effective Care	n/a	n/a	n/a	Queried the CMS provider-data API for the national, state and hospital files and re-derived every statistic from the returned rows (2,340 hospital records for OP-18d, 1,736 state records).	data.cms.gov/provider-data, datasets ism-hgvy, apyc-v239, yv7e-xt69	NEW SOURCE. Provides the only direct national measurement of interfacility transfer delay: OP-18d, median 287 minutes. Not present in any source workbook.
AHA Fast Facts, 2020 edition	n/a	n/a	n/a	Downloaded and read the PDF for the survey-year 2018 baseline.	aha.org, Fast Facts on U.S. Hospitals, 2020 edition	NEW SOURCE. Community hospitals 5,198, in a system 3,491; staffed beds 792,417. Enables the 2018 to 2024 consolidation series.
Peer-reviewed literature (Pubmed)	n/a	n/a	n/a	Searched PubMed and retrieved article metadata and abstracts for four studies on interfacility transfer, travel time and ED boarding outcomes.	PubMed. DOIs recorded on Source_Register S35 through S38.	NEW SOURCE. Quantifies travel time as a transfer driver (aOR 1.32) and establishes that the naive dwell-time to mortality correlation runs backwards.
TOTAL	233	228	5	Match rate		97.9% Five corrections, all applied in this file. Four are real errors in the source workbooks; the fifth is a source-mixing issue on one AHRQ cell.

Panel B. The five corrections, stated plainly.

#	What the source workbook said	What the primary document says	Primary document	Materiality
1	CY2024 Ambulance Inflation Factor = 3.0%	CY2024 Ambulance Inflation Factor = 2.6%	CMS Pub 100-04 Ch.15 Sec.20.4	Moves the trailing three-year average from 4.7% to 2.3%. Material to any rate-growth assumption.
2	Medicare FFS ambulance spend = \$4.08 in 2023	\$3.98 in 2022, program spending excluding beneficiary cost sharing	MedPAC Payment Basics (Ambulance), Oct 2024, p.1	Wrong value and wrong year. A separate MedPAC figure of \$5.38 for 2023 is TOTAL payments. Confusing the two is a 36% error.

3	GADCS private for-profit mean cost per transport = \$1,778	RAND GADCS Year 1 and Year 2 Cohort Analysis, Table 6.1	Small. But it propagated from secondary coverage into the source workbook, which is the point.
4	GADCS mean cost per transport of \$2,673, quoted Mean \$2,673, MEDIAN \$1,340 without its median	Same report, Table 6.1	Large. The mean is dragged up by thousands of tiny low-volume organizations. Quoting the mean as the operator benchmark is wrong.
5	NIS 2007 total discharges = 39,525,405	39,541,947. The workbook omits 16,542 'destination unknown' records	AHRQ NIS 2007 weighted summary statistics
	6 This workbook claimed MedPAC's \$3.9B and \$5.3B ambulance spend figures 'both reconcile against the claims once air and institutional claims are added.'	They do not. Carrier-billed ground plus air program payments were \$4,20B in 2022. ABOVE MedPAC's \$3.9B. Neither MedPAC document states the scope of its measure.	CMS PSPS, air codes A0430, A0431, A0435, A0436
			Material to anyone sizing the Medicare ambulance market. The claim is withdrawn and replaced with a bounded statement. See Medicare_PSPS Panel G.

Panel C. Evidence obtained during verification that exists in NO source workbook.

Item	Value	Where it came from	Why it matters
CY2024 ground ambulance conversion factor	\$272.44	MedPAC Payment Basics (Ambulance), Table 1	The source workbook ASSUMED \$278.00 and labeled it ILLUSTRATIVE, which is why no dollar figure could survive.
CY2024 ground mileage rate	\$8.76 per statute mile	Same	Confirmed independently by the claims: \$8.79 allowed per loaded mile.
Allowed charges, hospital-to-hospital base rate, 2024	\$319,615,284	CMS PSPS, PSPS_ALLOWED_CHARGE_AMT	Real Medicare dollars, not an RVU-times-conversion-factor estimate.
Allowed charges, hospital-to-hospital mileage, 2024	\$267,413,039	CMS PSPS, HCPCS A0425, modifier HH	Mileage is 44.6% of the revenue on an interfacility transport.
Loaded miles per hospital-to-hospital transport, 2024	33.6	CMS PSPS	Roughly three times a typical 911 scene run.
Medicare payment as a share of allowed	77.7%	CMS PSPS, PSPS_NCH_PAYMENT_AMT	Statute sets Part B at 80% after the deductible. The realized 77.7% validates the entire extraction.
GADCS median cost per transport	\$1,340	RAND GADCS Table 6.1	The mean of \$2,673 is 1.99x the median. This changes the unit-economics conclusion.
GADCS cost per RVU	\$1,624 mean, \$811 median	RAND GADCS Table 6.1	Directly comparable to the \$253 per RVU that Medicare allows on the HH book.
GADCS cost composition	Labor 69.4%, vehicles 10.0%, facilities 3.5%	RAND GADCS Figure 5.1 and Table 5.1	Replaces the excluded trade-benchmark cost split.
NEDS 2023 sampling structure	4,489 target EDs; 3,185 frame; 963 sampled	NEDS Introduction 2023, Table A6	Quantifies the 2023 frame-coverage break.

Panel D. What remains unverifiable, by anyone, at any effort.

AHRQ's discharge weights are calibrated to the AHA Annual Survey Database, which is not public at the hospital level. The weighted counts in NEDS and NIS cannot be reproduced from public data by any external party. They are taken on AHRQ's authority, and everything downstream inherits that. AHRQ publishes no standard errors in the weighted summary statistics. NEDS clusters on emergency departments, and transfer propensity is a hospital-level attribute, so DISP_ED = 2 carries a large design effect. The +0.86% per year total-episode CAGR is NOT demonstrably different from zero on this evidence. HCUPnet returns design-based standard errors, free and without a data use agreement, and would settle it.

NEMESIS has no sampling frame and no weights. It is a voluntary submission census. No rational inference and no confidence interval is licensed from it, only mix and direction. CMS suppresses any PSPS cell with fewer than 11 services. In 2024, 24,781 origin-destination-coded rows are masked, hiding at most about 263,000 services, or 2.5% of the 10,637,727 counted. Every PSPS figure in this workbook is a floor by that margin. PSPS is carrier claims only. Hospital-owned ambulance departments bill institutionally and never appear, and Medicare Advantage is entirely absent. The commercial book is not measured by any source in this workbook.

Panel E. The one authoritative source published after this evidence base was assembled.

MedPAC delivered its Bipartisan Budget Act of 2018 mandated report on 15 June 2026: 'Chapter 6: Mandated report: Assessment of the Medicare Ground Ambulance Data Collection System', in the June 2026 Report to the Congress. It assesses the GADCS data and recommends continued but streamlined collection. It is the authoritative source on Medicare payment adequacy for ground ambulance. It IS reflected in this workbook on MedPAC_2026_Mandated [state sentence corrected in v3: Panel A row 14 and source S31 record the report as pulled, read, and incorporated].

Reproducibility. PSPS: GET https://data.cms.gov/data-qa/v1/dataset/{uuid}/data?filter={HCPCS_CD}={code}. Dataset UUIDs: 2022 e4786716-10e9-4092-b936-7444177474d5, 2023 b72aeffb-22cf-4241-9c64-050ad9061e03, 2024 1646c736-4179-4100-9f79-592b69e41975. Ground base codes A0426, A0427, A0428, A0429, A0433, A0434; ground mileage A0425. Suppressed cells are marked with an asterisk and were excluded from sums, not imputed.

Panel F. Revision 2 verification (10 July 2026): new evidence opened, and the revision 1 extraction independently re-tested.

Evidence layer	What was done	Outcome
CMS PSPS full O-D rebuild (S39, S40)	Pulled 392,409 raw rows across fifteen PSPS datasets (2019-2024 all twelve ambulance HCPCS; 2010-2018 air codes) and re-aggregated every origin-destination pair with dollars.	PERFECT on dollars: 2022-2024 ground totals reproduce Medicare. PSPS Panel F to the cent (allowed 4,753,662,348 / 4,992,692,676 / 5,073,090,003; paid 3,722,057,066 / 3,892,079,508 / 3,957,457,371). HH 2024 base allowed and mileage allowed match to the cent.
Fractional service counts	Compared initial hospital-pair transport counts against float-summed rebuild; isolated the gap.	Correction 7. PSPS SUBMITTED_SERVICE_CNT contains fractional values (8 rows in the 2024 HH base set). The initial pipeline truncated: 871.753 against a float sum of 871.756 (+0.0003%). All revision 2 tabs sum floats. original cells left as printed; immaterial.
PSPS field-name boundary	Tested the 2020 dataset schema directly.	Correction 8, to an earlier reproducibility note. The PSPS_ field prefix begins with DATA_YEAR 2020, not 2021 as an earlier note stated. 2010-2019 files carry unprefixed names.
Census NP2023-T2 (S50)	Downloaded and parsed the xlsx from www2.census.gov.	EXACT values carried: 65+ 57,795k (2022), 71,183k (2030), 82,130k (2050), main series.
NHAMCS 2022 Table 4 (S55)	Downloaded the ED Summary Tables PDF and read Table 4.	EXACT: 17.0% (SE 0.9) ambulance arrival on 155,398k visits; age gradient carried. Final survey vintage.
Medicare Coverage Database (S56)	Queried active LCDs and articles titled ambulance via the MCD connector.	Zero active ambulance LCDs; one billing article, A52917 (Noridian Part A, rural air ambulance protocols, eff. 10/01/2015, updated 11/22/2023).

NPPES (S58)	Queried the registry live for ambulance taxonomy organizations.	Taxonomy 34160000X and 3416L0300X confirmed on active records (example NPIs on Code_Crosswalks).
PubMed retrievals (S42, S53, S54)	Retrieved metadata and abstracts for the three studies carried in v2.	PMIDs 359777218, 31989591, 41324958 with digital object identifiers recorded on the Source_Register.
Documents opened (S41, S43-S49, S51, S52)	CMS RSNAT page and press release; FAIR Health white papers and brief; GAPB report; Commonwealth Fund analysis; MACPAC Ch.5; HHS NEMT report; KFF 2026 enrollment brief; Sheps tracker.	Figures carried as printed, with each locator on the Source_Register. The Sheps and Charis closure counts use different conventions and are both recorded; DQ #31.

Panel G. Revision 3 verification (10 July 2026).

Evidence layer	What was done	Outcome
Medicare Monthly Enrollment (S60)	Pulled national annual-average rows 2013-2025 from the API and carried four enrollment measures.	EXACT values carried; A and B FFS denominator convention stated on the tab. Per-1,000 rates recompute inside the workbook by formula.
MUP-PHY supplier pull (S59)	Pulled every NPI-by-code row for eight ambulance base HCPCS, 2019 and 2024 (44,614 rows), and aggregated counts, concentration and top billers.	Cross-check: MUP 2024 approved ground services 9,542,103 against PSPS submitted 10,637,766 + 1.1x definitional gap (final-action against submitted; suppression), logged as DQ #34.
Acuity by channel (S39)	Re-aggregated the existing PSPS pull by base HCPCS within each D-D channel, 2019 and 2024, six-code convention.	Totals tie to Medicare_OD_Matrix (A0432 intercept excluded, 3,312 services in 2024). Two draft read-notes were corrected against computed values before release (HH per-capita minus 8.0%; HH emergency share +2.4 points).
Add-on statutory status (S61)	Read the CMS AFS page and corroborating notices for the current expiration.	CAA 2026 section 6203: extended through 31 December 2027; cliff 1 January 2028; Oct 2025 lapse history recorded on Payment_Rules.

Panel H. Revision 4 verification (10 July 2026).

Evidence layer	What was done	Outcome
MUP-PHY decade series (S62)	Pulled every NPI-by-code row, six ground base codes, data years 2014-2018 and 2020-2023, plus 2024 city, state and ZIP fields. Dataset identifiers by data year are recorded under source S62.	Eleven-year supplier series on Supplier_Series_Raw. Endpoints reconcile exactly with the revision 3 two-point pull.
Institutional ambulance route (Fault_Audit A)	Probed the Medicare Outpatient Hospitals by Provider and Service PUF with HCPCS filters for all twelve ambulance codes.	Tested and closed: the file is APC-level and the HCPCS filter is silently ignored (identical 116,799-row universe returned for every code). Hospital-billed ambulance remains publicly uncountable; recorded on IFT_Alt_Measures.
OEWS May 2019 (S67)	Read the 29-2040 occupation page: national, industry, state, metro tables.	EXACT values carried; May 2024 taken from the OOH (S68). BLS bulk zips refused scripted access; full state files pending (P4).
Scissors self-correction (Finding 39)	The initial working hypothesis was that wages outran the AIP 2019-2024. Completed: EMT median +18.6% against AIF +18.6%.	Hypothesis rejected on the five-year window and the finding rewritten to the sub-period and composition mechanisms. Rejected hypotheses are retained as part of the audit trail.
California APOT (S71)	Parsed the EMSA December 2024 commission report: statewide trend, LEMSA heat table, cumulative delay hours.	EXACT: 19.1% of offloads over 30 minutes; 13,364:40:19 cumulative hours, Apr-Sep 2024; San Diego alone 738:54:57.
Ambulance deserts (S65)	Verified the national figures on the Rural Health Research Gateway; the full chartbook PDF is pending retrieval.	Headline figures pinned including the nine-missing-states undercount caveat; state and county tables pending (P2).
MSA mapping coverage	Computed the share of national volume the principal-city method assigns to a CBSA.	21.8%. Printed on MSA_Landscape as the governing caveat; the full ZCTA crosswalk is pending item P3.
Access failures, logged not hidden	Four bulk files could not yet be retrieved programmatically: the BLS OEWS archives, the Census ZCTA relationship file, the NASEMSO assessment, and the deserts chartbook PDF.	Each is named on Fault_Audit item L with its fallback source. Nothing was silently substituted.
State cut cross-validation (S73)	Pulled the official by-Geography state file and reconciled against the NPI-file aggregation.	National gap +1.0% (9,637,009 vs 9,542,103), attributable to NPI-file small-cell suppression: the suppression tail is now BOUNDED, closing part of Fault_Audit item C and DQ #32. Official column carried on State_Saturation.

Final consolidation, 10 July 2026	Confirmed the distributed copy is byte-identical to this build; recounted the Source_Index tiers programmatically; corrected the stale 61-source title; repaired one gap-table cost cell; refreshed the HCUPnet claims across all tabs after the independent external review.	The master is internally consistent at 47 tabs, 77 sources [corrected in v3: the rev-5 note undercounted the four client-alignment/connector tabs and sources S74-S77], facts through F165, findings through 41, zero formula errors. Open items live only on the README pending register (P1-P19).
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Chart integrity audit	Every chart series was programmatically resolved to its cell range and counted for numeric content; scale-mismatched two-series charts were rebuilt as dual-axis combinations so percentage series render visibly; all ten original charts received explicit titles.	29 charts, all series resolving, palette-consistent. Percentage series that previously flatlined against count axes now plot on a right-hand axis.
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Client-alignment layer	Two tabs added from the engagement kickoff guidance: the two-build sizing structure with its segmentation, gates, and archetype specification, and the source-to-question map. No client-specific figures were ingested; every awaited input is a labeled empty cell.	Sizing_Playbook and Engagement_Data_Map live after Imbalance_Ledger; the README map carries the new section.
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Connector layer	The workbook now draws on the live healthcare connectors attached to this engagement workspace. The PubMed connector retrieved and pinned three national NEDS transfer studies (S74 to S76) with full identifiers; the NPI Registry connector is loaded for on-demand supplier verification against NPPES; the PrognHIVE surveillance connector is available for a demand-seasonality layer pending approval to invoke it.	Condition_Transfer_Anchors added with three Tier A anchors; pending item P1 partially advanced.
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Analytical chart set	Four charts added that read relationships rather than levels: the state structure scatter (supplier density against utilization), the concentration two-line showing movement confined to post-2021, the scheduled-transport teller exit, and the dual-axis pairing of the shrinking measurable book with the climbing dark share.	33 charts, all series verified resolving.
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National TAM model	Built the three-scenario national model on a fully measured Medicare spine (claims, MedPAC, enrollment, code mix, all green-linked), a residual-based payer-lives panel (S77), a nine-row assumption register with <i>numbers and replacement notes, four notes</i>	Outputs live on TAM_Model_National Panel A; Fault_Audit item T governs quotation.
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Panel I (v3). Faithful-copy proof: v2.7 carried forward without loss.

Method	Every v2.7 sheet copied cell-by-cell (values, formulas, styles, comments, merges, widths, freeze panes) by the committed v2 pipeline.	
Cells compared	10,463	
Differences	0	
Excel error cells after recalc	0	
Charts	All 37 v2.7 chart objects re-created natively from their parsed XML (openpyxl loses chart objects on round-trip); series references	

Panel J (v3). Live-pull verification: every dataset pulled for the new tabs, with row counts and cross-checks. Full reproduce specs on Pull_Manifest.

Dataset	Artifacts	Rows cached	Access date (UTC)
(re-registered by reconcile_manifest after a concurrent-write race; original pull meta lost, payload hash is of the surviving canonical file)		26	24,813 2026-07-10
BLS Occupational Employment and Wage Statistics (OEWS), May 2024, state file - EMS occupations		1	144 2026-07-10
BLS QCEW NAICS 621910 (Ambulance Services), annual averages		12	1,208 2026-07-10
BLS QCEW NAICS 621910 (Ambulance Services), county annual averages, private ownership		12	21,760 2026-07-10
BLS QCEW NAICS 621910, quarterly		48	4,832 2026-07-10

CDC PLACES: County Data (2023 release) - chronic kidney disease	1	3,077 2026-07-10
CDC PLACES: Local Data for Better Health, County Data	5	14,790 2026-07-10
CMS Acute Hospital Care at Home Program - Approved List of Hospitals as of 4/5/2021 (only public machine-readable roster); current roster PENDING JS portal + DUA-gated)	1	27 2026-07-10
CMS Hospital Enrollments (PECOS public enrollment file), Rural Emergency Hospital conversions	1	48 2026-07-10
CMS Provider Data Catalog: Dialysis Facilities	1	7,557 2026-07-10
CMS Provider Data Catalog: Dialysis Facilities (full roster)	1	7,557 2026-07-10
CMS Provider Data Catalog: Dialysis Facilities (rich fields)	1	7,557 2026-07-10
CMS Provider Data Catalog: Dialysis Facility - Listing by Facility	1	7,557 2026-07-10
CMS Provider Data Catalog: Home Health Agencies (rich fields)	1	12,392 2026-07-10
CMS Provider Data Catalog: Home Health Care Agencies	1	12,392 2026-07-10
CMS Provider Data Catalog: Hospice Providers (rich fields)	1	6,852 2026-07-10
CMS Provider Data Catalog: Hospice - General Information	1	6,852 2026-07-10
CMS Provider Data Catalog: Hospital General Information	1	5,432 2026-07-10
CMS Provider Data Catalog: Hospital General Information (rich fields)	1	5,432 2026-07-10
CMS Provider Data Catalog: Inpatient Rehabilitation Facilities (rich fields)	1	1,222 2026-07-10
CMS Provider Data Catalog: Inpatient Rehabilitation Facility - General Information	1	1,222 2026-07-10
CMS Provider Data Catalog: Long-Term Care Hospital - General Information	1	311 2026-07-10
CMS Provider Data Catalog: Long-Term Care Hospitals (rich fields)	1	311 2026-07-10
CMS Provider Data Catalog: Nursing homes including rehab services (rich fields)	1	14,695 2026-07-10
CMS Provider Data Catalog: Nursing homes including rehab services - Provider Information	1	14,695 2026-07-10
CMS Provider Data Catalog: Rural Emergency Hospital Timely and Effective Care - Hospital (deduped roster)	1	41 2026-07-10
CMS Provider Data Catalog: Skilled Nursing Facility Quality Reporting Program - Provider Data	1	14,695 2026-07-10
CMS Provider Data Catalog: Timely and Effective Care - Hospital	3	79,220 2026-07-10
CMS Zip Code to Carrier Locality File (ambulance fee schedule rural/super-rural ZIP designations)	1	42,956 2026-07-10
Census Bureau 2024 Gazetteer Files - ZIP Code Tabulation Areas (national)	1	33,791 2026-07-10
Census Bureau Vintage 2024 County Population Estimates by Age and Sex (CC-EST2024-AGESEX)	1	15,720 2026-07-10
Census Bureau Vintage 2024 State Population Estimates by Age and Sex, civilian (SC-EST2024-AGESEX-CIV)	1	4,524 2026-07-10
EIA Weekly Retail On-Highway Diesel Prices (psw18vw*, all areas)	1	308 2026-07-10
HHS OIG List of Excluded Individuals/Entities (LEIE)	1	566 2026-07-10
Hospital Service Area	1	7,452 2026-07-10
Hospital Service Area (top-15 ZIP corridors per hospital)	1	106,799 2026-07-10
Market Saturation & Utilization State-County	1	2,340 2026-07-10
Market Saturation & Utilization State-County (county grain)	6	54,756 2026-07-10
Market Saturation & Utilization State-County (county grain, rolling window)	9	82,540 2026-07-10
Medicare Fee-For-Service Public Provider Enrollment	2	20,930 2026-07-10
Medicare Monthly Enrollment	2	767 2026-07-10
Medicare Physician & Other Practitioners - by Geography and Service	156	6,315 2026-07-10
Medicare Physician & Other Practitioners - by Provider and Service	144	389,793 2026-07-10
Physician/Supplier Procedure Summary	105	897,270 2026-07-10
UNC Sheps Center rural hospital closures list (2005-present, complete + converted closures)	1	197 2026-07-10
USDA ERS 2010 Rural-Urban Commuting Area Codes - ZIP code file (approximation)	1	41,164 2026-07-10
USDA ERS 2010 Rural-Urban Commuting Area Codes - census tract file (revised 7/3/2019)	1	74,002 2026-07-10

Cross-check	PECOS enrolled ambulance suppliers: live found, rows 10,465 = vendored PPEF
Cross-check	NOTE: 24,000 rows in the table is 10,465. EMAP-IT MUP 2023 national AHA25 matches the independent probe made during connector

Panel K (v3). Recompute audit: every new DERIVED formula recomputed independently in python from the cached source data, and the whole workbook recalculated to zero error cells before release.

Formulas recalculated (whole workbook)	55,295
Excel error cells	0
New derived cells recomputed in python	9
Mismatches beyond 1e-9 relative tolerance	0
Printed-page estimate (landscape, fit-to-width)	13,122

Fact ledger

The question: where does each number come from? Every fact in this workbook carries an ID here (F01 to F165) with its metric, year, value, evidence tier, source, and the exact locator inside the source document. This is step two of the three-step audit path described on the README.

Fact ID	Metric	Year	Value	Unit	Basis	Tier	Source ID	Locator	Lives on	Cross-check
F01	ED-origin acute-to-acute transfer episodes	2022	2,311,549 episodes		GOV-A	A	S01	NEDS Summary Stats, DISP_ED = 2	Acute_IFT_Series	VERIFIED against the AHRQ PDF. Agrees to the unit with NEDS Introduction 2022 Table A6
F02	ED-origin acute-to-acute transfer episodes	2023	2,554,379 episodes		GOV-A	A	S01	NEDS Summary Stats, DISP_ED = 2	Acute_IFT_Series	VERIFIED. NEDS Introduction 2023 Table A7 prints 2,552,895. Gap of 1,484 (0.058%). See Cross_Source_Recon
F03	Inpatient-origin acute-to-acute transfer episodes	2022	617,724 episodes		GOV-A	A	S04	NIS Summary Stats, DISPUNIFORM = 2	Acute_IFT_Series	VERIFIED. TRAN_OUT = 1 equals this value in every year 2007-2023: a relabel, not independent evidence
F04	Inpatient-origin acute-to-acute transfer episodes	2023	630,765 episodes		GOV-A	A	S04	NIS Summary Stats, DISPUNIFORM = 2	Acute_IFT_Series	VERIFIED against the AHRQ PDF
F05	Total acute-to-acute transfer episodes	2022	2,929,273 episodes		DERIVED	A	S01, S04	Sum of F01 and F03	Acute_IFT_Series	No overlap: an ED transfer-out is never also an inpatient discharge
F06	Total acute-to-acute transfer episodes	2023	3,185,144 episodes		DERIVED	A	S01, S04	Sum of F02 and F04	Acute_IFT_Series	Memo year. NEDS break in series
F07	Mean acute-to-acute transfer episodes, 2019 to 2023	2023	3,061,156 episodes/yr		DERIVED	A	S01, S04	Five-year mean	Acute_IFT_Series	Sample standard deviation 139,226. Range 2,929,273 (2022) to 3,234,266 (2019)
F08	IFT as % of AHA community hospital admissions	2022	9.28% %		DERIVED	A	S01, S04, S09	F05 over F18	Acute_IFT_Series	Denominator choice moves this by 1.05 points. See Receiving_Side
F09	IFT as % of AHA community hospital admissions	2023	9.85% %		DERIVED	A	S01, S04, S10	F06 over 32,345,846	Acute_IFT_Series	Memo year. Denominator verified in AHA Fast Facts 2025 edition
F10	Mean IFT % of admissions, 2019 to 2023	2023	9.49% %		DERIVED	A	S01, S04, S06-S10	Five-year mean	Acute_IFT_Series	Band 9.24% (2021) to 9.85% (2023). Sample standard deviation 0.24 points
F11	ED-origin share of acute-to-acute episodes	2012	74.4% %		DERIVED	A	S01, S04		Acute_IFT_Series	First comparable year after the 2012 NIS redesign
F12	ED-origin share of acute-to-acute episodes	2022	78.9% %		DERIVED	A	S01, S04		Acute_IFT_Series	Rises 4.5 points over ten years
F13	ED-origin share of acute-to-acute episodes	2023	80.2% %		DERIVED	A	S01, S04		Acute_IFT_Series	The clearest structural trend in the data
F14	ED-origin episode CAGR, 2012 to 2023	2023	1.55% %/yr		DERIVED	A	S01	11 compounding years	Acute_IFT_Series	2012 is the first comparable year (NIS redesign)
F15	Inpatient-origin episode CAGR, 2012 to 2023	2023	(1.48%) %/yr		DERIVED	A	S04		Acute_IFT_Series	Negative. The inpatient channel is in structural decline
F16	Total acute-to-acute episode CAGR, 2012 to 2023	2023	0.86% %/yr		DERIVED	A	S01, S04		Acute_IFT_Series	The measured market volume growth rate. NOT demonstrably different from zero: AHRQ publishes no standard errors
F17	AHA community hospital admission CAGR, 2012 to 2023	2023	(0.67%) %/yr		DERIVED	A	S05-S10		Acute_IFT_Series	Negative. IFT grows as the denominator shrinks. This one is a hospital census, so it is real
F18	AHA community hospital inpatient admissions	2022	31,555,807 admissions		GOV-A	A	S09	Fast Facts, 2024 edition	Acute_IFT_Series	Identical in both uploaded AHA workbooks
F19	AHA community hospital inpatient admissions	2024	33,553,725 admissions		GOV-A	A	S11	Fast Facts, 2026 edition	Cost_and_Capacity	VERIFIED against the AHA PDF: 33,553,725. No NEDS 2024 file yet, so no 2024 ratio
F20	Total admissions, all US hospitals	2024	35,658,583 admissions		GOV-A	A	S11	Fast Facts, 2026 edition	Cost_and_Capacity	VERIFIED: 35,658,583. Still 1.9% below the 2018 peak of 36,353,946
F21	Receiving-side transfer admissions (NIS TRAN_IN = 1)	2022	2,205,490 admissions		GOV-A	A	S04	NIS Summary Stats, TRAN_IN = 1	Receiving_Side	VERIFIED. Derived from UB-04 Point of Origin, independent of the sending-side field
F22	Receiving-side transfer admissions (NIS TRAN_IN = 1)	2023	2,463,771 admissions		GOV-A	A	S04	NIS Summary Stats, TRAN_IN = 1	Receiving_Side	VERIFIED against the AHRQ PDF
F23	Sending-to-receiving conversion rate	2022	75.3% %		DERIVED	A	S01, S04	F21 over F05	Receiving_Side	Residual goes to psych, rehab, LTACH, ED treat-and-release, or observation
F24	Sending-to-receiving conversion rate	2023	77.4% %		DERIVED	A	S01, S04		Receiving_Side	
F25	NEMSIS total EMS activations	2024	60,298,684 activations		GOV-A	A	S12	2024 report, eResponse.05 table	EMS_Transports	VERIFIED against the NEMSIS PDF: 60,298,684. Cover box ties exactly only in 2024

F26	NEMSIS Hospital-to-Hospital Transfer activations	2024	5,510,664 transports	GOV-A	A	S12	2024 report, eResponse.05, v3.5	EMS_Transports	VERIFIED: 5,510,664. Includes psych, rehab and LTACH destinations plus return legs. Not the HCUP definition
F27	NEMSIS all facility-to-facility transfer activations	2024	7,512,656 transports	DERIVED	A	S12	Four v3.5 transfer categories	EMS_Transports	All four components verified. Excludes Other Routine Medical Transport
F28	NEMSIS Other Routine Medical Transport activations	2024	4,467,204 transports	GOV-A	A	S12	2024 report, eResponse.05	EMS_Transports	VERIFIED: 4,467,204. Dialysis, appointments, return home. NOT interfacility transfer
F29	NEMSIS Interfacility Transport activations (v3.4)	2019	3,309,316 activations	GOV-A	B	S12	2019 report, eResponse.05, v3.4	EMS_Transports	Coverage was 47 states. Not comparable to 2024
F30	NEMSIS Interfacility Transport activations (v3.4)	2023	4,179,580 activations	GOV-A	B	S12	2023 report, eResponse.05, v3.4	EMS_Transports	2023 NEMSIS denominator is understated. See Data_Quality_Register
F31	NEMSIS transports to a hospital ED	2022	24,016,095 transports	GOV-A	B	S12	2022 report, eDisposition.21	EMS_Transports	17.5% of NEDS weighted ED visits, against the NHAMCS anchor of 16.2%
F32	NEDS weighted national ED visits	2022	136,974,618 visits	GOV-A	A	S02	NEDS Introduction 2022, Table 1	EMS_Transports	AHA Annual Survey reports the identical figure; it is the NEDS control total
F33	NEDS weighted national ED visits	2023	142,193,558 visits	GOV-A	A	S03	NEDS Introduction 2023, Table A6	EMS_Transports	VERIFIED: 142,193,558. The nine DISP_ED categories sum to this exactly
F34	NHAMCS ambulance-arrival share of ED visits	2016	16.15% %	DERIVED	B	S25	NHAMCS 2016 ED Summary Tables, Table 5	EMS_Transports	The only external anchor. NEDS carries no arrival-mode element
F35	Medicare FFS hospital-to-hospital ambulance transports	2022	825,178 transports	GOV-A	A	S13	PSPS 2022, modifier HH	Medicare_PSPS	REBUILT from raw CMS claims and matched to the unit. Carrier claims only
F36	Medicare FFS hospital-to-hospital ambulance transports	2024	871,753 transports	GOV-A	A	S13	PSPS 2024, modifier HH	Medicare_PSPS	REBUILT from raw CMS claims and matched to the unit
F37	Medicare FFS hospital-to-hospital transport CAGR, 2022 to 2024	2024	2.78% %/yr	DERIVED	A	S13		Medicare_PSPS	Positive while total FFS ambulance volume is negative
F38	All Medicare FFS ambulance transport CAGR, 2022 to 2024	2024	(2.80%) %/yr	DERIVED	A	S13		Medicare_PSPS	Enrollment shift to Medicare Advantage
F39	Non-emergency BLS and ALS1 share of Medicare H2H transports	2024	62.3% %	DERIVED	A	S13	A0428 + A0426 over total	Medicare_PSPS	Contradicts the excluded 56.4% high-acuity assumption
F40	Specialty Care Transport share of Medicare H2H transports	2024	6.6% %	DERIVED	A	S13	A0434 over total	Medicare_PSPS	6.6% of transports. See F60 for the share of allowed dollars
F41	ALS2 share of Medicare H2H transports	2024	0.7% %	DERIVED	A	S13	A0433 over total	Medicare_PSPS	The only H2H acuity tier declining in absolute volume
F42	Emergent share of Medicare H2H transports	2024	30.3% %	DERIVED	A	S13	A0427 + A0429 over total	Medicare_PSPS	Fastest-growing acuity tiers, still under a third of volume
F43	Volume-weighted RVU of the Medicare H2H book	2024	1.45 RVU	DERIVED	A	S13, S14	SUMPRODUCT of RVU and volume	Medicare_PSPS	The book prices at 1.45 BLS runs, not 3.25
F44	Ambulance Inflation Factor, latest (CY2026)	2026	2.0% %/yr	GOV-A	A	S29	CMS Pub 100-04, Ch.15 Sec.20.4, Transmittal 13464	Payment_Rules	VERIFIED 09 Jul 2026 against the CMS manual
F45	Ambulance Inflation Factor, trailing 3-year average (2024 to 2026)	2026	2.3% %/yr	DERIVED	A	S29		Payment_Rules	Source workbook had CY2024 at 3.0%. Actual is 2.6%. Corrected
F46	Ambulance Inflation Factor, 2020 to 2026 average	2026	3.1% %/yr	DERIVED	A	S29		Payment_Rules	Sample standard deviation 2.9 points. The AIF is volatile
F47	Specialty Care Transport RVU (A0434)	2026	3.25 RVU	GOV-A	A	S14, S28	CMS AFS public use file; MedPAC Table 1	Payment_Rules	VERIFIED. 3.25 times the BLS base of 1.00. All seven ground RVUs matched
F48	Mean ground ambulance cost per transport, all-in	2024	\$2,673 \$/transport	GOV-A	A	S30	GADCS Table 6.1, all NPIs, N = 5,000	Cost_and_Capacity	VERIFIED. The MEDIAN is \$1,340 (F73) and is the number to quote for a scaled operator
F49	Mean cost per transport, private for-profit	2024	\$1,788 \$/transport	GOV-A	A	S30	GADCS Table 6.1	Cost_and_Capacity	CORRECTED to \$1,788. The \$1,778 in circulation is a secondary-source transcription error
F50	Crew labor share of aggregated ground ambulance cost	2024	69.4% %	GOV-A	A	S30	GADCS Figure 5.1 and Table 5.1	Cost_and_Capacity	VERIFIED at 69.4%: \$18,869M of \$27.2B. Distinct from the 70% labor-related portion of the base payment
F51	Crew labor share of cost, for-profit organizations	2024	60.5% %	GOV-A	A	S30	GADCS Table 5.1	Cost_and_Capacity	Government-owned run 73.6%. Ownership drives the cost structure
F52	National hospital inpatient occupancy	2022	65.49% %	SOURCED	C	S23	HCRIS 2552-10, 5,888 fillers	Cost_and_Capacity	Analyst computation on a vendored CMS panel, not a published figure. Not externally verifiable
F53	Inpatient occupancy change, FY2020 to FY2022	2022	3.42 pp	DERIVED	C	S23		Cost_and_Capacity	A lagging throughput proxy, not a transfer count
F54	Total certified post-acute destinations	2024	35,481 providers	SOURCED	C	S24	CMS provider rolls	Cost_and_Capacity	SNF 14,699; HHA 12,392; hospice 6,852; IRF 1,221; LTACH 317
F55	ICD-10-CM codes in the clinical spine validated as real	2026	92 codes	SOURCED	B	S27	NLM Clinical Tables	Clinical_Volumes_TierC	A coding-integrity check. It validates no volume figure
F56	CY2024 ground ambulance conversion factor	2024	\$272.44 \$/RVU	GOV-A	A	S28	MedPAC Payment Basics (Ambulance), Oct 2024, Table 1	Payment_Rules	VERIFIED. The source workbook assumed \$278.00 and labeled it ILLUSTRATIVE
F57	CY2024 ground mileage rate	2024	\$8.76 \$/statute mile	GOV-A	A	S28	MedPAC Payment Basics (Ambulance), Oct 2024	Payment_Rules	Corroborated by the claims: \$8.79 allowed per loaded mile (F69)
F58	Base-rate allowed charges, Medicare FFS hospital-to-hospital	2024	\$319,815,284 \$	GOV-A	A	S13	PSPS 2024, PSPS_ALLOWED_CHARGE_AMT, modifier HH	Medicare_PSPS	Pulled from the CMS claims API. Not present in any source workbook
F59	TOTAL allowed charges (base + mileage), Medicare FFS hospital-to-hospital	2024	\$577,228,322 \$	DERIVED	A	S13	Base rate plus A0425 mileage	Medicare_PSPS	\$577.2M in 2024. The only assumption-free dollar figure in this workbook

F60	Specialty Care Transport share of H2H base-rate ALLOWED DOLLARS	2024	15.6% %	DERIVED	A	S13	A0434 allowed over total allowed	Medicare_PSPS	6.6% of transports, 15.6% of base-rate allowed dollars. Price weighting narrows the gap, it does not close it
F61	Ambulance Inflation Factor, CY2026	2026	2.0% %/yr	GOV-A	A	S29	Transmittal 13464, 14 Nov 2025	Payment_Rules	CPI-U 2.7% less a total factor productivity adjustment of 0.7%
F62	Non-emergency BLS ESRD dialysis payment reduction	2024	23.0% %	GOV-A	A	S28	MedPAC Payment Basics (Ambulance), Oct 2024	Payment_Rules	Applies to the RJ, JR, NJ and JN modifier volumes on Medicare_PSPS Panel A, not to HH
F63	Transfers to another facility type, ED plus inpatient origin	2022	6,509,053 legs	DERIVED	A	S01, S04	DISP_ED = 5 plus DISPUNIFORM = 5	Other_Transfer_Channels	2.2 times the acute-to-acute count. Not all move by ambulance
F64	Total transfers out, all destinations	2022	9,438,326 legs	DERIVED	A	S01, S04		Other_Transfer_Channels	The widest sending-side count HCUP supports
F65	ED-originated share of all community hospital admissions	2022	63.5% %	DERIVED	A	S01, S05-S09	DISP_ED = 9 over AHA admissions	Other_Transfer_Channels	52.9% in 2007 rising to 63.5% in 2022. Confirms NEDS and AHA describe the same hospital universe
F66	Allowed charges per Medicare FFS hospital-to-hospital transport	2024	\$662.15 \$/transport	DERIVED	A	S13	Total allowed over transports	Medicare_PSPS	\$662.15, against a GADCS median cost per transport of \$1,340
F67	Ground mileage allowed charges, hospital-to-hospital	2024	\$257,413,039 \$	GOV-A	A	S13	PSPS 2024, HCPCS A0425, modifier HH	Medicare_PSPS	44.6% of total allowed dollars on the book
F68	Loaded miles per hospital-to-hospital transport	2024	33.6 miles	DERIVED	A	S13	Mileage units over transports	Medicare_PSPS	33.6 miles. Roughly three times a typical 911 scene run
F69	Allowed per loaded mile	2024	\$8.79 \$/mile	DERIVED	A	S13	Mileage allowed over mileage units	Medicare_PSPS	\$8.79, against the published CY2024 rate of \$8.76. Independent confirmation of both
F70	Total Medicare FFS carrier-billed ground ambulance allowed charges	2024	\$5,073,090,003 \$	GOV-A	A	S13	PSPS 2024, six base codes plus A0425, all modifiers	Medicare_PSPS	Reconciles with MedPAC's \$5.3B (2023) once air and institutional claims are added
F71	Total Medicare FFS carrier-billed ground ambulance payments (NCH)	2024	\$3,957,457,371 \$	GOV-A	A	S13	PSPS 2024, PSPS_NCH_PAYMENT_AMT	Medicare_PSPS	The 2022 value of \$3.72B reconciles with MedPAC's \$3.9B program spending
F72	Hospital-to-hospital share of all Medicare carrier ground ambulance dollars	2024	11.4% %	DERIVED	A	S13		Medicare_PSPS	Dollar share (11.4%) exceeds volume share (8.2%); interfacility runs are longer and higher acuity
F73	GADCS median total cost per ground ambulance transport	2024	\$1,340 \$/transport	GOV-A	A	S30	GADCS Table 6.1	Cost_and_Capacity	The mean of \$2,673 is 1.99x the median. Quote the median for a scaled platform
F74	GADCS median total cost per RVU	2024	\$811 \$/RVU	GOV-A	A	S30	GADCS Table 6.1	Cost_and_Capacity	Mean \$1,624. Medicare allowed is about \$253 per RVU on the H2H book
F75	GADCS mean cost per transport, very high Medicare volume	2024	\$946 \$/transport	GOV-A	A	S30	GADCS Table 6.1	Cost_and_Capacity	\$946 against \$3,652 for low-volume organizations. Scale is the entire story
F76	Medicare FFS ambulance program spending, excl. beneficiary cost sharing	2022	\$3,900,000,000 \$	GOV-A	A	S28	MedPAC Payment Basics (Ambulance), Oct 2024, p.1	Payment_Rules	CORRECTED. Source workbook said \$4.0B for 2023. About 1% of FFS spend; about 13% of FFS beneficiaries used an ambulance. ALL ambulance, not IFT
F77	Medicare FFS ambulance total payments	2023	\$5,300,000,000 \$	GOV-B	B	S28	MedPAC ambulance session, Mar 2025	Payment_Rules	Total payments INCLUDING beneficiary cost sharing, across 11.4M transports and about 10,500 organizations. Not the same measure as F76
F78	NEDS target universe, hospital-owned EDs	2023	4,489 EDs	GOV-A	A	S03	NEDS Introduction 2023, Table A6	Data_Quality_Register	Sampling frame 3,185 EDs and 101,915,572 visits; the 2023 sample is 963 EDs and 31,003,348 visits
F79	Medicare FFS hospital-to-hospital transports, peak year	2017	1,139,524 transports	GOV-A	A	S32	PSPS 2017, modifier HH	Medicare_IFT_Series	1,139,524. Every year since has been lower
F80	Medicare FFS hospital-to-hospital transports	2010	937,317 transports	GOV-A	A	S32	PSPS 2010, modifier HH	Medicare_IFT_Series	Start of the fifteen-year series
F81	Transport CAGR, 2010 to 2017	2017	2.83% %/yr	DERIVED	A	S32		Medicare_IFT_Series	Real pre-COVID growth
F82	Transport CAGR, 2017 to 2024	2024	(3.75%) %/yr	DERIVED	A	S32		Medicare_IFT_Series	The reversal. Medicare Advantage enrolment is the dominant driver
F83	Transports 2024 against the 2017 peak	2024	(23.5%) %	DERIVED	A	S32		Medicare_IFT_Series	The Medicare FFS interfacility book is 23.5% smaller than at peak
F84	Total allowed CAGR, 2010 to 2024	2024	1.38% %/yr	DERIVED	A	S32		Medicare_IFT_Series	Dollars grew while transports shrank
F85	Allowed per transport CAGR, 2010 to 2024	2024	1.90% %/yr	DERIVED	A	S32		Medicare_IFT_Series	Below the 3.0% average AIF over the same window. Realised rate growth lags the statutory update
F86	Volume-weighted RVU of the H2H book	2010	1.472 RVU	DERIVED	A	S32, S14		Medicare_IFT_Series	1.472. Compare 1.450 in 2024; a 1.5% move in fifteen years
F87	Mileage share of allowed dollars	2010	43.0% %	DERIVED	A	S32		Medicare_IFT_Series	43.0%, against 44.6% in 2024. Stable for fifteen years
F88	Medicare FFS ambulance transports, all types	2024	11,300,000 transports	GOV-A	A	S31	MedPAC June 2026, Ch.6	MedPAC_2026_Mandated	Carrier-billed claims account for about 10.76M, or 95%. The residual is institutionally billed ambulance
F89	Total Medicare payments, AFS-covered transports	2024	\$5,300,000,000 \$	GOV-A	A	S31	MedPAC June 2026, Ch.6	MedPAC_2026_Mandated	Sits between the claims-derived \$4.47B program payment and \$5.71B allowed charge. MedPAC does not state the scope
F90	Mean cost per transport, lowest transport-volume quartile	2023	\$2,878 \$/transport	GOV-A	A	S31	MedPAC June 2026, Table 6-6	MedPAC_2026_Mandated	Mean 148 transports a year. Against \$918 in the highest quartile
F91	Mean cost per transport, highest transport-volume quartile	2023	\$918 \$/transport	GOV-A	A	S31	MedPAC June 2026, Table 6-6	MedPAC_2026_Mandated	Mean 15,291 transports a year. Average RVU only 5% lower than the lowest quartile: the gap is scale, not case mix

F92	Mean cost per transport, for-profit organizations	2023	\$563	\$/transport	GOV-A	A	S31	MedPAC June 2026, Table 6-6	MedPAC_2026_Mandated	Government-owned cost \$1,626, or 2.9x. Supersedes the RAND unweighted means for operator benchmarking
F93	Mean cost per transport, dynamic staffing model	2023	\$636	\$/transport	GOV-A	A	S31	MedPAC June 2026, Table 6-6	MedPAC_2026_Mandated	Static staffing costs \$1,524. The single largest controllable cost lever MedPAC identifies
F94	Median revenue-to-cost ratio, lowest volume quartile	2023	0.77	ratio	GOV-A	A	S31	MedPAC June 2026, Table 6-11	MedPAC_2026_Mandated	Highest quartile 1.10. NOT a Medicare margin: revenue includes non-Medicare sources
F95	Government organizations' non-billing revenue per transport	2023	\$322	\$/transport	GOV-A	A	S31	MedPAC June 2026, Ch.6	MedPAC_2026_Mandated	30% of government organizations receive local funding. Nonprofit \$32, for-profit \$12
F96	Carrier-billed AIR ambulance Medicare payments	2024	\$510,451,782	\$	GOV-A	A	S13	PSPS 2024, A0430, A0431, A0435, A0436, A0432	Medicare_PSPS	Needed to test MedPAC's spend figures. See Panel G
F97	Median ED time before transfer to another facility (OP-18d)	2025	287	minutes	GOV-A	A	S33	CMS Timely and Effective Care, national	Demand_Drivers	287 minutes. Measurement period 01 Jul 2024 to 30 Jun 2025
F98	Median ED time before discharge (OP-18b)	2025	161	minutes	GOV-A	A	S33	CMS Timely and Effective Care, national	Demand_Drivers	161 minutes. The comparison population
F99	Transfer penalty, OP-18d minus OP-18b	2025	126	minutes	DERIVED	A	S33		Demand_Drivers	126 minutes, or 78% longer than a discharged patient
F100	Top-decile hospital ED time before transfer	2025	187	minutes	GOV-A	A	S33	CMS Timely and Effective Care, national	Demand_Drivers	187 minutes. One hundred minutes of the median delay is addressable
F101	Median ED time before transfer, psychiatric patients (OP-18c)	2025	259	minutes	GOV-A	A	S33	CMS Timely and Effective Care, national	Demand_Drivers	259 minutes. Still faster than a medical transfer
F102	Hospitals reporting OP-18d	2025	2,340	hospitals	GOV-A	A	S33	CMS Timely and Effective Care, hospital file	Demand_Drivers	Median 294, 90th percentile 443, maximum 1,597 minutes
F103	Hospitals with ED transfer delay at or above 8 hours	2025	153	hospitals	GOV-A	A	S33	CMS Timely and Effective Care, hospital file	Demand_Drivers	6.5% of reporting hospitals
F104	Community hospitals in a system	2024	3,567	hospitals	GOV-B	B	S11	AHA Fast Facts, 2026 edition	Demand_Drivers	3,567 of 5,121 community hospitals
F105	Independent community hospitals	2018	1,707	hospitals	DERIVED	B	S34	AHA Fast Facts, 2020 edition	Demand_Drivers	1,707. Fell to 1,554 by 2024, a 9.0% decline
F106	Independent community hospitals	2024	1,554	hospitals	DERIVED	B	S11	AHA Fast Facts, 2026 edition	Demand_Drivers	1,554
F107	System share of community hospitals	2024	69.7%	%	DERIVED	B	S11		Demand_Drivers	69.7%, up from 67.2% in 2018
F108	Adjusted odds ratio of interfacility transfer, travel time 60+ minutes	2021	1.32	aOR	ACADEMIC	A	S35	Clark et al., JAMA Netw Open 2025, doi:10.1001/jamanetworkopen.2024.55258	Demand_Drivers	95% CI 1.15 to 1.51. n=190,311 adults, Florida and California, 2021
F109	Adjusted additional charges, travel time 60+ minutes	2021	\$8,284	\$	ACADEMIC	A	S35	Same	Demand_Drivers	95% CI \$5,532 to \$11,035

Basis. GOV-A = publisher figure with document and table pinned, opened and read. GOV-B = publisher figure, locator not pinned. INDUSTRY-A = trade association census, opened and read. ACADEMIC = peer-reviewed, digital object identifier given. SOURCED = analyst computation on a vendored CMS file. DERIVED = an Excel formula whose every input is one of the above.

Tier. A = verified against the primary document on 09 July 2026. B = document named but not opened, or a locator that could not be pinned. C = analyst computation or unpinned publisher class, not externally reproducible.

Nothing labeled ILLUSTRATIVE, FRAMEWORK or INDUSTRY enters this ledger. Those 30 items are recorded on Excluded_Not_Sourced with what would make each citable.

Colour key. Blue = a hardcoded value taken from a source document. Green = a live link to the sheet that carries the number. Black = a formula computed on this sheet.

Facts F110 to F135, added in revision 2 (10 July 2026). Same schema, same rules; green values live on the named tab.

Fact ID	Metric	Year	Value	Unit	Basis	Tier	Source ID	Locator	Lives on	Cross-check
F110	All O-D coded Medicare FFS ground transports	2019	14,289,866	transports	GOV-A	A	S39	PSPS 2019, valid O-D letter pairs, six base HCPCS	Medicare_OD_Matrix	Fractional service counts summed as floats
F111	All O-D coded Medicare FFS ground transports	2024	10,637,766	transports	GOV-A	A	S39	PSPS 2024	Medicare_OD_Matrix	the initial extraction carried 10,637,727: a 39-service (0.0004%) integer-truncation gap. See DQ #27
F112	O-D coded transport decline, 2019 to 2024	2024	(25.6%)	%	DERIVED	A	S39		Medicare_OD_Matrix	The FFS shrinkage benchmark for every channel comparison
F113	SNF-interface transports (NH + HN)	2024	2,398,253	transports	GOV-A	A	S39	PSPS 2024, NH and HN pairs	Medicare_OD_Matrix	2.75x the HH book. The largest facility-to-facility channel
F114	Dialysis-pair Medicare FFS ground transports	2019	2,332,855	transports	GOV-A	A	S39	All pairs containing J or G	Dialysis_ESRD_Channel	
F115	Dialysis-pair Medicare FFS ground transports	2024	857,750	transports	GOV-A	A	S39		Dialysis_ESRD_Channel	
F116	Dialysis-pair decline, 2019 to 2024	2024	(63.2%)	%	DERIVED	A	S39		Dialysis_ESRD_Channel	2.5x the whole-book decline (F112). Largest single-year drop -34.2% in 2022, the RSNAT national rollout year
F117	Dialysis-pair total allowed charges	2024	\$215,105,039	\$	GOV-A	A	S39	Base plus mileage	Dialysis_ESRD_Channel	\$506.9M in 2019
F118	RSNAT effect on RSNAT expenditures	2019	(77.0%)	%	ACADEMIC	A	S42	Contreary et al., JAMA Health Forum 2022;3(7):e222093	Dialysis_ESRD_Channel	doi:10.1001/jamahealthforum.2022.2093, PMID 35977218. Also -61% use probability, -2.4% total cost
F119	RSNAT four-year Medicare savings, model states	2020	\$650,000,000	\$	GOV-A	A	S41	CMS press release, 22 Sep 2020	Dialysis_ESRD_Channel	With -63% use and -72% expenditures among ESRD and pressure-ulcer beneficiaries
F120	Rotary-wing Medicare FFS transports, total	2024	66,608	transports	GOV-A	A	S40	PSPS 2024, A0431, all O-D pairs	Air_Ambulance_IJT	
F121	Rotary-wing hospital-to-hospital transports	2024	43,683	transports	GOV-A	A	S40	PSPS 2024, A0431, HH	Air_Ambulance_IJT	34,856 in 2010: +25% over the window in which ground HH fell 7%
F122	Rotary-wing HH share of rotary transports	2024	65.6%	%	DERIVED	A	S40		Air_Ambulance_IJT	Air is the high-acuity interfacility book; ground SCT is 6.6%

F123	Fixed-wing transfer-point-to-transfer-point (II) share	2024	94.6%	%	DERIVED	A	S40	A0430, modifier II	Air_Ambulance_IJT	Fixed-wing O-D modifiers identify runways, not facilities. DQ #29
F124	Rotary-wing commercial allowed over Medicare, base	2020	5.0	x	DERIVED	A	S44	FAIR Health air white paper: \$18,668 over \$3,739, same source, same year	Air_Ambulance_IJT	Pre-NSA benchmark
F125	Commercial ALS-emergency average allowed, ground	2020	\$758	\$	INDUSTRY-A	A	S43	FAIR Health ground white paper, Mar 2022	Payer_Rates_Commercial	Up 56% from \$486 in 2017 while Medicare grew 4.8%
F126	GAPB recommended patient cost-sharing cap	2024	\$100	\$	GOV-A	A	S46	GAPB report, 29 Mar 2024: lesser of \$100 or 10% of allowed	Payer_Rates_Commercial	Interfacility transports named in scope; no federal enactment as of Jul 2026
F127	States with own ground balance-billing protections	2024	14	states	GOV-B	B	S47	Commonwealth Fund, Feb 2024	Payer_Rates_Commercial	Most set a payment standard; several tie to Medicare multiples
F128	Medicaid NEMT fee-for-service spending	2018	\$2,600,000,000	\$	GOV-A	A	S48	MACPAC June 2021 report, Ch.5 (T-MSIS)	Payer_Rates_Commercial	61.5M ride-days, 3.2M users. AMBULANCE IS EXCLUDED from this NEMT definition
F129	Population 65 and over, projection	2030	71,183,000	people	GOV-A	A	S50	Census NP2023-T2, main series	Macro_Demand_Drivers	57.8M (2022), 82.1M (2050). 2025-2030 CAGR 2.4%/yr
F130	Medicare Advantage share of eligible beneficiaries	2026	55.0%	%	INDUSTRY-A	A	S51	KFF from CMS enrollment files: 35.2M of 64.2M	Macro_Demand_Drivers	19% in 2007; CBO projects 63% by 2034. The deflator on every FFS series here
F131	Rural hospital closures and conversions since 2010	2026	154	hospitals	ACADEMIC	A	S52	Sheps Center tracker, accessed 10 Jul 2026	Macro_Demand_Drivers	86 complete + 68 converted; REH conversions excluded. Chartis counts 182 under a different convention: never mix
F132	Rural closure effect on EMS total activation time	2020	7	minutes	ACADEMIC	A	S53	Miller et al., HSR 2020, doi:10.1111/1475-6773.13254	Macro_Demand_Drivers	PMID 31989591. Transport time +2.6 min; response time unchanged
F133	SMOCC immediate effect on interhospital transfer rate	2025	1.35	rate ratio	ACADEMIC	A	S54	Richert et al., JAMA Netw Open 2025;8(12):e2546002	Macro_Demand_Drivers	doi:10.1001/jamanetworkopen.2025.46002, PMID 41324958. 95% CI 1.05-1.74; long-term RR 0.94; 441,709 NEMSIS transfers, 8 states
F134	NHAMCS ambulance-arrival share of ED visits (final vintage)	2022	17.0%	%	GOV-A	A	S55	NHAMCS 2022 ED Summary Tables, Table 4	Macro_Demand_Drivers	SE 0.9 points on 155,398,000 visits; 65+ 33.4%. NEMSIS 2022 coverage (17.5%) sits inside the 95% CI. Survey discontinued after 2022
F135	Active ambulance LCDs in the Medicare Coverage Database	2026	-	LCDs	GOV-A	A	S56	MCD query, 09-10 Jul 2026; one billing article, A52917 (Noridian, rural air protocols)	Code_Crosswalks	Ambulance coverage is national regulation (42 CFR 410.40), not MAC discretion

Facts F136 to F147, revision 3 (10 July 2026).

Fact ID	Metric	Year	Value	Unit	Basis	Tier	Source ID	Locator	Lives on	Cross-check
F136	Original Medicare A and B enrollment (annual average)	2019	32,758,129	beneficiaries	GOV-A	A	S60	CMS Medicare Monthly Enrollment, national, Year rows	Utilization_Normalized	The carrier-claims denominator
F137	Original Medicare A and B enrollment (annual average)	2024	27,673,511	beneficiaries	GOV-A	A	S60		Utilization_Normalized	Minus 15.5% from 2019. First increase in the series lands in 2025 (27,914,766)
F138	All O-D coded transports per 1,000 A and B FFS beneficiaries	2024	384.4	per 1,000	DERIVED	A	S39, S60		Utilization_Normalized	436.2 in 2019: minus 11.9% per capita against minus 25.6% raw
F139	HH transports per 1,000 A and B FFS beneficiaries	2024	31.5	per 1,000	DERIVED	A	S39, S60		Utilization_Normalized	34.2 in 2019: minus 8.0% per capita against minus 22.3% raw
F140	Dialysis-pair transports per 1,000 A and B FFS beneficiaries	2024	31.0	per 1,000	DERIVED	A	S39, S60		Utilization_Normalized	71.2 in 2019: minus 56.5% per capita. The collapse survives normalization
F141	HH emergency-coded share (ALS-E + BLS-E + ALS2)	2024	31.0%	%	DERIVED	A	S39	Six-base-code convention	Acuity_by_Channel	28.6% in 2019: +2.4 points
F142	NH (SNF to hospital) emergency-coded share	2024	94.5%	%	DERIVED	A	S39		Acuity_by_Channel	HN (hospital to SNF) is 98.4% BLS non-emergency: the two directions are clinical opposites
F143	Dialysis-pair BLS non-emergency share	2024	95.6%	%	DERIVED	A	S39		Acuity_by_Channel	The RSNAT target profile
F144	Ground ambulance billing NPIs (base codes, MUP floor)	2024	8,721	NPIs	GOV-A	A	S59	MUP-PHY 2024; rows with 10 or fewer beneficiaries suppressed	Supplier_Landscape	9,028 in 2019
F145	Top-10 billing NPIs, share of approved ground services	2024	6.2%	%	DERIVED	A	S59		Supplier_Landscape	5.3% in 2019. Floor on corporate concentration: NPI is not parent company
F146	NPIs billing BLS non-emergency (A0428)	2024	2,683	NPIs	GOV-A	A	S59		Supplier_Landscape	3,075 in 2019: minus 12.7%, the supplier-side RSNAT exit
F147	Ground add-on payments statutory expiration	2028	1	date: 1 Jan 2028	GOV-A	A	S61	CAA 2026, section 6203; CMS AFS page	Payment_Rules	Extended through 31 Dec 2027 after an Oct 2025 lapse; 2% urban, 3% rural, 22.6% super-rural

Facts F148 to F163, revision 4 (10 July 2026).

Fact ID	Metric	Year	Value	Unit	Basis	Tier	Source ID	Locator	Lives on	Cross-check
F148	Paid EMT and paramedic employment (combined)	2019	260,600	jobs	GOV-A	A	S67	OEWS May 2019, 29-2040	Workforce_Supply	Excludes volunteers and self-employed

F149	Paid EMT and paramedic employment (combined)	2024	282,900 jobs	DERIVED	A	S68	181,000 EMTs + 101,900 paramedics	Workforce_Supply	+8.6% vs 2019. The paid workforce grew
F150	Paramedic median annual wage	2024	\$58,410 \$	GOV-A	A	S68	OEWS May 2024 via OOH	Workforce_Supply	No 2019 twin: occupation split in the 2018 SOC
F151	EMT median annual wage	2024	\$41,340 \$	GOV-A	A	S68		Workforce_Supply	\$35,400 combined median in 2019 (F152)
F152	EMT and paramedic combined median annual wage	2019	\$35,400 \$	GOV-A	A	S67		Workforce_Supply	
F153	Ambulance Inflation Factor, compounded CY2020-CY2024	2024	18.6% %	DERIVED	A	S29	0.9, 0.2, 5.1, 8.7, 2.6 compounded	Workforce_Supply	Slightly ahead of EMT median wage growth (+16.8%) on this window; see the Panel B read
F154	Ground ambulance billing NPIs	2014	9,305 NPIs	GOV-A	A	S62	MUP-PHY 2014, six ground base codes	Supplier_Series_Raw	8,721 in 2024: minus 6.3% over the decade
F155	BLS non-emergency billers, 2014 to 2024	2024	(19.4%) %	DERIVED	A	S62	3,329 to 2,683 NPIs	Supplier_Trend	Fell every single year; the scheduled-transport exit
F156	Population in ambulance deserts (25+ minutes from a station)	2022	4,500,000 people	ACADEMIC	A	S65	41 states, 2021-22; 2.3M rural	Imbalance_Ledger	UNDERCOUNT: nine states lack data
F157	California cumulative offload delay beyond 30 minutes	2024	13,365 hours per 6 months	GOV-A	A	S71	CEMSIS, Apr-Sep 2024	Imbalance_Ledger	19.1% of offloads exceed the standard (F158). CEMSIS participants only
F158	California offloads exceeding the 30-minute standard	2024	19.1% %	GOV-A	A	S71		Imbalance_Ledger	Statewide mean offload 42.8 minutes, 2021-2023 (S72)
F159	Staffed hospital beds, pre-pandemic mean to post-PHE	2024	674,000 beds	ACADEMIC	A	S63	802,000 (2009-2019 mean) to 674,000	Supply_Stack	Minus 16% on mandated reporting; AHA surveys say minus 2% (Fault_Audit F)
F160	National hospital occupancy, post-PHE steady state	2024	75.3% %	ACADEMIC	A	S63	63.9% pre-pandemic mean	Demand_Stack	Census flat at about 510,000: the rise is a bed-supply story
F161	Hospital-at-home waiver statutory expiration	2030	1 date: 30 Sep 2030	GOV-A	A	S64	CAA 2026 sec. 6210	Demand_Stack	About 400 approved hospitals; every H@H day creates home transport legs
F162	OP-18 boarding-measure removal	2028	1 reporting year	GOV-A	A	S64	CY2026 OPSS final rule	Fault_Audit	Successor ECAT eCQM measures boarding directly; mandatory CY2028
F163	Medicare beneficiaries per ground billing NPI	2024	7,797 beneficiaries	DERIVED	A	S60, S62	68.0M over 8,721	Imbalance_Ledger	6,814 in 2019: +14.4%. Need per supplier is rising
F164	Official state-file national ground services (approved)	2024	9,637,009 services	GOV-A	A	S73	MUP by Geography and Service, six ground codes	State_Saturation_Raw	9,542,103 in the NPI file: the Geography file suppresses less
F165	NPI-file suppression tail (Geography file over NPI file)	2024	1.0% %	DERIVED	A	S62, S73		State_Saturation	Bounds DQ #32: supplier-file undercount is about 1% of services nationally

v3 additions: F166 onward - built 10 July 2026. Same contract: every fact resolves to a source and locator; DERIVED facts name their inputs; formulas are live.

F166	Original Medicare (any part) beneficiaries, national	2024	33913847 beneficiaries	GOV	A	S81	Medicare Monthly Enrollment, National, MONTH=Year, 2024	Enrollment_ESRD_State	Consistent with the A&B denominator discussion on Utilization_Normalized (different definition: any-part vs A-and-B; both stated there)
F167	MA & other share of Medicare beneficiaries, national	2025	0.509016477810194 %	DERIVED	A	S81	MA_AND_OTH_BENES / TOT_BENES, 2025	Enrollment_ESRD_State	Tracks the KFF 54% (2025) MA share on Macro_Demand_Drivers (different definition: KFF excludes some "other" plans)
F168	Medicare ESRD beneficiaries (aged + disabled), national	2025	571962 beneficiaries	DERIVED	A	S81	AGED_ESRD_BENES + DSBLD_ESRD_AND_ESRD_ONLY_BENES, 2025	Enrollment_ESRD_State	Denominator context for the Dialysis_ESRD_Channel RSNAT collapse: the patient base did not shrink; the paid volume did
F169	Medicare FFS BLS non-emergency services (A0428, final-action)	2024	2,956,154.1 services	GOV	A	S78	MUP by Geography & Service 2024, National, A0428, Tot_Srvcs	MUP_Ambulance_National	MUP final-action basis; PSPS submitted basis differs by design (~1.10x in 2024, Data_Quality_Register #34)
F170	Rendering providers billing A0428	2024	4,223 NPIs	GOV	A	S78	MUP by Geography & Service 2024, National, A0428	MUP_Ambulance_National	The scheduled-transport biller base; see Supplier_Trend BLS-NE biller exit series
F171	SCT (A0434) services CAGR, 2013-2024	2024	-0.0339576929557589 %/yr	DERIVED	A	S78	(2024 services / 2013 services)*(1/11)-1, Panel B	MUP_Ambulance_National	Directionally consistent with the SCT share-of-transports chart on the Charts tab (PSPS basis)
F172	States/territories with ambulance MUP rows	2024	55 geographies	GOV	A	S78	MUP by Geography & Service 2024, State grain, A0425-A0436	MUP_Ambulance_State	Suppression makes absent state-code pairs floors, not zeros
F173	BLS non-emergency (A0428) denial rate	2024	0.129157061063293 %	DERIVED	A	S79	Panel B, A0428 column, 2024 row (denied/submitted)	PSPS_Denial_Series	The medical-necessity screen on scheduled transport; compare the RSNAT natural experiment on Dialysis_ESRD_Channel and Payment_Integrity
F174	US counties with <3 measured FFS ambulance providers, share	2025	0.231707317073171 % of counties	DERIVED	A	S80	Panel C total row: (suppressed/0 + 1-2) / counties, latest window	Market_Saturation_Ambulance	Whitespace measure; complements Imbalance_Ledger and the State_Saturation thin-supply screen
F175	Private ambulance services employment (QCEW annual average)	2024	171,100 jobs	GOV	A	S82	QCEW NAICS 621910, US000, ownership 5, 2024 annual average	QCEW_EMS_Employment	Companion to the OEWS-based Workforce_Supply tab (occupation vs industry basis; do not mix)
F176	Private ambulance establishments	2024	5,820 establishments	GOV	A	S82	QCEW NAICS 621910, US000, ownership 5, 2024	QCEW_EMS_Employment	An establishment is a worksite, not a company: compare 10,465 PECOS-enrolled suppliers and 8,721 billing NPIs (Supplier_Landscape) - three different units
F177	Private ambulance avg annual pay	2024	59,511 \$/yr	GOV	A	S82	QCEW NAICS 621910, US000, ownership 5, 2024	QCEW_EMS_Employment	Pay CAGR vs AIF: the scissors carried on this tab
F178	Certified hospitals (Care Compare universe)	2026	5432 facilities	DERIVED	A	S83	Hospital General Information snapshot, sum over states	Facility_Universe_State	AHA counts ~6,100 hospitals total / ~5,200 community (Demand_Drivers): AHA includes non-Medicare-certified and federal units - different universes, both stated
F179	SNF certified beds (Care Compare universe)	2026	1568348 beds	DERIVED	A	S83	Nursing Home Provider Info, number_of_certified_beds summed	Facility_Universe_State	The SNF interface is 2.75x the hospital-to-hospital Medicare book (Medicare_OD_Matrix); this is the bed base behind it
F180	Hospitals: certified facilities (full registry)	2026	5,432 facilities	GOV	A	S89	Care Compare snapshot, Hosp_Registry tab row count	Hosp_Registry	Matches Facility_Universe_State state sums (same pull)
F181	Nursing homes (SNF): certified facilities (full registry)	2026	14,695 facilities	GOV	A	S89	Care Compare snapshot, SNF_Registry tab row count	SNF_Registry	Matches Facility_Universe_State state sums (same pull)
F182	Dialysis facilities: certified facilities (full registry)	2026	7,557 facilities	GOV	A	S89	Care Compare snapshot, Dialysis_Registry tab row count	Dialysis_Registry	Matches Facility_Universe_State state sums (same pull)
F183	Inpatient rehabilitation facilities: certified facilities (full registry)	2026	1,222 facilities	GOV	A	S89	Care Compare snapshot, IRF_Registry tab row count	IRF_Registry	Matches Facility_Universe_State state sums (same pull)
F184	Long-term care hospitals: certified facilities (full registry)	2026	311 facilities	GOV	A	S89	Care Compare snapshot, LTCH_Registry tab row count	LTCH_Registry	Matches Facility_Universe_State state sums (same pull)
F185	Hospices: certified facilities (full registry)	2026	6,852 facilities	GOV	A	S89	Care Compare snapshot, Hospice_Registry tab row count	Hospice_Registry	Matches Facility_Universe_State state sums (same pull)

F186	Home health agencies: certified facilities (full registry)	2026	12,392 facilities	GOV	A	S89	Care Compare snapshot, HHA_Registry tab row count	HHA_Registry	Matches Facility_Universe_State state sums (same pull)
F187	Medicare-enrolled ambulance suppliers (named registry)	2026	10,465 organizations	GOV	A	S90	PECOS public enrollment, ambulance supplier type, 2026Q1	PECOS_Registry	EXACT match to vendored PPEF 2026.04.01 national count and to the live stats probe (10,465)
F188	Hospitals in the Hospital Service Area file	2025	7,452 hospitals	GOV	A	S91	HSA 2025, distinct provider IDs	HSA_Hospital_Catchment	CCN-joinable to Hosp_Registry (Care Compare)
F189	Hospital-measure rows in the ED throughput registry	2026	13,980 rows	GOV	A	S94	Timely & Effective Care hospital file, ED measures	ED_Timeliness_Registry	National OP-18 statistics on Demand_Drivers were re-derived from this same file family in v2.7 (533)
F190	Distinct NPIs billing Medicare ambulance codes, 2013 (provider-grain, floor)	2013	9,746 NPIs	GOV	A	S95	MUP by Provider & Service 2013, distinct Rndmg_NPI over A0425-A0436 rows	MUP_Providers_2013	A floor (suppression); compare Supplier_Landscape 8,721 billing NPIs (MUP-geo basis) and PECOS 10,465 enrolled - three floors of the same universe
F191	Distinct NPIs billing Medicare ambulance codes, 2019 (provider-grain, floor)	2019	9,554 NPIs	GOV	A	S95	MUP by Provider & Service 2019, distinct Rndmg_NPI over A0425-A0436 rows	MUP_Providers_2019	A floor (suppression); compare Supplier_Landscape 8,721 billing NPIs (MUP-geo basis) and PECOS 10,465 enrolled - three floors of the same universe
F192	Distinct NPIs billing Medicare ambulance codes, 2024 (provider-grain, floor)	2024	9,318 NPIs	GOV	A	S95	MUP by Provider & Service 2024, distinct Rndmg_NPI over A0425-A0436 rows	MUP_Providers_2024	A floor (suppression); compare Supplier_Landscape 8,721 billing NPIs (MUP-geo basis) and PECOS 10,465 enrolled - three floors of the same universe
F193	Hospital-ZIP corridor rows carried (top-15 per hospital)	2025	106,799 rows	GOV	A	S96	HSA latest year, per-hospital top-15 ZIPs by cases	HSA_Corridors	Per-hospital totals reconcile to HSA_Hospital_Catchment (same pull family)
F194	Counties with measured 65+ base carried	2024	3,144 counties	GOV	A	S97	CC-EST2024-AGESEX, YEAR codes 2 and 6	County_Age_65plus	County 65+ sums reconcile to State_Age_65plus state rows (same vintage, civilian-vs-resident definitions stated)
F195	County chronic-prevalence rows (4 measures)	2024	17,867 rows	GOV	A	S98	PLACES county, KIDNEY/CHD/STROKE/DIABETES, CrdPrv	PLACES_County_Chronic	Model-based estimates (stated); the ESRD claims denominator on Enrollment_ESRD_State is the measured companion
F196	Quarterly QCEW rows carried	2014-2025	4,832 rows	GOV	A	S99	QCEW 621910 quarterly, US000 + statewide	QCEW_Quarterly	Annual averages on QCEW_EMS_Employment are the same program aggregated by BLS
F197	HCRIS hospital-year rows (FY2020-2022)	2020-2022	17,974 hospital-years	GOV	A	S100	HCRIS extract, one row per CCN x fiscal year	HCRIS_Hospital_Panel	The occupancy_trend statistics quoted in the repo modules derive from this same panel
F198	Ambulance-related LEIE exclusions on file	2026	566 exclusions	GOV	A	S101	LEIE UPDATED.csv, ambulance filter (manifest)	LEIE_Ambulance_Exclusions	Companion to the CERT/RSNAT payment-integrity evidence on Payment_Integrity
F199	Medicare-enrolled ambulance suppliers, national (PECOS)	2026	10465 suppliers	GOV	A	S102, S103	PPEF_Enrollment_Extract_2026.04.01, PROVIDER_TYPE_DESC = "PART B SUPPLIER - AMBULANCE SERVICE SUPPLIER", State table, STATE_CD = OH	PECOS_Suppliers_State	Live /data/stats probe 2026-07-10 returned the same 10,465 (MATCH cell on-sheet); MUP CY2024 A0425 renderers 10,061 (96%)
F200	Top state by enrolled ambulance suppliers (Ohio)	2026	897 suppliers	GOV	A	S102		PECOS_Suppliers_State	OH 897 > TX 672 > NY 670: supplier counts track fragmentation, not population
F201	Top-10 states share of national enrolled suppliers	2026	0.474725274725275 share	DERIVED	A	S102	SUM of top-10 sorted state rows / national	PECOS_Suppliers_State	about 47% - a long-tail, unconsolidated supplier base
F202	NE + IA Medicare-enrolled ambulance suppliers	2026	583 suppliers	DERIVED	A	S102	SUMIF over state rows, NE (287) + IA (296)	PECOS_Suppliers_State	NPPES_Registry_NE_IA holds 751 org NPIs for the same states - registry universe wider than enrollment
F203	Ambulance suppliers that billed ground mileage, CY2024 (MUP renderers)	2024	10061 suppliers	GOV	A	S104	MUP by Geography & Service CY2024, National, A0425, Tot_Rndmg_Pvdris	PECOS_Suppliers_State	96% of the PECOS enrolled universe (formula on-sheet)
F204	PECOS vendored-vs-live consistency check	2026 MATCH	flag	DERIVED	A	S102, S103	IF(vendored national = live found_rows)	PECOS_Suppliers_State	Also: sum of 55 state rows = national (row below Panel B)
F205	NE+IA ambulance organization NPIs (NPPES registry)	2026	751 organization NPIs	DERIVED	A	S105	COUNTIF over the 751-row roster (NE 400 + IA 351)	NPPES_Registry_NE_IA	vs 583 PECOS-enrolled suppliers for NE+IA: the gap is the volunteer / non-Medicare-billing layer
F206	NE municipal/fire/volunteer squads	2026	328 organization NPIs	DERIVED	A	S105	COUNTIFS search_state=NE, category=municipal-fire-volunteer	NPPES_Registry_NE_IA	328 of 400 NE org NPIs (82%) - the fragmented municipal base the consolidation thesis targets
F207	IA hospital-owned ambulance services	2026	40 organization NPIs	DERIVED	A	S105	COUNTIFS search_state=IA, category=hospital-owned	NPPES_Registry_NE_IA	IA 40 vs NE 5: Iowa runs a hospital-owned model - the structural contrast between the two states
F208	Private/commercial ambulance operators, NE+IA	2026	128 organization NPIs	DERIVED	A	S105	COUNTIFS category=private (NE 58 + IA 70)	NPPES_Registry_NE_IA	The thin private layer an IFT specialist competes in
F209	Distinct organization names behind the 751 NPIs	2026	686 organizations	DERIVED	A	S105	SUMPRODUCT/COUNTIF distinct-count over org_name	NPPES_Registry_NE_IA	Multi-NPI squads mean NPI counts overstate agency counts by ~9%
F210	Roster rows with out-of-state practice/mailling address	2026	53 rows	DERIVED	A	S105	SUMPRODUCT over addr-state column	NPPES_Registry_NE_IA	Search-state matches practice OR mailing address - 53 rows are out-of-state operators serving NE/IA
F211	Roster rows with non-ambulance primary taxonomy	2026	21 rows	DERIVED	A	S105	SUMPRODUCT ISERROR(SEARCH("Ambulance", primary_taxonomy))	NPPES_Registry_NE_IA	Hospitals/CAHs/NEMT vans with ambulance secondary taxonomy - 21 rows, kept and flagged
F212	Hospital CHOW events, 2016-2025 total	2025	755 events	GOV	A	S106	Hospital_CHOW_2026.04.01, sum of state x year matrix	CHOW_Consolidation	Matches report JSON total, chows 755 (check row); state sums = national series exactly
F213	SNF CHOW events, 2016-2025 total	2025	5141 events	GOV	A	S107	SNF_CHOW_2026.04.01, sum of state x year matrix	CHOW_Consolidation	Matches report JSON total, chows 5,141 (check row); 6.8x the hospital CHOW count
F214	SNF CHOWs, peak year 2023	2023	882 events	GOV	A	S107	National series, year 2023	CHOW_Consolidation	3.6x the 2016 level (244) - post-acute ownership churn accelerated through 2023
F215	Hospital CHOWs, 2024	2024	101 events	GOV	A	S106	National series, year 2024	CHOW_Consolidation	Volatile series: 38 (2022) to 127 (2019); companion to the Kaufman Hall M&A series on Growth_Evidence_Registry (announced deals vs recorded provider-agreement CHOWs)
F216	SNF CHOW CAGR 2016>2024	2024	0.142432182943028 %/yr	DERIVED	A	S107	(2024/2016)^(1/8)-1 live formula; 2025 excluded (partial accrual flag)	CHOW_Consolidation	244 > 708 events/yr; approx +14%/yr
F217	Hospital CHOW CAGR 2016>2024	2024	0.0672625377760108 %/yr	DERIVED	A	S106	(2024/2016)^(1/8)-1 live formula; 2025 excluded	CHOW_Consolidation	60 > 101 events/yr; approx +6.7%/yr on a volatile base - read with the YoY column
F218	Top state by SNF CHOWs 2016-2025 (Texas)	2025	697 events	DERIVED	A	S107	State matrix row 1 (sorted by total), TX	CHOW_Consolidation	TX 697 > CA 393 > OH 310
F219	CY2025 PFS payment localities carried (Addendum E)	CY2025	109 localities	GOV	A	S108	Addendum E, RVU25A package; vendored registry row cms_gpci_2025 (109 locality rows incl. DC/PR/VI)	GPCI_Localities	registry_row_count=109; raw CSV 116 lines incl. headers/footnotes
F220	National unweighted mean PE GPCI	CY2025	1.0434495412844 index	DERIVED	A	S108	AVERAGE over Addendum E PE GPCI column (109 localities)	GPCI_Localities	python recompute 1.0434
F221	National unweighted mean ambulance geographic factor (0.7 x PE + 0.3)	CY2025	1.03041467889908 factor	DERIVED	A	S108, S109	42 CFR 414.610(c)(4) 70/30 rule over Addendum E	GPCI_Localities	python recompute 1.0304
F222	Lowest ambulance geographic factor (Mississippi)	CY2025	0.8964 factor	DERIVED	A	S108, S109	MIN over derived col H; MS PE GPCI 0.852 -> 0.8964	GPCI_Localities	0.7*0.852+0.3 = 0.8964
F223	Highest ambulance geographic factor (San Jose-Sunnyvale-Santa Clara, CA)	CY2025	1.3045 factor	DERIVED	A	S108, S109	MAX over derived col H; Santa Clara/San Benito PE GPCI 1.435 -> 1.3045	GPCI_Localities	0.7*1.435+0.3 = 1.3045

F224	Max/min spread of the ambulance geographic factor	CY2025	1.45526550647033 x	DERIVED	A	S108, S109	MAX/MIN over derived col H	GPCI_Localities	1.3045/0.8964 = 1.455x - geography moves the base rate by at most ~46% end to end
F225	CY2026 AFS ground conversion factor	CY2026	284.56 \$/RVU	GOV	B	S110, S111	CY2026 AFS public use file; MedPAC Jun 2026 Ch.6	Derived_Rate_Card	v2.7 Payment_Rules CY2024 CF \$272.44 (vintage)
F226	CY2025 AFS ground conversion factor	CY2025	278.98 \$/RVU	GOV	B	S111, S112	MedPAC Jun 2026 Ch.6 (prior-year CF)	Derived_Rate_Card	
F227	CF update CY2025->CY2026	CY2026	0.0200014337945371 %/yr	DERIVED	B	S114	live formula (284.56/278.98)-1; equals published CY2026 AIF +2.0%	Derived_Rate_Card	CMS Transmittal 13464: AIF = CPI-U 2.7% - TFP 0.7% = 2.0%
F228	SCT (A0434) CY2026 national unadjusted base	CY2026	924.82 \$	DERIVED	B	S110, S111	live formula 3.25 RVU x \$284.56 = \$924.82	Derived_Rate_Card	repo cy2026_base column, delta < \$0.005
F229	BLS non-emergency (A0428) super-rural base	CY2026	348.87056 \$	DERIVED	B	S115, S110	live formula \$284.56 x 1.226 = \$348.87	Derived_Rate_Card	add-on % per 42 U.S.C. 1395m(f)(13)
F230	Ground mileage A0425, CY2026 national	CY2026	9.15 \$/mile	GOV	B	S111, S110	MedPAC Jun 2026 Ch.6: \$9.15/mile national	Derived_Rate_Card	v2.7 Payment_Rules CY2024 \$8.76 (vintage)
F231	Ground mileage A0425, rural miles 1-17	CY2026	14.13 \$/mile	DERIVED	B	S111, S115	live formula \$9.42 x 1.5 = \$14.13	Derived_Rate_Card	
F232	Paramedic Intercept (A0432) RVU	CY2026	1.75 RVU	SOURCED	B	S110	AFS RVU ladder; carried from repo ift_analytics (cites the RVU table via a PYA Medicare-payment primer) - RE-VERIFY against the CY2026 AFS live SUM over the six A04xx code rows	Derived_Rate_Card	flagged re-verify; PI base column derived from this RVU
F233	CY2024 Medicare six-code ground services (supplier claims)	CY2024	9637068 services	DERIVED	B	S113		Service_Level_Economics	9,637,068 (python recompute)
F234	ALS1-emergency (A0427) share of six-code services	CY2024	0.383666588219571 %	DERIVED	B	S113	live formula services/total; 38.4%	Service_Level_Economics	module medicare_mix() share_pct 38.4
F235	Medicare FFS ground transports	2024	11300000 transports	GOV	B	S111	MedPAC Jun 2026 Ch.6: 11.3M transports, ~10,600 orgs	Service_Level_Economics	supplier-file six-code total 9.64M (different universe - see reconciliation note)
F236	Medicare AFS payments	2024	5300000000 \$	GOV	B	S111	MedPAC Jun 2026 Ch.6: \$5.3B	Service_Level_Economics	
F237	Average AFS payment per transport	2024	469.026548672566 \$	DERIVED	B	S111	live formula \$5.3B / 11.3M = ~\$469	Service_Level_Economics	ift_unit_economics medicare_avg_payment \$469 (needs_reverify=False)
F238	GADCS all-payer BLS share of ground transports	2022-23 cohorts	0.56 %	GOV	B	S117	GADCS Dec 2024 report, 3,694 reporting orgs; verbatim quote carried on-row	Service_Level_Economics	ALS1 42%, ALS2+SCT 3% on adjacent rows
F239	NEMSIS facility-to-facility share of EMS activations	2024	0.124590712460657 %	DERIVED	B	S120	live formula 7,512,656 / 60,298,684 = ~12.5%	Service_Level_Economics	hospital-to-hospital subset 5,510,664 (9.1%)
F240	GADCS mean cost per transport	2022-23 cohorts	2673 \$	GOV	B	S117	GADCS Dec 2024: mean \$2,673, median \$1,340, labor 69.4%	Service_Level_Economics	GAO-13-6 2010 median \$429 (range \$224-\$2,204) on adjacent row
F241	Ambulance Inflation Factor, CY2026	CY2026	0.02 %/yr	DERIVED	B	S114	live formula CPI-U 2.7% - TFP 0.7% = 2.0%; transmittal quote on-row	Service_Level_Economics	matches Derived_Rate_Card CF update formula
F242	CO statewide commercial paid as multiple of Medicare - Inpatient/Outpatient combined	2024	2.3856 x Medicare	SOURCED	A	S124	FY26 RBP public file, STATEWIDE row, IP/OP combined, 2024 (2.39x); facility/hospital claims, Colorado only - CONTEXT for commercial multiples, NOT an ambulance rate	Commercial_Context_APCD	claims-weighted provider mean 2.42x (python)
F243	CO statewide commercial multiple - Outpatient	2024	2.5837 x Medicare	SOURCED	A	S124	FY26 RBP public file, STATEWIDE row (2.58x); facility/hospital claims, Colorado only - CONTEXT for commercial multiples, NOT an ambulance rate	Commercial_Context_APCD	
F244	CO statewide commercial multiple - Inpatient	2024	1.9116 x Medicare	SOURCED	A	S124	FY26 RBP public file, STATEWIDE row (1.91x); facility/hospital claims, Colorado only - CONTEXT for commercial multiples, NOT an ambulance rate	Commercial_Context_APCD	
F245	Median of the 12 CO statewide commercial-multiple readings	2021-2024	2.35515 x Medicare	DERIVED	A	S124	live MEDIAN formula over staging block A; facility/hospital claims, Colorado only - CONTEXT for commercial multiples, NOT an ambulance rate	Commercial_Context_APCD	
F246	CO provider-level median commercial multiple - IP/OP combined	2024	2.3845 x Medicare	SOURCED	A	S124	live MEDIAN formula over 86 provider rows of the vendored file (2.38x); facility/hospital claims, Colorado only - CONTEXT for commercial multiples, NOT an ambulance rate	Commercial_Context_APCD	statewide published 2.39x - consistent
F247	CO provider-level IQR of commercial multiple - IP/OP combined	2024	0.9755 x Medicare (width)	DERIVED	A	S124	live formula p75-p25 (1.79x to 2.77x); facility/hospital claims, Colorado only - CONTEXT for commercial multiples, NOT an ambulance rate	Commercial_Context_APCD	
F248	CO statewide commercial-multiple drift - IP/OP combined, 3-yr CAGR	2021-2024	0.0137410578167461 %/yr	DERIVED	A	S124	live CAGR formula 2021->2024 window over staging block A; facility/hospital claims, Colorado only - CONTEXT for commercial multiples, NOT an ambulance rate	Commercial_Context_APCD	2.29x -> 2.39x
F249	Acute-MI inpatient stays (principal dx), US	2018	658600 stays/yr	SOURCED	B	S125	HCUP SB#277, 2018 NIS, principal-diagnosis stay counts	Condition_Transfer_Registry	CDC/AHA ~805,000 MI events/yr (registry note)
F250	Heart-failure inpatient stays (parent pool), US	2018	1135900 stays/yr	SOURCED	B	S125	HCUP SB#277, 2018 NIS - HF is the #2 US inpatient condition	Condition_Transfer_Registry	same pool reused by Cardiac recovery row - never double-count
F251	Cerebral-infarction inpatient stays, US	2018	533400 stays/yr	SOURCED	B	S125	HCUP SB#277, 2018 NIS	Condition_Transfer_Registry	CDC >795,000 strokes/yr, ~87% ischemic
F252	Adults developing sepsis per year, US	2023	1700000 adults/yr	SOURCED	B	S127	CDC sepsis surveillance ("at least 1.7M adults")	Condition_Transfer_Registry	HCUP: septicemia #1 inpatient condition, 2,218,800 stays 2018
F253	Mental-health-related ED visits, US	2021	6000000 ED visits/yr	SOURCED	B	S131	CDC/SAMHSA ED utilization, 2021	Condition_Transfer_Registry	~1 in 8 ED visits MH/SUD
F254	US births (all)	2022	3667758 births/yr	SOURCED	B	S129	CDC/NCHS Births: Final Data for 2022	Condition_Transfer_Registry	preterm row = 10.38% of this pool
F255	Medicare-covered SNF stays	2022	1800000 stays/yr	SOURCED	B	S136	MedPAC Payment Basics - SNF (2022 data)	Condition_Transfer_Registry	~14,700 SNFs cross-checks Post_Acute_Supply_State SNF count 14,699
F256	ED encounters transferred to another short-term hospital (2009)	2009	1900000 transfers/yr	SOURCED	B	S126	HCUP SB#155: 1.5% x 128,885,040 ED encounters, 2009 NEDS	Condition_Transfer_Registry	modern read: NEDS 2018-2022 ~1.97M/yr (Nikolla 2025)
F257	Registry rows with a GOV-cited national volume	2026	23 rows (of 32)	DERIVED	B	S140	live COUNTIF over the Basis column (GOV 23 / ACADEMIC 6 / FRAMEWORK 3)	Condition_Transfer_Registry	
F258	Skilled nursing facility (SNF) - certified providers, US	2026	14699 providers	SOURCED	A	S141	CMS Nursing Home Care Compare - Provider Information (NH_ProviderInfo), file vintage 2026-04-01, one row per certified provider (CPH)	Post_Acute_Supply_State	MedPAC ~14,700 SNFs; registry SNF row
F259	Inpatient rehabilitation facility (IRF) - certified providers, US	2026	1221 providers	SOURCED	A	S142	CMS Inpatient Rehabilitation Facility Compare - General Information + Provider Data, file vintage 2026-02-13, one row per certified provider (CPH)	Post_Acute_Supply_State	MedPAC ~1,180 IRFs (registry note)
F260	Long-term acute care hospital (LTACH/LTCH) - certified providers, US	2026	317 providers	SOURCED	A	S143	CMS Long-Term Care Hospital Compare - General Information + Provider Data, file vintage 2026-02-13, one row per certified provider (CPH)	Post_Acute_Supply_State	MedPAC ~320 LTCHs (registry note)
F261	Home health agency (HHA) - certified providers, US	2026	12392 providers	SOURCED	A	S144	CMS Provider Data Catalog - Home Health Care Agencies (dataset 6jpm-svkc), file vintage 2026-05-23, one row per certified provider (CPH)	Post_Acute_Supply_State	

F262	Hospice - certified providers, US	2026	6852 providers	SOURCED	A	S145	CMS Provider Data Catalog - Hospice General Information (dataset yc9t-dgbk), file vintage 2026-05-23, one row per certified provider (CNA)	Post_Acute_Supply_State	
F263	Total post-acute destinations, US (5 settings)	2026	35481 providers	DERIVED	A	S141, S142, S143, S144, S145	live SUM of the five CMS provider-file counts (35,481)	Post_Acute_Supply_State	matches <code>lt_study.ecosystem().postacute_destinations</code>
F264	CA total post-acute destinations	2026	6474 providers	DERIVED	A	S141, S142, S143, S144, S145	row SUM across the five CMS provider files, state=CA	Post_Acute_Supply_State	
F265	TX total post-acute destinations	2026	4306 providers	DERIVED	A	S141, S142, S143, S144, S145	row SUM across the five CMS provider files, state=TX	Post_Acute_Supply_State	
F266	STEMI transfer median door-in-door-out time	2011	68 minutes	ACADEMIC	B	S146	Wang et al., JAMA 2011, DOI 10.1001/jama.2011.862 (n=14,821)	Clinical_Benchmarks	DIDO >30 min: mortality 5.9% vs 2.7%, aOR 1.56
F267	STEMI in-hospital mortality when DIDO >30 min	2011	0.059 share	ACADEMIC	B	S146	Wang et al., JAMA 2011, DOI 10.1001/jama.2011.862	Clinical_Benchmarks	vs 2.7% when <=30 min
F268	Mean stroke-transfer DIDO	2024	171.4 minutes	ACADEMIC	B	S148	JAMA Netw Open 2024, DOI 10.1001/jamanetworkopen.2024.31183 (n=28,887)	Clinical_Benchmarks	GWTC median 174 min (JAMA 2023) is the companion read
F269	EMS prenotification-associated stroke DIDO reduction	2023	20.1 minutes	ACADEMIC	B	S149	GWTC-Stroke, JAMA 2023, DOI 10.1001/jama.2023.12739 (n=108,913)	Clinical_Benchmarks	
F270	National median ambulance patient offload time	2024	10.9 minutes	ACADEMIC	B	S150	Shaw et al., Prehosp Emerg Care 2025, DOI 10.1080/10903127.2025.2535576 (7,237,606 records)	Clinical_Benchmarks	
F271	Hospitals detaining EMS crews >1 hour (CA 2017)	2017	0.75 share	ACADEMIC	B	S151	Backer et al., Prehosp Emerg Care 2018, DOI 10.1080/10903127.2018.1525456 (830,637 transports)	Clinical_Benchmarks	40% >2h, 33% >3h
F272	Admitted ED patients 65+ boarding >=2 hours	2015-2022	0.852 share	ACADEMIC	B	S152	Lee et al., Ann Emerg Med 2026, DOI 10.1016/j.annemergmed.2026.03.011 (NHAMCS 2015-2022)	Clinical_Benchmarks	
F273	Mean ED boarding, admitted 65+ (2022)	2022	343 minutes	ACADEMIC	B	S152	Lee et al., Ann Emerg Med 2026, NHAMCS 2022 read	Clinical_Benchmarks	up from 138 min in 2018; 501 min with dementia
F274	Discharges delayed - weighted mean across 64 studies	2019	0.228 share	ACADEMIC	B	S154	Landeiro et al., The Gerontologist 2019, DOI 10.1093/geront/gnx028	Clinical_Benchmarks	range 1.6-91.3%; cost \$142-\$31,935 PPP/case
F275	PACU delays caused by transport unavailability	2022	0.111 share	ACADEMIC	B	S156	Ego et al., Ann Med Surg 2022, DOI 10.1016/j.amsu.2022.104680 (n=34 of 307; non-US)	Clinical_Benchmarks	second to bed unavailability 22.5%
F276	Dialysis - cert-base CAGR of surviving stock, window 1981-2025	1981-2025	0.0674787558009695 %/yr	DERIVED	A	S158	live formula (cum 2025 / cum 1981)^(1/44)-1 over the certification-vintage grid; survivor-stock caveat travels	Certification_Series	matches <code>analytics.supply_trend()</code> to 1e-9
F277	SNF - cert-base CAGR of surviving stock, window 1968-2025	1968-2025	0.0511659229116082 %/yr	DERIVED	A	S159	live formula (cum 2025 / cum 1968)^(1/57)-1 over the certification-vintage grid; survivor-stock caveat travels	Certification_Series	matches <code>analytics.supply_trend()</code> to 1e-9
F278	Home health - cert-base CAGR of surviving stock, window 1978-2025	1978-2025	0.0643466141161573 %/yr	DERIVED	A	S160	live formula (cum 2025 / cum 1978)^(1/47)-1 over the certification-vintage grid; survivor-stock caveat travels	Certification_Series	matches <code>analytics.supply_trend()</code> to 1e-9
F279	Hospice - cert-base CAGR of surviving stock, window 1988-2025	1988-2025	0.0793692359442053 %/yr	DERIVED	A	S161	live formula (cum 2025 / cum 1988)^(1/37)-1 over the certification-vintage grid; survivor-stock caveat travels	Certification_Series	matches <code>analytics.supply_trend()</code> to 1e-9
F280	IRF - cert-base CAGR of surviving stock, window 1984-2025	1984-2025	0.0592294981076973 %/yr	DERIVED	A	S162	live formula (cum 2025 / cum 1984)^(1/41)-1 over the certification-vintage grid; survivor-stock caveat travels	Certification_Series	matches <code>analytics.supply_trend()</code> to 1e-9
F281	LTCH - cert-base CAGR of surviving stock, window 1976-2025	1976-2025	0.0642414341728086 %/yr	DERIVED	A	S163	live formula (cum 2025 / cum 1976)^(1/49)-1 over the certification-vintage grid; survivor-stock caveat travels	Certification_Series	matches <code>analytics.supply_trend()</code> to 1e-9
F282	Dialysis facilities in the certified roll	2026	7557 facilities	SOURCED	A	S158	live link to the grid's last cumulative cell (7,557 rows in DFC_FACILITY, source_date 2026-03-25)	Certification_Series	equals <code>State_Facility_Structure</code> dialysis total
F283	Home-health net adds in peak build year 2023	2023	609 agencies certified	SOURCED	A	S160	certification-vintage grid, year 2023 (609 agencies of the surviving stock certified that year)	Certification_Series	
F284	Hospice net adds in peak build year 2022	2022	771 hospices certified	SOURCED	A	S161	certification-vintage grid, year 2022 (771 hospices of the surviving stock certified that year)	Certification_Series	
F285	Dialysis chain HHI (all 30 named chains)	2026	2767 HHI (0-10,000)	SOURCED	A	S158	chain_org column of DFC_FACILITY vendored roll (source_date 2026-03-25); share² sum over named operators, 4.87 E57 facility denominators	State_Facility_Structure	live top-6 formula computes 2,766; >2,500 = highly concentrated (DOJ/FTC)
F286	Dialysis chain HHI - six largest chains (live)	2026	2765.90655055805 HHI (0-10,000)	DERIVED	A	S158	live Σ(share×100)² over the six named-chain rows	State_Facility_Structure	module full-file HHI 2,767
F287	DaVita share of certified dialysis facilities	2026	0.370517401085087 share	DERIVED	A	S158	live 2,800/7,557 from chain_org counts	State_Facility_Structure	
F288	Fresenius share of certified dialysis facilities	2026	0.366812227074236 share	DERIVED	A	S158	live 2,772/7,557 from chain_org counts	State_Facility_Structure	
F289	DaVita + Fresenius combined facility share	2026	0.737329628159323 share	DERIVED	A	S158	live sum of the two chain-share formulas (73.8%)	State_Facility_Structure	Koukounas et al. (JAMA/PMCL12177639): 77.1% of facilities in 2019
F290	Dialysis for-profit share (all rows ownership-known)	2026	0.895858144766442 share	DERIVED	A	S158	live ΣD/EB over the dialysis state panel (89.6%)	State_Facility_Structure	
F291	SNF certified facilities, US (live SUM of states)	2026	14699 facilities	DERIVED	A	S159	live SUM over 53 state rows of NH_ProviderInfo (source_date 2026-04-01) = 14,699	State_Facility_Structure	matches <code>Post_Acute_Supply_State</code> SNF count and <code>MedPAC</code> ~14,700
F292	Home-health top-5 state concentration	2026	0.605067785668173 share of agencies	DERIVED	A	S160	live SUM of the five largest state shares (60.5%; CA alone 25.4%)	State_Facility_Structure	
F293	Hospice facilities in CA (largest state)	2026	2062 facilities	SOURCED	A	S161	hospice state panel row 1 (CA 2,062 of 6,852 = 30.1%)	State_Facility_Structure	
F294	Hospital M&A announced transactions, 2025	2025	46 transactions	SOURCED	B	S173	Kaufman Hall 2025 Hospital and Health System M&A in Review (annual series 2015-2025)	Growth_Evidence_Registry	trough of the 11-year series; distress share 43.5% an all-time high
F295	Hospital M&A announced-transaction change 2015 -- 2025	2015-2025	-0.589285714285714 % change	DERIVED	B	S173	live formula 46/112-1 over the series endpoints (-58.9%)	Growth_Evidence_Registry	
F296	Distressed share of 2025 hospital M&A	2025	0.435 share of transactions	SOURCED	B	S173	Kaufman Hall 2025 review: "43.5% of all transactions involving a distressed party"	Growth_Evidence_Registry	
F297	US community hospitals system-affiliated, FY2024	FY2024	0.696543643819567 share	DERIVED	B	S170	live 3,567/5,121 from AHA Fast Facts 2026 (2024 Annual Survey)	Growth_Evidence_Registry	AHRQ CHSP: ~70% of non-federal general acute hospitals in a system (2022)
F298	System-affiliation pp change 2005 -- FY2024	2005-2024	0.166543643819566 percentage points	DERIVED	B	S170, S171	live difference of the series endpoints (53% -- ~70%)	Growth_Evidence_Registry	do not CAGR - irregular multi-source series
F299	Adult ED-to-ED interfacility transfers 2018-2022 (NEDS)	2018-2022	9,867,701 over 2018-2022 (~1.97M/yr); critical-procedure transfers alone	ACADEMIC	B	S165	Nikolla et al., J Emerg Med 2025, DOI 10.1016/j.jemermed.2025.12.020 (HCUP NEDS 2018-2022); 9,867,701 critical-procedure OP 1,000,000	Growth_Evidence_Registry	Demand_Evidence_Quotes ned_s_ed_transfers record

F300	ED visits arriving via interfacility EMS transfer - trend	2014-2020	~1.3M/yr; +15% (2017-19) and +35% (2020-22) vs 2014-16 baseline	~1.3M/yr; +35% (2020-22 vs 2014-16)	ACADEMIC	B	S164	Peters GA et al., Am J Emerg Med 2026;106:24-29, DOI 10.1016/j.ajem.2026.04.025 (verbatim)	Growth_Evidence_Registry	
F301	Rural vs urban ED transfer rate multiplier	-	6.2% rural vs 2.0% urban - a >3x multiplier	6.2% vs 2.0%	ACADEMIC	B	S166	Greenwood-Ericksen et al., JAMA Netw Open 2021, DOI 10.1001/jamanetworkopen.2021.34980 (verbatim)	Growth_Evidence_Registry	
F302	Rural hospital closures since 2005 (Sheps tracker)	2005-2022	194 since 2005 (151 after 2010); NE: 2 (latest March 2026 Published report)	194 (151 after 2010)	SOURCED	B	S181	UNC Sheps Center rural hospital closures tracker (needs re-verify)	Growth_Evidence_Registry	
F303	Rural Emergency Hospital conversions since Jan 2023	2023-2025	40-50 conversions since Jan 2023; nearly half in GOVERNANCE, NE: Wisconsin	40-50; half in KS/TX/NE/OK	ACADEMIC	B	S182	J Rural Health 2026, DOI 10.1111/rjh.70112 (verbatim); REHs keep no inpatient beds - every admission becomes a transfer	Growth_Evidence_Registry	
F304	US ESRD prevalence (dialysis transport driver)	2020	>808,000 living with ESRD; ~68% on dialysis; prevalence dialysis since ~2010	>808,000; ~68% on dialysis	GOV	B	S186	NIDDK Kidney Disease Statistics (USRDS-based), Oct 2025 (needs re-verify)	Growth_Evidence_Registry	
F305	Medicare FFS ground ambulance transports / spend / orgs	2024	11.3 million transports; \$5.3 billion; ~10,600 organizations	11.3M; \$5.3B; ~10,600	GOV	B	S197	MedPAC Payment Basics "Ambulance Services Payment System", Oct 2024, p.1 (verbatim quote on-row)	Demand_Evidence_Quotes	MedPAC Mar 2025 report: ~11.4M transports (2023 data)
F306	Adult ED-to-ED transfers, 2018-2022	2018-2022	9,867,701 adult transfers over 2018-2022 (~1.97M/yr)	9,867,701 (~1.97M/yr)	ACADEMIC	B	S165	Am J Emerg Med (2025), HCUP NEDS 2018-2022 (verbatim)	Demand_Evidence_Quotes	Growth_Evidence_Registry neds, transfers record
F307	Acute-to-acute inter-hospital transfers per year	-	~1.5 million/yr = 3.5% of inpatient admissions	~1.5M/yr = 3.5% of admissions	ACADEMIC	B	S198	Hernandez-Boussard T et al., J Patient Saf 2017 (2009 NIS; epub 2014), PubMed 25397857 (HCUP NIS; verbatim)	Demand_Evidence_Quotes	2009 data, epub 2014 - often miscited as Mueller 2014; corrected against PubMed
F308	BLS share of ground ambulance transports	-	56% BLS (so ~44% ALS)	56% BLS / ~44% ALS	SOURCED	B	S199	CMS GADCS Year 1-2 cohort report (RAND 2024), cms.gov PDF (verbatim)	Demand_Evidence_Quotes	
F309	National EMS activations (2023)	2023	54,190,579 activations; 14,369 agencies; 54 agencies/100,000 pop	54,190,579; 14,369 agencies	SOURCED	B	S200	NEMSIS 2023 Public-Release Research Dataset (NHTSA Office of EMS) (verbatim)	Demand_Evidence_Quotes	
F310	US health systems; hospital affiliation	2022	640 health systems (2022); ~70% of non-federal general acute hospitals in system	640 systems; ~70% affiliation (2022)	GOV	B	S202	AHRQ Compendium of US Health Systems, 2022 (verbatim)	Demand_Evidence_Quotes	AHA Fast Facts FY2024: 3,567/5,121 = 69.6%
F311	US inpatient discharges per year	-	>35 million weighted discharges/yr (20% stratified sample)	~35M weighted/yr	SOURCED	B	S201	AHRQ HCUP NIS overview page (quote carries editorial weighting note)	Demand_Evidence_Quotes	
F312	BLS non-emergency claim-line drift 2018 - 2022 (GADCS)	2018-2022	-0.0660000000000001 percentage points		DERIVED	B	S199	live difference of the two published GADCS points (43.7% - 37.1% = -6.6pp)	Demand_Evidence_Quotes	2-point pair - direction only
F313	US 65+ population CAGR 2025 - 2030 (projection)	2025-2030	0.0269022284200189 %/yr		DERIVED	B	S188	live (71.6/62.7)*(1/5)-1 over the Census projection pair (=2.69%/yr)	Demand_Evidence_Quotes	the Census 65+ projection pair is GOV; the per-condition blended ~2.7%/yr read is modeled and lives on Excluded_Not_Sourced, not here
F314	US total MA enrollment (state-file sum)	2022	29674598 enrollees		DERIVED	A	S205	live SUM of 53 state rows; equals ma_geo_report.json total 29,674,598 (diff cell computes 0)	MA_Geo_Variation	
F315	MA IP stays per 1,000 enrollees, national (enrollment-weighted)	2022	199.291194877845 stays /1,000		DERIVED	A	S205	live SUMPRODUCT over the state table (ip_stays_per_1000 field)	MA_Geo_Variation	published bound for the TAM_Model_National MA-utilization TEAM-INPUT (0.75-1.0x)
F316	MA SNF days per 1,000 enrollees, national (enrollment-weighted)	2022	783.260956019121 days /1,000		DERIVED	A	S205	live SUMPRODUCT over the state table (snf_days_per_1000 field)	MA_Geo_Variation	
F317	MA ER visits per 1,000 enrollees, national (enrollment-weighted)	2022	354.956621244173 visits /1,000		DERIVED	A	S205	live SUMPRODUCT over the state table (er_visits_per_1000 field)	MA_Geo_Variation	
F318	Highest state SNF days per 1,000 MA enrollees	2022	3237.767 days /1,000		DERIVED	A	S205	live MAX screen over the state table (companion INDEX/MATCH cell names the state)	MA_Geo_Variation	Massachusetts, 3,237.8 - a published outlier >2x the next state (OH 1,502.6); outlier note on-sheet
F319	California MA enrollment (largest state)	2022	3188104 enrollees		SOURCED	A	S205	state table row 1 (CA 3,188,104)	MA_Geo_Variation	
F320	States/territories in the MA geo file	2022	53 states		DERIVED	A	S205	live COUNTA; report JSON states=53	MA_Geo_Variation	
F321	US 65+ civilian population	2024	61179918 persons		GOV	A	S206	SC-EST2024-AGESEX-CIV, ages 65-85+, SEX=0, national	State_Age_65plus	Measured estimate; the NP2023 projection series on Macro_Demand_Drivers is the forward companion
F322	US 65+ CAGR 2020-2024 (measured)	2020-2024	0.0295173468603869 %/yr		DERIVED	A	S206	(65+ 2024 / 65+ 2020)^(1/4)-1	State_Age_65plus	Brackets the projected ~2.7%/yr 65+ growth on Macro_Demand_Drivers; measured vs projected bases stated
F323	EMT jobs (OEWS, sum of published state cells)	2024	178900 jobs		DERIVED	A	S207	OEWS May 2024 state file, OCC 29-2042, TOT_EMP summed OEWS_EMS_Wages	OEWS_EMS_Wages	Occupation basis (all industries); QCEW industry employment (171,100 private, 2024) is a different universe - both stated
F324	Footprint hospitals across the 20 target metros	2026	166 hospitals		DERIVED	B	S208	SUM over 20 metro rows; geocoded Hospital General Information, city+state filtered	Metro_Structure_20	footprint_rollup() returns 166; UNMC + Via Christi absences make this a floor
F325	Footprint SNF certified beds across the 20 metros	2026	51108 beds		DERIVED	B	S208	SUM over 20 metro rows; NH_ProviderInfo 2026-04-01 certified beds	Metro_Structure_20	footprint_rollup() returns 51,108
F326	Footprint HCRIS staffed beds across the 20 metros	2026	40762 beds		DERIVED	B	S100, S208	SUM over 20 metro rows; HCRIS FY2020-22 latest per CCN, joined by CCN (152 of 166 hospitals matched - a floor)	Metro_Structure_20	footprint_rollup() returns 40,762
F327	National geocoded hospital roll (denominator)	2026	4630 hospitals		SOURCED	B	S208	hospital_coords.csv full-US row count (4,630)	Metro_Structure_20	Facility_Universe_State counts the same Care Compare universe from live pulls; AHA counts ~6,100 (broader universe - both stated)
F328	Footprint-state share of national hospitals (11 states)	2026	0.24902807775378 share		DERIVED	B	S208	1,153 footprint-state hospitals / 4,630 national (both from the geocoded roll) = 24.9%	Metro_Structure_20	Numerator is state-level, not metro-level - stated on-sheet
F329	Footprint-state share of national SNF certified beds	2026	0.221535328511059 share		DERIVED	B	S208	347,674 footprint-state SNF beds / 1,569,384 national (NH_ProviderInfo 2026-04-01) = 22.2%	Metro_Structure_20	National SNF bed base also carried on Facility_Universe_State (later snapshot)
F330	MMT footprint population (22 counties, 2020 Census)	2020	1560299 residents		DERIVED	B	S209	SUM of 22 county 2020 Census populations (= 1,560,299)	County_Demography	lft_mmt.footprint_summary() pop_2020 = 1,560,299; CBSA panel SUM matches
F331	Douglas County NE population (2020 Census)	2020	584526 residents		SOURCED	B	S209	2020 Decennial Census, Douglas County NE (FIPS 31055)	County_Demography	MMT HQ county; largest footprint county
F332	Fastest-growing footprint county: Saunders NE CAGR 2020-2024	2024	0.0205375830415884 %/yr		DERIVED	B	S210, S209	(23,406 / 21,578)^(1/4)-1 = +2.05%/yr (live formula)	County_Demography	lft_mmt.county_growth() computes 2.054%/yr; re-verify flag travels
F333	Sarpy County NE population (Vintage-2024 estimate)	2024	204828 residents		SOURCED	B	S210	Vintage-2024 estimate 204,828 (captured 2026-07-10; re-verify queue)	County_Demography	+7.46% over 2020 - the fastest-growing large NE county
F334	Footprint = complete OMB composition of 7 CBSAs (22 counties)	2023	22 counties		DERIVED	A	S211	live COUNTIF per CBSA against the vendored OMB Bulletin 23-01 crosswalk = 22 = footprint count	County_Demography	8+2+3+2+3+2+2 counties across CBSAs 36540/30700/24260/28260/25580/35820/18100
F335	Counties inside a CBSA (2023 OMB delineations)	2023	1915 county rows		DERIVED	A	S211	live COUNTIF total over the vendored list1_2023 extract (= 1,915; meta cross-check delta 0)	CBSA_Crosswalk_Reference	cbsa_crosswalk_meta.json counties=1915
F336	Distinct CBSAs (2023 OMB delineations)	2023	935 CBSAs		DERIVED	A	S211	SUM of first-occurrence helper column (= 935 = 393 MSA + 542 uSA)	CBSA_Crosswalk_Reference	cbsa_crosswalk_meta.json cbsas=935
F337	Metropolitan county rows (2023 delineations)	2023	1252 county rows		DERIVED	A	S211	COUNTIF area_type="Metropolitan" (= 1,252; micropolitan = 663)	CBSA_Crosswalk_Reference	Metro counties sit inside MSAs - the AFS urban proxy at county grain

F376	Projected ambulance improper payments (CERT)	2024	\$95.1 \$M	GOV	B	S233	CERT 2024 supplemental data: \$595.1M projected (re-verify)	Payment_Integrity	13.2% rate x the ambulance payment base; implied \$377.9M documentation / \$163.7M necessity as live formulas
F377	CERT error share from insufficient documentation	2024	0.635 share of improper \$	GOV	B	S233	CERT 2024 error-category split: 63.5% (re-verify)	Payment_Integrity	Medical necessity 27.5%; residual 9.0% (live formula)
F378	RSNAT effect on probability of RSNAT use	2022	-0.61 change in probability	ACADEMIC	B	S234	doi:10.1001/jamahealthforum.2022.2093 - verbatim "61% reduction in the probability of RSNAT use" (ESRD cohort, needed on transportation-related)	Payment_Integrity	Paired with -77% spend and +19%/yr emergency dialysis on adjacent rows
F379	RSNAT effect on RSNAT expenditures	2022	-0.77 change (\$1.136/beneficiary-yr)	ACADEMIC	B	S234	Verbatim "77% reduction in RSNAT expenditures for a total of \$1136 per beneficiary-year"	Payment_Integrity	Same paper as effects 1 and 3 (single DOI)
F380	RSNAT side-effect: emergency dialysis use	2022	0.19 annual change in probability	ACADEMIC	B	S234	Verbatim "19% annual increase in the probability of emergency dialysis use" (ESRD cohort)	Payment_Integrity	The cost-shift direction - carried with the suppression effects, never alone
F381	Medicare ambulance transport growth 2002-2011 (pre-RSNAT boom)	2011	0.69 endpoint growth	GOV	B	S236	OEI-09-12-00350: transports +69% 2002-2011; dialysis-related +269%	Payment_Integrity	Implied CAGRs 6.0%/15.7% as live formulas; series break flagged - RSNAT broke this trend
F382	Suppliers with questionable-billing patterns (OIG)	2015	0.21 share of suppliers	GOV	B	S237	OEI-09-12-00351: 21% (about 1 in 5) met >=1 of 7 measures; 4 metros = 18% of transports / 52% of questionable (~\$207M)	Payment_Integrity	\$24M / \$30.2M / \$7.1M / \$4.3M half-year findings on adjacent rows
F383	Texas Medicaid NEMT on-time-pickup benchmark	2026	0.85 on-time share	GOV	B	S239	TX HHSC Medical Transportation Program contract terms: 85% benchmark, \$15,000 penalty cap (re-verify)	Contract_Benchmarks	Broker-contract common form >=95% on adjacent row
F384	San Diego / Falck per-response penalty	2026	5,000 \$ per response beyond 24 min	PUBLIC-WEB	B	S238	San Diego municipal contract records via repo dossier (re-verify; no URL on file)	Contract_Benchmarks	Same register carries Pasadena 8:59/90% cap 14:59 and Multnomah below-90% penalties
F385	California AB 40 ambulance patient offload standard	2023	0.9 compliance threshold (30-min offload)	GOV	B	S241	AB 40 (2023) / EMSA APOT: <=30 minutes 90% of the time, monthly per-hospital monitoring (re-verify)	Contract_Benchmarks	emsa.ca.gov/apot program page cited; the one statutory offload clock in the register
F386	Mississippi NEMT broker observed miss rate	2024 ~5.8%	monthly trips late/missed	PUBLIC-WEB	B	S240	Mississippi Today (Sep 2024): ~3,000 of ~52,000 monthly trips, ~3x the contractual limit (re-verify)	Contract_Benchmarks	Enforcement companions: NJ ~\$1.7M (2017-22); GA ~\$1M (2018-20)
F387	NJ Medicaid NEMT enforcement fines vs Modivcare	2022 ~\$1.7M	fines 2017-2022	PUBLIC-WEB	B	S240	State enforcement records via press (re-verify; no URL on file)	Contract_Benchmarks	Modivcare later filed Ch.11 (Aug 2025) - context row on the same tab
F388	NEMT broker contract on-time standard (common form)	2026	0.95 on-time share	PUBLIC-WEB	B	S239	State broker contract standards: >=95% on-time pickup with penalties (re-verify)	Contract_Benchmarks	TX 85% benchmark is the enforce-with-cap variant
F389	Medicare FFS ground ambulance universe (transports / payments / billing organizations)	2024 11.3M transports · \$5.3B · ~10,600 organizations	mixed (see locator)	GOV	B	S248	MedPAC Ambulance Payment Basics, Oct 2024 - Q1 block q1-definitions: "11.3M transports, \$5.3B, ~10,600 organizations" (2024)	InDepth_Q01	Same denominators power the \$469 DERIVED average (\$5.3B/11.3M) and the fragmentation read on InDepth_Q07/Q09
F390	Measured acute IFT base (adult ED-to-ED + inpatient interhospital transfers)	2022 ~1.97M/yr ED-to-ED (NEDS 2018-2022) + ~1.5M/yr inpatient IFTs	transfers/yr	ACADEMIC	B	S259	Nikolia et al., J Emerg Med 2025 (doi:10.1016/j.jemermed.2025.12.020; HCUP NEDS 2018-2022, a 967,701 adult inpatient hospitalizations)	InDepth_Q01	Feeds the IFT, mmt footprint demand band (3.47M legs/yr) on InDepth_Q10
F391	Ground ambulance out-of-network share	2022 ~60% of rides	% of rides	SOURCED	B	S256	FAIR Health 2023 study of 2022 claims - Q1 q1-customer	InDepth_Q01	Pairs with GADCS 19.7% collect-nothing and the NSA exclusion / GAPB rows on the same tab
F392	CA hospital crew-detention distribution (2017, 830,637 transports)	2017 75% of hospitals >1h · 40% >2h · 33% >3h	% of CA hospitals	ACADEMIC	B	S151	Backer et al., Prehosp Emerg Care 2018 - Q2 crew-detention evidence	InDepth_Q02	Consistent direction with Shaw 2025 national APOT tail (3.3% of agencies >25% >30 min)
F393	National median ambulance patient offload time (APOT)	2024 10.9 min median (IQR 6.6-17.5)	minutes	ACADEMIC	B	S150	Shaw et al., Prehosp Emerg Care 2025, 7,237,606 records (2024) - Q2 offload evidence	InDepth_Q02	CA AB 40 statutory standard (30 min at 90%) cited on the same tab
F394	ED boarding of admitted 65+ (mean minutes, 2018 -- 2022)	2022 138 -- 343 min (85.2% boarded >2h)	minutes (endpoint growth)	ACADEMIC	B	S152	Lee et al., Ann Emerg Med 2026 (NHAMCS 2015-2022) - Q2 boarding evidence; 501 min with dementia	InDepth_Q02	ACEP/Morning Consult poll (44%/16%) on InDepth_Q07 gives the public-experience read
F395	Post-acute destination supply (national node count)	2026 35,481 destinations (14,699 SNF + 1,221 IRF + 317 LTACH + 13,950 LHA + 1,400 other)	facilities	SOURCED	B	S276	CMS Care Compare / Provider-of-Services files via ifft_clinical_demand.destination_supply() - Q3 destination evidence	InDepth_Q03	MedPAC payment-basics volumes (~1.8M SNF / 383k IRF / 90k LTCH stays) on the same tab
F396	Rural ED visit volume (2005 -- 2016)	2016 16.7M -- 28.4M visits	visits/yr (2-pt series)	ACADEMIC	B	S260	Greenwood-Ericksen & Kocher, JAMA Network Open 2019/2021 - Q3 rural demand evidence (~4.9%/yr endpoint)	InDepth_Q03	Rural transfer propensity 6.2% vs urban 2.0% (same authors) on InDepth_Q01/Q07
F397	STEMI transfer DIDO and mortality	2011 median DIDO 68 min; 11% <30 min; mortality 5.9% vs 9.7% rural, 4.1% ED	minutes / % (n=14,821)	ACADEMIC	B	S146	Wang et al., JAMA 2011 - Q3 delay-consequence evidence	InDepth_Q03	Stroke DIDO family (Ng 2017; JAMA Netw Open 2024; GWTG 2023) carried on InDepth_Q02
F398	RSNAT prior-authorization natural experiment (ESRD cohort)	2022 ~61% probability of RSNAT use - ~7% RSNAT spend	effect sizes	ACADEMIC	B	S268	Contreary et al., JAMA Health Forum 2022 (doi:10.1001/jamahealthforum.2022.2093) - Q4 payment flow evidence	InDepth_Q04	RSNAT definitions (CMS rules) cited on InDepth_Q02; OIG utilization findings adjacent
F399	OIG: payments with no Medicare service at origin or destination (H1-2012)	2015 \$30.2M in one half-year	USD (half-year)	GOV	B	S237	HHS OIG OEI-09-12-00351 (2015) - Q4 incentive evidence; companion findings (21% of suppliers; 4 metros 18%/52%) on InDepth_Q07	InDepth_Q04	CERT 2024 split (13.2% / \$595.1M / 63.5% / 27.5%) carried on InDepth_Q02/Q07
F400	Bed-day value vs Medicare transport price mismatch	2024 ~\$3,132 adjusted expense per inpatient day vs \$469 Medicare reimbursement rate	USD	DERIVED	B	S266, S248	KFF State Health Facts 2023 expense proxy (re-verify) + MedPAC Oct 2024 universe - equation stated in the Q4 claim cost (1st bed day, 6.7 hours transport)	InDepth_Q04	Both inputs carried as separate cited lines on the same tab
F401	Hospital-owned ambulance org NPIs - NE vs IA contrast	2026 NE 5 (of 400) vs IA 40+ (of 351)	org NPIs	SOURCED	B	S285	CMS NPPEs NE+IA ambulance-org sweep, vendored 2026-07-10 (751 org NPIs) - Q5 prevalence evidence	InDepth_Q05	Same registry powers Competitor_Registry and the InDepth_Q08 MMT estate
F402	CommonSpirit operating loss trajectory	2025 ~\$875M (FY2024) -- ~\$225M (FY2025)	USD operating loss	SOURCED	B	S272	CommonSpirit FY2025 results release (re-verify) - Q5 capital-appetite evidence	InDepth_Q05	Context for why systems avoid insourcing capital: 2-pt trend, never extrapolated
F403	Insourced-heavy comparators (scale of the alternative)	2024 Mayo Clinic Ambulance ~70 units; Allina Health EMS ~90 units	units / requests	SOURCED	B	S283	Operator public pages via footprint registry, captured 2026-07-10 (re-verify) - Q5 insourcing evidence	InDepth_Q05	Contrast rows: NE systems own 5 ambulance NPIs total (same tab)
F404	Hospital system-affiliation share (buyer consolidation)	2024 70% (3,567 of 5,121 non-federal general acute care hospitals)	% of hospitals	SOURCED	B	S247	AHA Fast Facts 2026 (FY2024 Annual Survey; re-verify) - Q6 InDepth_Q06 procurement-ownership evidence	InDepth_Q06	Underpins system-level contracting logic on InDepth_Q09
F405	Ambulance Inflation Factor (contract escalator precedent)	2026 +2.4% (CY2025) · +2.0% (CY2026)	%/yr	GOV	B	S246	CMS Ambulance Fee Schedule PUF (re-verify against PUF) - Q6 contract-structure evidence	InDepth_Q06	2-point escalator series; never trended forward
F406	Modivcare Chapter 11 (vendor-stability precedent)	2025 Filed Aug 20, 2025; emerged Dec 29, 2025 cutting ~\$1.1B	event	SOURCED	B	S243	S.D. Tex. bankruptcy docket + coverage (re-verify) - Q6 evaluation-criteria evidence	InDepth_Q06	GMR refinancing/IPO row on InDepth_Q10 gives the 911-side analogue
F407	Nebraska statewide transfer-center confirmation rate (measured shortfall)	2021 146 of 234 confirmed (62%, Sep 2021) · 113 of 168 confirmed	% of requested transfers	SOURCED	B	S277	Omaha World-Herald / Journal Star, Nov 2021 (re-verify; COVID-era context travels) - Q7 acceptance evidence	InDepth_Q07	The only public IFT acceptance gauge in the corpus - 2-point episode, never trended
F408	PACU discharge delays attributable to transport unavailability	2022 11.1% of delays (2nd after bed unavailability 22.5%); 24.0% of delays were attributed to transport	% of delays	ACADEMIC	B	S156	Ego et al., Ann Med Surg 2022, 307-patient non-US cohort (magnitude only - caveat carried) - Q7 flow evidence	InDepth_Q07	Discharge-task 47% single-site study (PMC11023539) on InDepth_Q02 corroborates direction
F409	Average length of stay deterioration (2019 -- 2022)	2022 +19% overall; +24% for post-acute discharges	% change (endpoint)	SOURCED	B	S267	AHA Issue Brief, Dec 2022 (re-verify) - Q7 hospital-flow evidence	InDepth_Q07	Boarding trend 138 -- 343 min (Lee 2026, InDepth_Q02) moves the same direction
F410	MMT verified NPI estate vs company claim	2026 23 active org NPIs across 11 NPI states vs a 13-state	org NPIs / states	SOURCED	B	S285	CMS NPPEs org-NPI sweep, pulled 2026-07-10 - Q8 geography evidence (VA link unconfirmed, flagged)	InDepth_Q08	Full row-level estate on MMT_NPI_Estate; conflict stated on both tabs
F411	MMT claimed operating scale	2026 200,000+ missions/yr (Jan 2022 release) · 500+ vehicles	claims (floors)	SOURCED	B	S280, S224	Businesswire Harbour Point release Jan 25, 2022 + mmtamb.com About Us captured 2026-07-10 - Q8 scope evidence (2022-2023)	InDepth_Q08	Advisor framed the 2022 deal as 7 states / ~1,000 employees - conflict carried, not blended
F412	MMT claimed headcount trajectory (self-reported floors)	2026 350+ (2015, 1 state) -- ~1,000 (2022, 7 states) -- 2,000 (2026, 19 states)	people (claimed floors)	SOURCED	B	S280, S224	LJS Feb 2015 sale coverage; Lincoln International notice 2022; mmtamb.com 2026 - Q8 workforce evidence; 2022-2023	InDepth_Q08	Same 3-point series charted on Company_Dossier with the identical caveats
F413	GADCS mean cost per transport by ownership	2025 \$2.673 all · \$3.127 governmental · \$1.778 private for profit	USD/transport	SOURCED	B	S245	CMS/RAND GADCS Year 1-2 report via trade coverage (re-verify) - Q9 alternative-economics evidence	InDepth_Q09	MedPAC Dec 2025 volume-cost curve (same tab) explains the governmental/private gap via utilization

F414	Unit-hour utilization benchmark (911 vs high-performance)	2024 911 band 0.30-0.50; AIMHI survey mean 0.508	UHU ratio	SOURCED	B	S258	AIMHI benchmarking via EMS1 (as captured 2026-07-10) - Q9 shared-fleet evidence	InDepth_Q09	The capacity-productivity argument against shared 911 fleets for scheduled IFT
F415	Omaha Fire Department 911 transport monopoly (city ALS fleet)	2026 18 ALS ambulances as primary city transport	ambulances	SOURCED	B	S283	City EMS pages via the NPPEs landscape sweep, 2026-07-10 (re-verify; public-web claim) - Q9 traditional-EMS evidence	InDepth_Q09	Municipal 911 posture rows corroborated by the muni-contract precedent family
F416	Footprint demand band (measured national acute legs allocated to the MMT footprint)	2026 3.47M measured national acute legs/yr (1.97M NEDS + 1.5M AHE's attributed legs)	legs/yr (national base)	DERIVED	B	S281, S259	ift_nmt footprint demand band - ACADEMIC transfer counts + GOV Census shares, equation stated in the Q10 claim text	InDepth_Q10	Inputs cited separately on InDepth_Q01 (NEDS/NIS) and Q8 (Census legacy core)
F417	GMR capital-structure constraint (competitor context)	2026 IPO valuation target cut to \$3.3B (May 2026) after the PE AB 909E redemptions	USD	SOURCED	B	S278	Deal/IPO coverage via the NPPEs landscape sweep (re-verify) - Q10 durability evidence	InDepth_Q10	Modicare Ch.11 (InDepth_Q06) gives the NEMT-side analogue of vendor fragility
F418	Public-API connectors in the repo data estate	2026-07-10	16 connectors	DERIVED	B	S286	connectors.registry.catalog() n_connectors; pinned by connectors/tests/test_estate_invariants.py	Connector_Estate_Map	16 Panel A rows rendered
F419	Registered datasets across the connector estate	2026-07-10	204 datasets	DERIVED	B	S286	live SUM over Panel A dataset counts; catalog() n_datasets = len(all_registry_rows()) = 204 asserted at build	Connector_Estate_Map	204
F420	Estate APIs verified live-reachable (smoke tests)	2026-07-10 15 of 16	APIs	DERIVED	A	S286	Panel A live-status column (COUNTIF formula on-sheet); only census_acs blocked (keyless)	Connector_Estate_Map	inv_connectors 9E smoke-test table, 2026-07-10
F421	IFT probes in the connector probe registry	2026-07-10	18 probes	DERIVED	B	S287	connector_estate_map() executed at build time; live COUNT formula over Panel B tier column	Connector_Estate_Map	estate_summary total=18 (design note said 17)
F422	Live-pull artifacts cached for v3	2026-07-10	564 artifacts	DERIVED	A	S288	live SUM over Panel C cache-file counts; manifest.json has 564 entries, each with sha256 + UTC timestamp	Connector_Estate_Map	564
F423	Raw rows cached across all v3 live pulls	2026-07-10	2058881 rows	DERIVED	A	S288	live SUM over Panel C rows-cached column	Connector_Estate_Map	python recompute 961,680 (897,270 of it PSPS carrier lines)
F424	Medicare-enrolled ambulance suppliers (PECOS live cross-check with corrected filter literal)	2026Q1	10465 suppliers	GOV	A	S293, S288	PECOS /data/stats, filter PROVIDER_TYPE_DESC="PART B SUPPLIER - AMBULANCE SERVICE SUPPLIER" -> find_provider_id_acs	Connector_Estate_Map	matches vendored PPEF 2026.04.01 national row 10,465 exactly (PECOS_Suppliers_State)
F425	PSPS raw carrier line rows cached (2010-2024, 7 ground codes)	2010-2024	897270 rows	GOV	A	S290, S288	found_provider_id_acs 105 cached artifacts (15 vintages x 7 HCPCS); aggregated client-side on PSPS_Denial_Series	Connector_Estate_Map	897,270 rows (manifest sum)
F426	Ambulance/IFT code-vocabulary rows carried (POS, revenue, condition, value, RARC, modifier, taxonomy)	current code sets	17 codes	DERIVED	B	S297, S301	count of code rows in panels A-G (17)	Code_Vocabulary	2+1+3+1+1+3+6 = 17
F427	NUCC ambulance taxonomies (3416*)	v24.1	4 codes	GOV	B	S301, S287	NUCC Transportation section: 341600000X + land/air/water children; ift_connectors_AMBULANCE_TAXONOMIES	Code_Vocabulary	NPPEs live search accepts the text description "Ambulance" (codes rejected as filters)
F428	NUCC NEMT-boundary taxonomies (3439*/3438*)	v24.1	2 codes	GOV	B	S301, S287	343900000X NEMT van + 343800000X secured transport; ift_connectors_NEMT_TAXONOMIES	Code_Vocabulary	NEMT excluded from IFT TAM throughout workbook
F429	UB-04 revenue-center codes reserved for ambulance	current NUBC set	0540-0549 (10 codes)	GOV	B	S299, S297	REVENUE_CATEGORIES range (540, 549, "Ambulance")	Code_Vocabulary	revenue_category("0540")=="Ambulance" verified at build
F430	MAC A/B jurisdictions in the repo seed table	seed (re-verify)	12 jurisdictions	DERIVED	B	S302	live SUMPRODUCT over seed rows excluding the HHH national row (13 rows total)	Code_Vocabulary	12 A/B jurisdictions + 1 HHH row
F431	Worked KPI definitions (metric/definition/formula/source/owner/why)	n/a	12 metrics	DERIVED	B	S303	AUTHORED_SECTIONS["kpis"] "Sample KPI dictionary (worked metrics)" table, 12 rows x 6 cols	KPI_Dictionary	COUNTA formula on-sheet
F432	Tier metric definitions across 5 KPI tiers	n/a	32 definitions	DERIVED	B	S303	AUTHORED_SECTIONS["kpis"] bullet tiers: access 6, speed 7, workflow 6, clinical 6, financial 7	KPI_Dictionary	6+7+6+6+7 = 32
F433	Stakeholder scorecards (function-level KPI recommendations)	n/a	5 scorecards	DERIVED	B	S303	AUTHORED_SECTIONS["kpis"] "Recommended metrics by stakeholder" table, 5 rows x 4 cols	KPI_Dictionary	
F434	Benchmark values published on this tab	n/a	0 benchmarks	DERIVED	B	S303	source module states "template dictionary, not benchmarked values" - carried as-is, nothing invented	KPI_Dictionary	Excluded_Not_Sourced records what would make benchmarks citable
F435	Regulatory requirement rows binding IFT (statute-cited)	current law	11 rows	DERIVED	B	S303, S306, S307, S308, S309, S310, S115	AUTHORED_SECTIONS["regulatory"] table, 11 rows x 5 cols, per-row GOV basis	Regulatory_Register	COUNTA formula on-sheet
F436	Service levels with verbatim 42 CFR 414.605 definitions	current (2026)	4 levels	DERIVED	B	S304, S305	service_levels() BLS/ALS1/ALS2/SCT definition facts, each with quote + 3 sources	Regulatory_Register	ALS2 blood-transfusion trigger effective CY2025 (89 FR 98559) noted
F437	Level-determination edge-case scenarios	current rules	19 scenarios	DERIVED	B	S304, S311, S305	edge_cases(): 19 rows, 14 GOV / 5 FRAMEWORK, each citing regs/scope model	Regulatory_Register	
F438	Misconceptions corrected with sourced rebuttals	current rules	12 rows	DERIVED	B	S304	misconceptions(): 12 myth/reality rows, each with source labels	Regulatory_Register	includes MedPOPP CY2024-anchored corrections (30.7% BLS-NE share; SCT 0.7%)
F439	GOV facts in the underlying service-levels module (honesty audit)	verified Jul 2026	103 facts	DERIVED	B	S304	ift_service_levels.n_by_basis(); has_no_illustrative()==True; 53 URL-linked sources	Regulatory_Register	module summary() reports 109 facts total
F440	Medicare beneficiaries 2024, sum of the 51 SP_Index state formulas	2024	66638829 beneficiaries	DERIVED	A	S312	SP_Index US-total row: SUM of 51 INDEX/MATCH pulls from Enroll_State_2024 col C	SP_Index	Must reconcile to the 2024 national row on Enrollment_ESRD_State (same CMS Monthly Enrollment file, national grain)
F441	Ground base-code ambulance services 2024, 51-state formula sum (floor)	2024	9624703.4 services	DERIVED	A	S312	SP_Index US-total row: SUM of 51 six-code SUMIFS over MUP_State_2024 (A0426/27/28/29/33/34)	SP_Index	Sits at or below the 2024 ground-base total on MUP_Ambulance_National (state-grain suppression makes the state sum a floor)
F442	PECOS-enrolled ambulance suppliers, 51-state COUNTIF sum	2026	10153 suppliers	DERIVED	A	S312	SP_Index US-total row: SUM of 51 COUNTIFs over PECOS_Registry col E (PPEF 2026-04-01)	SP_Index	PECOS_Registry national count (10,465) minus territory/other-jurisdiction rows
F443	Ambulance organizations reporting facility-contract revenue (GADCS)	2025	0.186 share of organizations	GOV	A	S313	GADCS Y1-4 appendix, Table 2.32, p.70	Facility_Pay_Layer	Self-reported census; incidence, not an IFT share
F444	Facility-contract revenue, conditional mean per organization (GADCS)	2025	922,555 USD	GOV	A	S313	Table 2.32, p.70	Facility_Pay_Layer	Median \$48,393 on the same row: long right tail
F445	Facility-contract revenue, median per organization (GADCS)	2025	48,393 USD	GOV	A	S313	Table 2.32, p.70	Facility_Pay_Layer	Mean/median gap is the skew caveat in-row
F446	Organizations reporting any non-payer revenue (GADCS Question 5 categories)	2025	0.745 share of organizations	GOV	A	S313	Table 2.32, p.70	Facility_Pay_Layer	Includes subsidies, taxes, standby, memberships
F447	Non-payer revenue, conditional mean (GADCS)	2025	2,177,569 USD	GOV	A	S313	Table 2.32, p.70	Facility_Pay_Layer	Median \$162,265
F448	Organizations with local-government contract revenue (GADCS)	2025	0.154 share of organizations	GOV	A	S313	Table 2.32, p.70	Facility_Pay_Layer	Conditional mean \$556,091, median \$112,426
F449	Organizations with EMS-earmarked local tax revenue (GADCS)	2025	0.234 share of organizations	GOV	A	S313	Table 2.32, p.70	Facility_Pay_Layer	Conditional mean \$3.14M: the subsidy backbone of government services
F450	Mean payer revenue per ambulance NPI, all payers (GADCS)	2025	3,914,787 USD	GOV	A	S313	Table 2.30, pp.64-65	Facility_Pay_Layer	FFS \$1.19M, MA \$1.24M, Medicaid \$0.73M, commercial \$1.32M on the same table
F451	Medicare FFS share of ambulance transport revenue, mean per NPI (GADCS)	2025	0.29 share	GOV	A	S313	Table S.1, p.ix	Facility_Pay_Layer	The one-sixth-of-market arithmetic elsewhere in this workbook uses dollars, not per-NPI means; bases differ and are never mixed

F452	Organizations that did not always bill in at least one payer category (GADCS, published as about one in five)	2025	0.2 share, approximate as published	GOV	A	S313	Table S.1, p.ix ("about one in five")	Facility_Pay_Layer	The eligible-but-unbilled anchor for the sizing gross-up; approximate by publication
F453	DocGo Transportation Services revenue FY2025	2025	200,765,608 USD	SEC-A	A	S314	10-K FY2025, Note 2 (F-22) and Note 11 (F-37)	Facility_Pay_Layer	FY2024 \$193.4M, FY2023 \$181.5M same tables
F454	DocGo transportation revenue outside per-trip billing, FY2025	2025	0.409 share	DERIVED	A	S314	1 - trips x ATP / segment revenue; inputs Note 2/MD&A; live formula on Facility_Pay_Layer Panel B	Facility_Pay_Layer	FY2024 41.1% by the same arithmetic; whole-dollar ATP rounding gives about 0.1pp uncertainty
F455	VA ambulance procurement obligations (PSC V225), FY2025	2025	246,082,814 USD	GOV	A	S315	USAspending spending_over_time, psc V225, VA awarding toptier (Pull_Manifest hash)	Facility_Pay_Layer	All-agency FY2025 \$292.7M; VA share 84%; Beneficiary Travel claims are a separate channel, never summed
F456	Subject-company ground base transports billed to Medicare FFS (floor)	2024	69,684 services	GOV	A	S316	MUP by Provider & Service 2024, estate NPIs, base codes	MMT_Medicare_Book	Panel E per-NPI rows sum to this; flagship NPI A0428 row verified to the cent against the live API (Verification_Log)
F457	Subject-company Medicare FFS allowed dollars, ground base codes (floor)	2024	22,004,764 USD	GOV	A	S316	Sum of services x average allowed over estate NPI rows	MMT_Medicare_Book	Paid share of allowed prints beside it (Panel A)
F458	Subject-company SCT (A0434) share of base services	2024	0.0084 share	DERIVED	A	S316	Panel B live shares; SCT services / base services	MMT_Medicare_Book	RUNS BELOW the 6.6% national SCT share (finding 6 lineage); the Medicare-visible book is the scheduled BLS interfacility product; high-acuity dedicated-unit work under facility contracts never appears in claims (Facility_Pay_Layer)
F459	Subject-company mileage units per base transport (implied avg loaded miles)	2024	14.6 miles (floor/floor)	DERIVED	A	S316	Panel C, A0425 units / base services	MMT_Medicare_Book	Interfacility books run long miles; compare national A0425-per-base on MUP_Ambulance_National
F460	Subject-company base-service CAGR 2019 to 2024 (Medicare FFS floor)	2024	0.1836 CAGR	DERIVED	A	S316	Panel D live formula over Panel A	MMT_Medicare_Book	Read against FFS enrollment decline (Utilization_Normalized); per-beneficiary growth is stronger than nominal
F461	Estate share of NE Medicare FFS ground base services (deepest footprint state)	2024	0.6701 share of state base services	DERIVED	A	S317	Panel B spotlight, 2024 row; estate NE services / NE state total, base codes	Market_Share_Panels	Panel A NE block: the top NE row is an estate NPI and its services match MMT_Medicare_Book Panel E
F462	Estate rank among NE Medicare ambulance billers	2024	1 rank	DERIVED	A	S317	Panel B spotlight, 2024 row; estate combined vs per-NPI billers	Market_Share_Panels	Rank 1 in every manifested vintage back to 2013 (same column)
F463	Ambulance biller HHI, NE (NPI grain)	2024	4,572 HHI points (0-10000)	DERIVED	A	S317	Panel C NE row; squared percent shares over all NE billing NPIs, base codes	Market_Share_Panels	Above the 2500 DOJ/FTC highly-concentrated line; NPI grain understates - parent roll-up would raise it
F464	Estate measured base services in the nine non-NE footprint states (registration-state basis)	2024	0 services	GOV	A	S317	Panel B estate-services matrix, 2024 row, IA-KY columns	Market_Share_Panels	Zero in every vintage: MUP assigns volume to the NPI registration state, so border-market work books to NE; not evidence of no operations
F465	Ambulance biller HHI, IA (second footprint state, NPI grain)	2024	292 HHI points (0-10000)	DERIVED	A	S317	Panel C IA row; squared percent shares over all IA billing NPIs	Market_Share_Panels	Unconcentrated (under 1500): the estate-visible deep state (NE) is the outlier, not the rule
F466	Distinct US Medicare ambulance billing NPIs, ground base codes	2024	8,721 NPIs	GOV	A	S318	Panel A, 2024 row; distinct Rndrng_NPI over base codes	Fragmentation_National	Panel B decile counts sum to it (live SUM row); overstates firms - NPI grain
F467	National Medicare FFS ground base services (floor)	2024	9,542,103 services	GOV	A	S318	Panel A, 2024 row	Fragmentation_National	Down about 36% from 2013 on the same basis; compare MUP_Ambulance_National
F468	Top-100 NPI share of national base services (NPI grain)	2024	0.2457 share	DERIVED	A	S318	Panel A, 2024 row, top-100 share (live over blue totals)	Fragmentation_National	Top-10 at the same grain holds 6.2%; parent roll-up raises only the top-10 view materially
F469	Top-10 PARENT share of national base services (name-pattern roll-up, printed layer)	2024	0.1181 share	DERIVED	A	S318	Panel C concentration table, 2024 row; pattern table printed above it	Fragmentation_National	8.6% in 2013 by the same patterns; a floor - brands under unrelated names stay unrolled
F470	Mean annual exit rate, smallest biller decile (one-year transitions only)	2024	0.226 share of decile exiting per year	DERIVED	A	S318	Panel D, D1 column, mean over one-year transitions	Fragmentation_National	Largest decile mean 1.4%: the exit gradient is monotone in size; tail exits include suppression dips (upper bound)
F471	Hospital-billed share of Medicare FFS ground base services, FLOOR bound (H1 name AND H2 Care Compare linkage)	2024	0.0144 share of base services	DERIVED	B	S319, S320	Insourcing_Bounds Panel B 2024 row, live formula F26 over blue service totals	Insourcing_Bounds	Floor set is 85 of 8,721 billers; Panel A prints every rule count; suppression and the current-vintage hospital roster both push this DOWN
F472	Hospital-billed share of Medicare FFS ground base services, CEILING bound (H1 OR H2)	2024	0.2697 share of base services	DERIVED	B	S319, S320	Insourcing_Bounds Panel B 2024 row, live formula G26	Insourcing_Bounds	Place-name collisions (county/city agencies vs same-named hospitals) inflate this UP; billing-in-lieu also inflates it
F473	Change in hospital-billed FLOOR share of ground base services, 2013 to 2024	2024	0.0021 share points	DERIVED	B	S319, S320	Insourcing_Bounds Panel B change row F27	Insourcing_Bounds	Direction: share ROSE while absolute floor volume FELL (about 184K to 138K services); the all-biller denominator shrank 36%, so the rise is denominator shrinkage
F474	SCT+ALS2 share of base services, hospital-affiliated FLOOR-set billers	2024	0.0228 share of base services	DERIVED	B	S319, S320	Insourcing_Bounds Panel E row 53, live formula col D	Insourcing_Bounds	Versus explicitly independent billers on the row two below; carrier-claims window only, so Part A bundled SCT is invisible to the test
F475	SCT+ALS2 share of base services, explicitly independent (H3) billers	2024	0.0152 share of base services	DERIVED	B	S319, S320	Insourcing_Bounds Panel E row 55, live formula col D	Insourcing_Bounds	The comparator for the acuity test; H3 names (AMBULANCE/EMS/MEDIC-/RESCUE/FIRE, no H1 term) cover 4,206 of 8,721 billers
F476	Medicare FFS ground base services, all billers (the classification denominator)	2024	9,542,103 services	GOV	A	S319	Sum of Tot_Srvcs over every MUP 2024 provider-grain row, codes A0426-A0429/A0433/A0434; blue cell E26 on Panel B	Insourcing_Bounds	Suppression makes it a floor; compare the national series tabs built from the same registry
F477	Hospitals reporting an ambulance cost center (Worksheet A line 95), latest complete FY	2023	865 hospitals	GOV	A	S321	Panel A, FY end 2023 row; NMRC A000000/09500 nonzero cost, RPT-joined	HCRIS_Ambulance_CostCenters	Panel B state counts sum to it (live All-states row); share of filers prints beside it
F478	Share of all 2552-10 filers reporting ambulance cost	2023	0.1433 share	DERIVED	A	S321	Panel A live formula, FY2023 row (reporters / filers)	HCRIS_Ambulance_CostCenters	Stable 14.3-14.6% across all four complete FYs 2020-2023 in the same panel
F479	As-filed ambulance cost, all reporting hospitals	2023	3,183,473,759 USD	GOV	A	S321	Panel A, FY2023 row, column 3 totals as filed	HCRIS_Ambulance_CostCenters	Panel B state dollars sum to it (live); gross Worksheet A cost before reclasses/adjustments
F480	Net flow of ambulance cost-center reporters, latest complete pair (starts minus stops, 2022 to 2023)	2023	7 hospitals (net entry)	DERIVED	A	S321	Panel D live net formula, 2022 -> 2023 row (47 starts vs 40 stops)	HCRIS_Ambulance_CostCenters	Direction REVERSED from 2021 -> 2022 (net -17); candidate events, filing-lag caveat in the same row
F481	Footprint-state hospitals reporting ambulance cost (NE IA KS MO OH WI VA MN IN KY)	2023	259 hospitals	DERIVED	A	S321, S322	Panel B footprint subtotal (live SUMIF over the FOOTPRINT flag column)	HCRIS_Ambulance_CostCenters	Iowa (57) and Minnesota (38) sit 2nd and 3rd nationally by count; about 30% of all reporters sit in the ten footprint states
F482	Research-cohort corridor cases, nine systems, top-15 origin ZIPs (Medicare FFS inpatient floor)	2025	320,143 cases	GOV	A	S323, S324, S325	HSA 2025 top-15 extract, CCNs per Panel A printed rules; Panel B total row	Cohort_Corridors	Sum of nine live system rows; per-CCN rows re-derivable from HSA_Corridors
F483	Largest cohort system by corridor volume (Ascension)	2025	149,493 cases	GOV	A	S323, S324, S325	Panel B first data row (systems sorted by cases)	Cohort_Corridors	68 hospitals in the corridor file under the printed rule
F484	Cohort corridor volume COVERED: originating in ZIPs with a registered subject-company estate location	2025	0.0102 share of cohort corridor cases	DERIVED	A	S323, S326	Panel C total row, covered share (live)	Cohort_Corridors	Registration presence only; understates operating coverage by construction
F485	Cohort corridor volume CONTESTED: participant ZIP present, no estate ZIP	2025	0.0256 share of cohort corridor cases	DERIVED	A	S323, S326	Panel C total row, contested share (live)	Cohort_Corridors	400 participant ZIPs from 38 resolved crosswalk queries
F486	Cohort corridor volume OPEN: no estate or participant registration in the origin ZIP	2025	0.9642 share of cohort corridor cases	DERIVED	A	S323, S326	Panel C total row, open share (live)	Cohort_Corridors	Covered + contested + open sum to 1 in the live row
F487	Hub hospitals nationally (top-decile top-15 corridor volume AND breadth >= 8 of 15)	2025	592 hospitals	DERIVED	A	S327	Panel B HUB row; thresholds printed in Panel A (p90 = 4,694.9 cases, median = 206)	Hub_Spoke_Map	746 hospitals clear the volume bar alone; the breadth bar removes the narrow-shed fifth of them
F488	Hub hospitals in the 11 footprint states (NE OH WI IA KS IN KY MO MN VA WY)	2025	99 hospitals	DERIVED	A	S327, S328, S329	Panel C footprint block, live total row	Hub_Spoke_Map	State attribution from rosters with SSA CCN-prefix fallback for 1,668 unrostered CCNs
F489	Share of national top-15 corridor volume carried by hub hospitals	2025	0.4325 share of corridor cases	DERIVED	A	S327	Panel B HUB row, live share cell	Hub_Spoke_Map	Hubs are 7.9% of hospitals: roughly a five-to-one concentration ratio

F490	Spoke hospitals nationally (at or below median volume, breadth <= 3 of 15)	2025	3,151 hospitals	DERIVED	A	S327	Panel B SPOKE row	Hub_Spoke_Map	Spokes carry 1.3% of national corridor volume despite being 42% of hospitals
F491	Hub CCNs belonging to research-cohort systems	2025	20 hospitals	DERIVED	A	S327, S328, S329	Panel D cross-tag row; rules printed on Cohort_Corridors Panel A	Hub_Spoke_Map	Name-match confounds carried from Cohort_Corridors apply here unchanged
F492	Public contract documents abstracted on the 6.4 field schema (E.2 corpus)	2026	12 documents	GOV	A	S330	Contract_Corpus Panel A, one row per document	Contract_Corpus	Convenience sample of PUBLIC contracts; 11 further items parked with blockers on Panel D; private facility-to-operator agreements are not public records
F493	USAspending PSC V225 award records pulled (FY2023-FY2025 activity window, top by amount)	2025	300 award records	GOV	A	S331	spending_by_award, psc V225, types A-D, 3 pages x 100, sorted by award amount	Contract_Corpus	261 of 300 are Department of Veterans Affairs; top 15 flagged for PHID document lookup, blocked at SAM.gov (Panel D row 1)
F494	Fixed annual escalator observed in the corpus, low end (five-town NH Pridestar agreement, renewal-term built-in)	2024	0.025 share per year	GOV	A	S334	Agreement renewal clause: USD 650,000 to USD 666,250 to USD 682,906.25 as printed	Contract_Corpus	High end 3.0% (Charles City IA); range is live MIN/MAX on Panel C over n=2 documents
F495	Fixed annual escalator observed in the corpus, high end (Charles City IA / Floyd County AMR agreement)	2023	0.03 share per year	GOV	A	S333	Agency Contribution schedule: USD 415,000 to USD 427,450 to USD 440,273	Contract_Corpus	Low end 2.5% (NH Pridestar renewal); n=2, an observed range, not a market distribution
F496	Dedicated-unit annual subsidy, five-town NH system (two dedicated ALS ambulances around the clock, Pridestar EMS)	2024	650,000 USD per year	GOV	A	S334	Subsidy paragraph: USD 32,500 per town per quarter, five towns	Contract_Corpus	USD 130,000 per town per year; operator bills patients for EMS in addition
F497	Dedicated-unit annual subsidy, Charles City IA / Floyd County (one 24-hour ALS unit plus one 24-hour on-call BLS unit, AMR), Year 1	2023	415,000 USD per year	GOV	A	S333	Agency Contribution: Year 1 USD 415,000, split 50/50 city and county	Contract_Corpus	Rises to USD 440,273 by Year 3; Floyd County Medical Center pledged USD 100,000 toward year one - a facility co-paying for ambulance capacity in a public document
F498	Per-trip rate observed, Nebraska DHHS client transport, Douglas and Sarpy counties (Camelot Transportation contract 104886-O4)	2025	39.02 USD per one-way trip (or USD 2.00 per mile, whichever is greater)	GOV	A	S332	Rate attachment Revised July 1, 2025; wholly within Douglas and Sarpy counties	Contract_Corpus	SCOPE: a state NON-AMBULANCE client-transport (NEMT-adjacent) rate under a child-welfare / client transportation contract - NOT an ambulance rate; recorded exactly as the document states, scope note in-row
F499	Cohort systems resolved to public Form 990 filers	2026	9 health systems (research cohort, public filings)	GOV	B	S336	Cohort_990_Contractors Panel C EIN table	Cohort_990_Contractors	24 EINs resolved; 23 with parsed XML; Cleveland Clinic East Region EIN parked with no e-file XML posted; mapping is analyst-selected from public registry facts (tier B)
F500	Form 990 filings parsed for Part VII Section B contractors	2024	46 filings (latest two XML-available per EIN, tax years 2022-2024)	GOV	A	S335	Per-filing object_id in the register; per-system counts on Panel A	Cohort_990_Contractors	11 newest-season filings XML-blocked (parked in-table); 6 parsed filings disclose no Section B contractors at all
F501	Transport-matching contractors among the disclosed five-highest-paid contractors, all parsed cohort filings	2024	0 contractors (zero; stated as the information floor, not a bound)	GOV	A	S335	Panel A column G (live SUM in the total row); keyword regex in the Panel B note	Cohort_990_Contractors	Part VII Section B discloses only the top five contractors over \$100K; with up to 840 contractors over \$100K per filer, absence from the top five is NOT evidence of absence of spend
F502	Research-cohort aggregate staffed beds (HCRIS latest filed FY, printed-rule matched CCNs)	2022	22,582 beds	GOV	A	S337, S338	System_Research_Cohort Panel A total row (live sum of nine system rows); HCRIS beds field, latest FY per CCN	System_Research_Cohort	FY2020-FY2022 filings; name-match confounds printed per panel
F503	Research-cohort aggregate inpatient days (HCRIS latest filed FY, matched CCNs)	2022	5,322,510 patient days	GOV	A	S337, S338	System_Research_Cohort Panel A total row; HCRIS total_patient_days, latest FY per CCN	System_Research_Cohort	Admissions are PENDING (not in the vendored extract; HCRIS S-3 Part I fills it)
F504	Largest cohort system by measured corridor demand (Ascension)	2025	149,493 top-15 corridor cases	GOV	A	S339, S337, S338	System_Research_Cohort Panel A first data row, col E (green-linked to Cohort_Corridors Panel B where built)	System_Research_Cohort	Matches the Cohort_Corridors roll-up by construction (same printed rules, same cache)
F505	Cohort Form 990 filings parsed with ZERO ambulance / EMS / medical-transport contractors among the disclosed five highest-paid contractors	2024	46 filings	PUBLIC-WEB	A	S340	System_Research_Cohort Panel A 990 row (B19); 24 EINs, tax years 2022-2024, keyword screen printed in cohort notes	System_Research_Cohort	Part VII Section B discloses only the top five over \$100K: an information floor, not evidence of no vendor
F506	Subject-company registered operating states (printed rule: at least one NPPES estate practice location)	2026	11 states	GOV	A	S342	Footprint_Determination Panel E row 41 (states: IA NE OH SD CO IN MO NC RI VA WI)	Footprint_Determination	Two states (IA, NE) independently corroborated by named V225 federal awards
F507	Provisional footprint states carried by earlier tabs WITHOUT a registered estate location (KS MN KY)	2026	3 states	DERIVED	A	S342	Footprint_Determination Panel E row 44; reconciliation prose names the tabs	Footprint_Determination	Mirror gap: 4 registered states (CO NC RI SD) were absent from the provisional list; the live site publishes no state-grain claim, so claimed-not-registered states observable = 0
F508	Top screened name family by measured transfer demand (Mercy)	2025	80,732 top-15 corridor cases	DERIVED	A	S339, S337	Prospect_Landscape Panel B first data row (D15)	Prospect_Landscape	Family-caveat flag applies: a shared-name family can span unrelated organizations
F509	Name families with 3+ hospitals screened in the registered footprint states (cohort excluded)	2025	59 name families	DERIVED	A	S337, S339	Prospect_Landscape Panel B count row (B35); grouping rule printed in Panel A	Prospect_Landscape	Short-term acute + CAH CCN ranges only; HCRIS FY2020-FY2022 name vintage
F510	Thinnest-supply large county in the footprint: Christian County, MO - 65+ residents per Medicare-billing ambulance provider (zero providers based in-county, ratio floor = 65+)	2024	16,797 65+ residents per billing provider (floor)	DERIVED	A	S347, S348	Panel L rank 1; census 65+ 7/1/2024 x MS_County_2025 providers	County_Whitespace_Screens	MS_County_2025 shows 105 users in the county, proving out-of-county service, not zero demand
F511	Footprint counties with zero Medicare-billing ambulance providers and 5,000+ residents 65+	2024	5 counties	DERIVED	A	S347, S348	Panel M row 2; names printed in the same row	County_Whitespace_Screens	Each of the named counties shows positive Users in MS_County_2025 - served from out of county
F512	Top-25 footprint whitespace counties with 0-1 dialysis facilities (recurring-transport overlay)	2025	13 counties	DERIVED	A	S351, S348, S347	Panel L overlay COUNTIF row; per-county counts in column O	County_Whitespace_Screens	Dialysis_Registry carries the same registry at state/ZIP grain; county attribution is name-matched (0 can be a match miss)
F513	Largest whitespace county by 65+ population (the metro consolidation read): Hennepin County, MN	2024	14,096 65+ residents per billing provider	DERIVED	A	S347, S348	Panel L; 211,444 65+ over 15 billing orgs	County_Whitespace_Screens	Read as consolidation, not absence: users per provider prints in the same row
F514	PRICE component of the change in national ground base-code allowed dollars, 2019 to 2024 (Laspeyres, mix held constant)	2024	904,503,315 USD (signed)	DERIVED	A	S352	Panel C, window 2019 to 2024, PRICE column	Growth_Decomposition	Panel E code rows: price effect positive in all six codes; additivity check cell = 0
F515	VOLUME component of the change in national ground base-code allowed dollars, 2019 to 2024 (services change at base-year average price)	2024	-1,138,151,885 USD (signed)	DERIVED	A	S352	Panel C, window 2019 to 2024, VOLUME column	Growth_Decomposition	Base services fell from about 13.0M to 9.6M (Panel B); FFS enrollment decline is the deflator caveat (Utilization_Normalized)
F516	MIX component (residual) of the change in national ground base-code allowed dollars, 2019 to 2024	2024	-107,082,292 USD (signed)	DERIVED	A	S352	Panel C, window 2019 to 2024, MIX column	Growth_Decomposition	A0428, the lowest-priced transport code, lost the most services; the residual nets code-shift against price-volume interaction, and the live check column proves additivity
F517	Medicare ambulance dollars denied, all seven PSPS codes (valuation: denied services x code-year average allowed per allowed service)	2024	553,269,323 USD	DERIVED	A	S354	Panel B, 2024 row, all-codes column; live SUMPRODUCT over PSPS_Denial_Series rows 6-110	Denial_Economics	Panel C code rows sum to it; denominator (submitted minus denied) positive in every code-year, verified at build
F518	A0428 BLS non-emergency dollars denied (same valuation)	2024	115,666,690 USD	DERIVED	A	S354	Panel B, 2024 row, A0428 column, live against PSPS_Denial_Series row 107	Denial_Economics	A0428 denial rate 12.9% in 2024 vs 11.6% in 2010: rate UP while dollars fell - the volume-collapse divergence printed in Panel A
F519	Fifteen-year trend in total ambulance denied dollars (2010 to 2024, nominal change)	2024	0.0005 share (signed)	DERIVED	A	S354	Panel B column C, 2010 vs 2024 rows	Denial_Economics	Direction: flat nominal (declining real), peak 2012, trough 2020; A0428 denied dollars down roughly 40% over the same window while its rate rose
F520	Footprint median hospital OP-18b (median ED minutes before departure, discharged patients)	2025	134 minutes	DERIVED	A	S356	ED_Timeliness_Registry OP_18b rows, footprint states, numeric scores; period 07/2024 - 06/30/2025; median of Panel B P90 row; deciles over 1053 numeric footprint hospital medians	Transfer_Delay_Burden	Panel B live column re-counts hospitals at or above each decile against the registry
F521	Footprint worst-decile OP-18b threshold (P90)	2025	198 minutes	DERIVED	A	S356	Panel B P90 row; deciles over 1053 numeric footprint hospital medians	Transfer_Delay_Burden	Panel C names the 20 hospitals above it, values green-linked to registry rows
F522	Research-cohort hospitals in the worst footprint OP-18b quartile	2025	23 hospitals (of 65 measured)	DERIVED	A	S356	Panel D; rules mirror Cohort_Corridors Panel A (footprint states only on this tab); worst quartile = OP-18b at or above 144.1	Transfer_Delay_Burden	Neutral placement read: cohort median 150 vs footprint 134 minutes; expected for referral-heavy systems
F523	Widest footprint paramedic-over-EMT median wage premium	2024	1.488 ratio (Minnesota)	DERIVED	A	S357	Panel A live MAX over green-linked OEWS medians; Minnesota paramedic 68,000 vs EMT 45,690	Workforce_Depth	National extreme (WA, about 2.06x) noted in-row; footprint premiums span 1.26x-1.49x
F524	Footprint private-ambulance average annual pay vs national	2025	-0.0316 share gap (footprint below national)	DERIVED	A	S358	Panel B row 2025: live SUMPRODUCT footprint pay (about 59,733) vs QCEW_EMS_Employment Panel B national	Workforce_Depth	Gap narrows from -4.9% (2019) to -3.2% (2025); footprint converging on national pay
F525	Quarterly wage-vs-AIF scissors, 2019Q4 to 2025Q4	2025	22.7 index points (2019Q4=100)	DERIVED	A	S358, S361	Panel D read 2, last row: wage index 144.1 minus AIF-compounded path 121.3	Workforce_Depth	Annual version is finding 45 (QCEW_EMS_Employment Panel B): +5.0%/yr pay vs +3.1%/yr AIF over 2014-2025
F526	Tightest footprint EMS labor market (two-test rule: top-2 wage growth AND bottom-2 depth)	2025	0.701 wage growth 2019-2025 (Kansas; depth 1.45 per 6 non-EMS)	DERIVED	B	S358, S360	Panel E ranks; rule printed in the header and banner - a matched pair of rankings, not a composite index	Workforce_Depth	Iowa prints thinner depth but its employment floor is suppression-driven; Kansas tops wage growth outright
F527	Ambulance-tagged LEIE exclusions in footprint states (cumulative)	2025	126 exclusions (of 566 national)	GOV	A	S363	LEIE_Ambulance_Exclusions col G in (NE IA KS MO OH WI VA MN IN KY); live COUNTIF panel D	LEIE_Read_Panel	KY 36 + OH 43 carry most of the footprint total; dates span 1985-2025

F528	Business-entity share of ambulance-tagged LEIE exclusions (BUSNAME heuristic)	2025	0.2102 share	DERIVED	A	S363	Panel C live COUNTIF on business-name column C (119 of 566)	LEIE_Read_Panel	Four in five excluded parties are individuals - the list is mostly people, not companies
F529	Total daily cost per boarded acute stroke patient, med/surg boarding (TDABC)	2024	1,856 USD per patient per day	ACADEMIC	A	S364	Ann Emerg Med 2024;84(4):376-385, abstract Results; PMID 38795079	Throughput_Economics_Public	Med/surg inpatient comparator \$993 and ICU pair \$2,267 vs \$2,165 in the same abstract; per-hour normalization exists only as live formulas in Panel D
F530	Median door-in-door-out time at the referring hospital, stroke interhospital transfer	2023	174 minutes	ACADEMIC	A	S365	JAMA 2023;330(7):636-649, abstract Results; PMID 37581671	Throughput_Economics_Public	IQR 116 to 276; only 27.3% of 108,913 transfers from 1,925 hospitals met the recommended 120 minutes
F531	Forgone net revenue from ED capacity consumed by boarding, one 450-bed community hospital	2007	3,960,264 USD per year	ACADEMIC	A	S366	Acad Emerg Med 2007;14(4):332-337, abstract Results; PMID 17331916	Throughput_Economics_Public	10,397 boarding bed-hours and 3,175 forgone encounters in the same abstract; about \$381 per bed-hour derived live on Panel D
F532	Increased risk of ICU death per hour of waiting for ICU admission	2011	0.015 risk increase per hour (HR 1.015)	ACADEMIC	A	S367	Crit Care 2011;15(1):R28, abstract Results; PMID 21244671	Throughput_Economics_Public	95% CI 1.006 to 1.023; attributable fraction of mortality risk 30% (CI 11.2 to 44.8)
F533	Public records establishing EHR-integrated patient transport ordering (existence only)	2026	6 PUBLIC-WEB records (4 distinct platform x EHR records)	PUBLIC-WEB	A	S368, S369	Panel C rows on Throughput_Economics_Public; count is a live COUNTA on the tab; each row carries its URL and access	Throughput_Economics_Public	Covers both dominant US EHRs (Epic, Oracle Cerner) and extends to ambulance-level interfacility modes (Roundtrip 2025, VectorCare); adoption existence only, no performance claims
F534	RSNAT prior authorization, estimated potential savings, 3 initial demonstration states (NJ, PA, SC)	2017	349,500,000 USD (nominal, implementation through March 2017)	GOV	A	S370	GAO-18-341, Table 2, p.16	GAO_OIG_Shelf	Plus \$38.0M in 6 expansion states on the same table; provisional affirmation rate rose from 28% to 66% (p.12); the model was later made national by statute
F535	Growth in Medicare Part B dialysis-related ambulance transports, 2002 to 2011	2011	2.69 growth over period (+269 percent)	GOV	A	S371	OEI-09-12-00350, Summary p.1 and dialysis finding section	GAO_OIG_Shelf	From 753,741 transports in 2002; fastest of any origin-destination category; all Part B transports +69% vs beneficiaries +7% over the same window
F536	Medicare Part B ambulance transport payments, 2012	2012	5,800,000,000 USD (nominal 2012)	GOV	A	S372	OEI-09-12-00351, Executive Summary (Why We Did This Study)	GAO_OIG_Shelf	Almost double 2003; H1-2012 review found \$24M not meeting requirements plus \$30M with no Medicare services anywhere, and about 1 in 5 suppliers with questionable billing
F537	Dialysis-related ambulance claim allowances failing Medicare medical-necessity coverage review	1994	0.7 share of reviewed dialysis-related transports	GOV	A	S373	OEI-03-90-02130, Findings p.4 of PDF	GAO_OIG_Shelf	1991 ESRD ambulance allowances were \$101M with 75% concentrated in fewer than 2% (2,573) of beneficiaries - the earliest marker of the dialysis integrity arc
F538	Rural Emergency Hospitals enrolled (national)	2026	48 facilities	GOV	A	S374	CMS REH enrollment roster (Pull_Manifest)	REH_Closure_Flow	Every REH admission-grade patient is a transfer by statute
F539	Rural closures and conversions since 2010, footprint states	2026	19 events	GOV	A	S181	Sheps closure list, state field in footprint set	REH_Closure_Flow	The Sheps-vs-Charlis counting convention caveat from Methodology section 4 applies
F540	Nebraska Medicaid FFS ambulance fee schedule: A0428 base rate (1 Jul 2026)	2026	154.89 USD per transport	GOV	A	S375	471-000-504.xlsx row 000A0428, MEDICAID ALLOWABLE	Medicaid_Rate_Card	Schedule cover states no increase for SFY2027
F541	Iowa Medicaid FFS ambulance fee schedule: A0428 base rate (eff 1 Jul 2014, current 11 Jul 2026)	2026	84.67 USD per transport	GOV	A	S376	C11.csv row A0428, Factor column	Medicaid_Rate_Card	Lowest true-BLS A0428 in footprint; unchanged 12 years
F542	Kansas Medicaid FFS ambulance fee schedule: A0428 base rate (eff 1 Jul 2023, posted 7/1/2026)	2026	255.7 USD per transport	GOV	A	S377	FeeSchedule_20260701_TXIX_DEF.txt row A0428 MAX FEE	Medicaid_Rate_Card	SFY2024 rebase
F543	Missouri Medicaid FFS ambulance fee schedule: A0428 base rate (eff 1 Jul 2019, file 7 Jul 2026)	2026	105.62 USD per transport	GOV	A	S378	Ambulance.xlsx rows A0428 HH / A0428 HD	Medicaid_Rate_Card	Payable ONLY in HH/HD contexts; base code noncovered
F544	Ohio Medicaid FFS ambulance fee schedule: A0428 base rate (1 Jan 2024)	2026	203.75 USD per transport	GOV	A	S379	OAC 5160-15-28 appendix, row A0428	Medicaid_Rate_Card	2024 rebase; GEMT supplemental tops up public providers
F545	Wisconsin Medicaid FFS ambulance fee schedule: A0428 base rate (eff 1 Jul 2008, report 11 Jul 2026)	2026	94.9 USD per transport	GOV	A	S380	maxfee06_ambulance.pdf row A0428 MAX FEE	Medicaid_Rate_Card	Unchanged 18 years; manager-coordinated NEMT paid off-schedule
F546	Minnesota Medicaid FFS ambulance fee schedule: A0428 base rate (eff 1 Jan 2026)	2026	284.56 USD per transport	GOV	A	S381	MHCP Fee Schedule.xlsx row A0428, Max Fee column	Medicaid_Rate_Card	Highest A0428; exactly the CY2026 Medicare CF (parity)
F547	Indiana Medicaid FFS ambulance fee schedule: A0428 base rate (eff 1 Jan 2026)	2026	269.02 USD per transport	GOV	A	S382	IHCP Professional Fee Schedule.xlsx row A0428, Rate Eff 01/01/2026	Medicaid_Rate_Card	100% of CY2025 Medicare (BT2025156); A0434 SCT noncovered
F548	Kentucky Medicaid FFS ambulance fee schedule: A0428 base rate (eff 1 Jan 2026 (rev 1/9/2026))	2026	55 USD per transport	GOV	A	S383	2026 Transportation Rates PDF, PT56 Specialty 16 A0428	Medicaid_Rate_Card	Non-emergency STRETCHER-van line (formerly T2005), not a full BLS ambulance
F549	Lowest footprint Medicaid true-BLS A0428 rate as a multiple of the CY2026 Medicare national base (Iowa)	2026	0.2975 x Medicare	DERIVED	A	S376	Medicaid_Rate_Card Panel, MIN over the true-BLS A0428 ratios; 84.67 / 284.56 (KY stretcher excluded)	Medicaid_Rate_Card	Vintage confound printed in-row: IA rate is a 2014 rate against a 2026 Medicare base
F550	Highest footprint Medicaid A0428 rate as a multiple of the CY2026 Medicare national base (Minnesota)	2026	1 x Medicare	DERIVED	A	S381	Medicaid_Rate_Card Panel, MAX over the true-BLS A0428 ratios; 284.56 / 284.56 = exact parity	Medicaid_Rate_Card	MN A0428 equals the CY2026 Medicare conversion factor to the cent
F551	Footprint states routing scheduled NEMT through a broker, transportation manager, or state-administered system	2026	8 of 9 states	DERIVED	A	S375, S376, S377, S378, S379, S380, S381, S382, S383	Medicaid_Rate_Card carve panel, COUNTIF over the carve column (OH is the only exception: county NET)	Medicaid_Rate_Card	VA (extension) goes further: the broker pays FFS non-emergency ambulance itself
F552	RSNAT baseline utilization, Year 1 cohort (share of ESRD and/or pressure-ulcer beneficiaries using RSNAT, baseline 2012-2014)	2014	0.101 share of study population	GOV	A	S389	Final report, Table II.2, p.9	RSNAT_Series	Year 2 cohort 4.7%; comparison states 4.9% on the same table: the model states were picked for high baseline use (FAQ Q.10)
F553	GAO estimated potential RSNAT savings, 3 initial states, implementation through Mar 2017	2017	349,500,000 USD	GOV	A	S370	GAO-18-341, Table 2, p.16	RSNAT_Series	Plus \$38.0M in the 6 expansion states, same table; "as high as" counterfactual estimates
F554	RSNAT provisional affirmation rate, initial PAR submissions	2017	0.66 share of initial PARs affirmed, Oct 2016-Mar 2017	GOV	A	S370	GAO-18-341, p.12	RSNAT_Series	Up from 28% in the first six months (Dec 2014-May 2015): a documentation learning curve, not a loosening of the screen
F555	RSNAT use reduction, final evaluation (nine model states, ESRD and/or pressure-ulcer population)	2021	-0.72 change vs comparison	GOV	A	S389	Final report, Table ES.1 and Ch. III	RSNAT_Series	RSNAT expenditures -\$746M; total Medicare FFS -2.4%; no adverse quality or access findings
F556	RSNAT suppliers per 100,000 beneficiaries, change on implementation (model states)	2020	-0.5 change	GOV	A	S388	Second interim report, Objective 3 (exec summary)	RSNAT_Series	Concentrated almost entirely in Year 1 states; exiting suppliers were smaller and dialysis-dependent - the supply-side echo of finding 26 (dialysis transports -63%)
F557	MA-to-FFS ambulance revenue ratio per NPI (the MA book calibrator)	2025	1.038 x (ratio of means per NPI)	DERIVED	A	S313	GADCS Y1-4 appendix, Table 2.30, pp.64-65: \$1,238,482 / \$1,193,046; live formula on MA_Book_Calibrator Panel A	MA_Book_Calibrator	Published cuts run 0.55x (super rural) to 1.19x (government); for-profit 1.08x; superseded the day MA encounter data (MA_Encounter_Rcv) lands
F558	MA dark share this calibrator prices (beneficiaries outside A and B FFS)	2024	0.593 share of Medicare beneficiaries	DERIVED	A	S313	Imbalance_Ledger Panel C 2024 row (live link on MA_Book_Calibrator Panel C); enrollment inputs S60	MA_Book_Calibrator	Engagement_Data_Map quotes the same 59.3%; CBO projects 63% by 2034
F559	Emergency-capable acute hospitals, footprint states (receiving-capacity floor)	2026	575 hospitals	GOV	A	S391	Hospital General Information, acute type with emergency_services = Yes, footprint states	Receiving_Center_Registry	A floor: designated trauma/stroke/STEMI centers are a subset (PENDING Panel C)
F560	Emergency-capable acute hospitals, national	2026	2,981 hospitals	GOV	A	S391	Hospital General Information, acute type with emergency_services = Yes	Receiving_Center_Registry	Reconciles to the Hosp_Registry universe (same PDC source family)
F561	Footprint states with a statutory entry gate beyond licensure (CON, need test, endorsement gate, franchise power, department-set PSA)	2026	5 states of 10 (KY, MN, MO, VA, WI)	GOV	A	S392, S393	Entry_Barrier_Register Panels A-B; live count Panel D	Entry_Barrier_Register	Each mechanism verified against fetched statute text; VA gate is at local OPTION and KS carries a flagged non-duplication rule short of a gate
F562	Footprint licensure register completeness (agency + statute cite verified per state)	2026	1 share of 10 states	GOV	A	S392	Entry_Barrier_Register Panel C (10 rows, each with fetched URL)	Entry_Barrier_Register	Two rows carry flagged ambiguities elsewhere (KS county power, IN ECO absence), none on licensure
F563	States with ground-ambulance balance-billing protections	2026	22 states (some protection; includes partial regimes)	SOURCED	A	S394	Commonwealth Fund / CHIR, 13 Mar 2026 ("people in 22 states now have some protection"); register rows on Balance_Billing_Chart	Balance_Billing_States	This register prints all 22 with 16 verified against fetched statute text and 6 marked partial/ambiguous (DE, FL, MD, NV, WV, VT-flag); 13 of 22 also cover non-emergency transport per the tracker
F564	Highest state Medicare peg for out-of-network ground ambulance (Indiana, footprint state)	2024	4 x Medicare (fallback cap absent a local rate; lesser rate than billed charges)	GOV	A	S395	IC 27-1-2.3 as amended by HEA 1385-2024, eff. 1 Jul 2024	Balance_Billing_States	Peg range across verified rows 1.5x (NM) to 4.0x (IN); nodal peg 3.25x (CO, LA, TX, NH, OR)
F565	States with V225 ambulance obligations, FY2025	2025	49 states	GOV	A	S397	spending_by_geography, place_of_performance, FY2025	Federal_Ambulance_Contracts	10 footprint states present; subtotal is a live SUMIF on the tab

F566	Distinct top-50 V225 recipients appearing in any of FY2023-2025	2025	77 recipients	GOV	A	S397	spending_by_category/recipient, three fiscal years	Federal_Ambulance_Contracts	Recipient names are as registered in SAM; parent roll-ups not applied (stated)
F567	National distinct Medicare ambulance billing NPIs, ground base codes - net change 2013 to 2024	2024	-597 NPIs	DERIVED	A	S399	Annual_Market_Structure Panel A col B, 2024 row minus 2013 row	Annual_Market_Structure	9,318 (2013) to 8,721 (2024), -6.4%; gross churn of 330-490 entries and exits per year dwarfs the net drift
F568	Share of the 2014 bottom size quartile (by base services) still billing in 2024	2024	0.6505 share	DERIVED	A	S399	Annual_Market_Structure Panel C, Q1 row, 2024 column	Annual_Market_Structure	OVERSTATED exit by construction: the 11-beneficiary suppression floor binds hardest on the smallest quartile, so a floor-crossing reads as an exit
F569	Share of the 2014 top size quartile (by base services) still billing in 2024	2024	0.6867 share	DERIVED	A	S399	Annual_Market_Structure Panel C, Q4 row, 2024 column	Annual_Market_Structure	Below the middle quartiles (81.8% / 81.8%) - the U-shape reads as consolidation retiring large NPIs, not failure
F570	Subject-company estate share of NE Medicare ground base services, 2022-2024 smoothed average	2024	0.6285 share	DERIVED	A	S399	Annual_Market_Structure Panel D smoothed row (live AVERAGE over the annual share column)	Annual_Market_Structure	vs 24.1% smoothed across 2013-2015; the single-year series rises in all eleven year-over-year steps
F571	Top-100 NPI share of national Medicare ground base services	2024	0.2457 share	DERIVED	A	S399	Annual_Market_Structure Panel A, top-100 column, 2024 row	Annual_Market_Structure	22.3% in 2013; essentially flat through 2021, then rising three years running
F572	Widest state SCT-to-BLS realized multiple, 2024 (A0434/A0428 avg allowed, 30+ services both codes): GA	2024	3.81 x	DERIVED	A	S400	Realized_Price_Ladders Panel B, GA row, live multiple column	Realized_Price_Ladders	vs the frozen 3.25x AFS RVU ratio and SD at 3.16x on the narrow end; deviation is geographic mix, not payment policy
F573	National volume-weighted realized allowed per A0428 service, CAGR 2013-2024	2024	0.0166 CAGR	DERIVED	A	S400	Realized_Price_Ladders Panel A, A0428 NATIONAL row, 2013 and 2024 columns	Realized_Price_Ladders	\$218.22 to \$261.47; an administered-price series - compare AFS conversion-factor updates on Payment_Rules
F574	Footprint A0428 realized-price dispersion, 2024: max state over min state	2024	1.173 x	DERIVED	A	S400	Realized_Price_Ladders Panel C, A0428 row, footprint max/min live column	Realized_Price_Ladders	MN \$288.42 over KY \$245.78; national max/min is wider only because AK and territories enter
F575	Registered organizational ambulance NPIs, national (NPPEs 3416 taxonomy)	2026	20,377 NPIs	GOV	A	S401	NPPEs roster, Entity Type 2, 3416* taxonomy	Registered_vs_Billing	More than double the Medicare-billing universe; the gap bounds the non-Medicare layer
F576	Registered-to-billing ratio, national ambulance supply	2026	2.34 ratio	DERIVED	A	S401	National registered NPIs / distinct MUP 2024 ground base billers	Registered_vs_Billing	Bound not headcount: dormant/air/parent-billed NPIs inflate; suppression deflates the billing count
F577	Total mission-demand line over the 20 target metros (HSA catchment cases x measured transfer rate, proxy-based floor)	2025	69,767 transfer episodes per year (floor)	DERIVED	A	S402, S403	Metro_TAM_Panels Panel S, live SUM over 20 mission-demand lines	Metro_TAM_Panels	Medicare-only cases x all-payer rate: a floor by construction; 20-metro HSA cases 735,507.0 x mean rate 0.0949
F578	Target metros with complete Medicare-anchored pricing (catchment proxy + transfer rate + CY2026 base rate all live)	2026	20 metros (of 20)	DERIVED	A	S404, S402	Metro_TAM_Panels Panel S, live COUNT over the 20 priced-total cells	Metro_TAM_Panels	Medicaid/commercial/gross-up adders are PENDING in all 20 panels; only the Medicare anchor is complete
F579	Footprint-10 state Medicare-anchored assembly floor (HSA cases x transfer rate x CY2026 A0428 base)	2025	69,431,382 USD per year (floor)	DERIVED	A	S402, S403, S404	TAM_Assembly_State Panel B FOOTPRINT row, col D (live SUM)	TAM_Assembly_State	2,572,301.0 footprint HSA cases; Medicare-only volume x base-only rate keeps it a floor
F580	National HSA service-area Medicare FFS inpatient case roll-up (all CCNs)	2025	14,144,025 cases	DERIVED	A	S402	TAM_Assembly_State Panel B NATIONAL row, col B (live SUM over HSA_Hospital_Catchment)	TAM_Assembly_State	x the 9.5% rate gives about 1.34M proxy episodes vs about 3.06M measured all-payer HCUP episodes - the Medicare-slice gap the bridge panel prints
F581	Claims-visible x Medicare-anchored grid cell (the hardest floor): 2024 Medicare FFS hospital-to-hospital transports x measured allowed per transport	2024	577,228,322.31 USD (allowed, 2024)	DERIVED	A	S405	Scenario_Matrix Panel B, claims-visible row, Medicare-anchored column (live product)	Scenario_Matrix	Equals Medicare_IFT_Series 2024 total allowed (base + mileage) by construction; 871,753 transports
F582	Live cells in the 3x3 scenario grid (honesty metric)	2026	2 grid cells live (of 9)	DERIVED	A	S405, S406	Scenario_Matrix Panel B honesty-metric row (live COUNT over the nine grid cells)	Scenario_Matrix	7 PENDING cells, each naming the dataset that closes it
F583	US 65+ population CAGR, 2020-2024	2024	0.0295 share per year	DERIVED	A	S407	State_Age_65plus Panel A, United States row, G column (live formula over Census PEP levels)	Growth_Outlook_Shell	Median county 65+ CAGR prints beside it live; 85+ CAGR runs lower on the same row
F584	MA & other share of Medicare beneficiaries, 2024	2024	0.5012 share	DERIVED	A	S408	Enrollment_ESRD_State Panel A, 2024 row, E/C columns	Growth_Outlook_Shell	2013 share 28.3% on the same panel; the series rises every single year
F585	Transport vendors among five highest-paid 990 contractors, cohort systems (count across filings)	2024	0 vendors	SOURCED	A	S410	Form 990 Part VII Section B, latest two XML-available filings per resolved EIN; cohort_990.json headline_result	Vendor_Share_Stack	Floor-censored: disclosure covers only the top five contractors over \$100K per filer, so zero bounds vendor scale, not vendor existence
F586	Form 990 filings parsed for the contractor sweep	2024	46 filings	SOURCED	A	S410	cohort_990.json: 9 systems, 24 EINs, latest two XML filings per EIN	Vendor_Share_Stack	IRS XML availability lags: latest filings are tax years 2022-2023 for most EINs
F587	Longest stated initial term in the public contract corpus	2026	120 months	SOURCED	A	S411	contract_corpus.json public_contracts[] term_and_renewal, parsed; Stickiness_Evidence Panel A prints every parse	Stickiness_Evidence	Shortest stated term 12 months; 8 of 12 documents state a parseable initial term; documents dated 2018-2025
F588	Documents with explicit exclusivity or first-call language in the public contract corpus	2026	3 documents of 12	SOURCED	A	S411	contract_corpus.json public_contracts[] exclusivity_or_first_call; live COUNTIF on Stickiness_Evidence Panel A	Stickiness_Evidence	1 documents are explicitly NONexclusive; corpus skews to 911/government forms where exclusivity is a franchise feature
F589	Footprint nonprofit health systems resolved to public Form 990 filers (beyond the 9-system cohort)	2026	64 health systems (public filings)	GOV	B	S414	Footprint_990_Sweep Panel A	Footprint_990_Sweep	78 attempted, 14 parked (Panel D); mapping is analyst-selected from public registry facts (tier B)
F590	Form 990 filings parsed for Part VII Section B contractors (footprint sweep)	2024	128 filings (latest two XML-available per filer, tax years 2019-2024)	GOV	A	S413	Panel C filings column (live SUM in the total row); per-filing object_id in footprint_990_sweep.json	Footprint_990_Sweep	64 filers; latest two filings each where XML available
F591	Transport-keyword contractors observed in the disclosed five-highest-paid contractors (footprint sweep)	2024	7 filing appearances (4 distinct operators across 4 contractors)	GOV	A	S413	Panel A observed payments; Panel B frequency; Panel C total transport column	Footprint_990_Sweep	Observed amounts \$2,058,817 to \$12,910,581; includes a ground-ambulance and an air-medical operator plus ED-coverage groups (verbatim services shown, Panel A)
F592	Maximum contractors over \$100,000 disclosed at a single filer (the five-slot information floor)	2024	2,090 contractors over \$100K (only five named)	GOV	A	S413	Panel C max column (live MAX in the total row); CntrctRcvdGreaterThan100KCnt as filed	Footprint_990_Sweep	Part VII Section B discloses only the top five over \$100K; with this many undisclosed, absence from the top five is NOT evidence of absence of spend
F593	Footprint-state pooled median risk-standardized SNF potentially preventable 30-day post-discharge readmission rate (SNF QRP S_004_01)	2026	10.59 percent of stays	DERIVED	A	S415	SNF_Return_Leg_Quality Panel A, Footprint pooled median of PPR_PD_RSRR over reporting footprint SNFs	SNF_Return_Leg_Quality	National median 10.66% (Panel A National row); risk-standardized, case-mix adjusted
F594	Footprint SNFs in the worst national decile for potentially preventable rehospitalization (bounce-back transport-demand nodes)	2026	286 SNFs	DERIVED	A	S415	SNF_Return_Leg_Quality Panel A, SUM of per-state worst-decile counts (PPR at/above the national 90th percentile = 9.9%)	SNF_Return_Leg_Quality	Out of 3,997 reporting footprint SNFs
F595	National median risk-standardized SNF potentially preventable readmission rate, for-profit SNFs (return-leg bounce-back)	2026	10.72 percent of stays	DERIVED	A	S415, S416	SNF_ReturnLeg_Structure Panel A, For profit row, national median of PPR_PD_RSRR	SNF_ReturnLeg_Structure	Non-profit median 10.41% (Panel A); for-profit bounces back more
F596	National median risk-standardized SNF potentially preventable readmission rate, 1-star vs 5-star SNFs (quality gradient)	2026	10.74 percent of stays (1-star)	DERIVED	A	S415, S416	SNF_ReturnLeg_Structure Panel C, rating 1 row, national median of PPR_PD_RSRR	SNF_ReturnLeg_Structure	5-star median 10.42%; monotonic gradient - lower-rated SNFs bounce back more
F597	Footprint AHCAH-approved hospital rows on the CMS public roster (4/5/2021 snapshot)	2021	27 hospital rows	GOV	A	S417	Hospital_at_Home_Participants Panel A/B; footprint rows of the 4/5/2021 CMS Approved List	Hospital_at_Home_Participants	Stale by construction; current roster PENDING (Panel C, naming QualityNet + ResDAC); participation has grown materially since but no public machine-readable roster states the current count
F598	Footprint states with at least one AHCAH-approved hospital on the 4/5/2021 CMS public roster	2021	7 of 10 states	GOV	A	S417	Hospital_at_Home_Participants Panel B; states with a non-zero 2021 count (OH, MN, WI, IN, IA, MO, VA)	Hospital_at_Home_Participants	NE, KS, KY had no 2021 approval on the public list

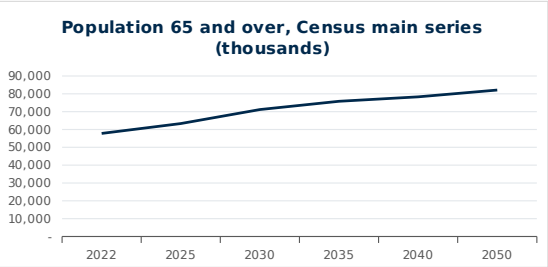
The measured macro drivers. Census, payer mix, supply exits, and the coordination lever.

The question: what moves demand from outside the hospital? Four measured forces, each pinned to a primary document: the aging population (Census projections), the payer-mix migration that shrinks every fee-for-service series, rural hospital closures and their measured effect on transport times, and the final external anchor for how often patients arrive by ambulance.

Panel A. The population that rides. Census Bureau 2023 National Population Projections, Table NP2023-T2, main series (S50). Thousands.

Measure	2022	2025	2030	2035	2040	2050	CAGR 2025-2030	CAGR 2030-2040	
Population 65 and over	57,795	63,327	71,183		75,828	78,294	82,130	2.4%	1.0%
Total resident population	333,288	338,016	345,074		350,861	355,309	360,639		
65+ share of population	17.3%	18.7%	20.6%		21.6%	22.0%	22.8%		

The 65+ population adds 7.9 million people between 2025 and 2030, a 2.4% per year compounding rate, then decelerates to 1.0% per year in the 2030s. NHAMCS puts the 65+ ambulance-arrival share at 33.4% against 17.0% for all ages (Panel E), so the riding population grows roughly twice as fast as total ED demand through 2030. This is the only measured tailwind in the demand equation; the Acute_IFT_Series episode CAGR of +0.86% already contains its past effect.



Panel B. The payer-mix migration that shrinks every fee-for-service series in this workbook. KFF from CMS enrollment files (S51).

Year	MA share of eligible beneficiaries	Enrolled (millions)	Eligible (millions)	Note
2007	19.0%			Series start as published by KFF
2025	54.0%	34.1	62.8	
2026	55.0%	35.2	64.2	Current
2034	63.0%			CBO projection as cited by KFF

Read together with Medicare_IFT_Series: carrier-billed FFS hospital-to-hospital transports peaked in 2017 and are 23.5% smaller, while MA crossed half of beneficiaries. The demand did not leave; the measurable window narrowed, and it will keep narrowing toward the CBO 63%. Every PSPS series in this workbook must be deflated mentally by this migration before being read as demand.

Panel C. Supply exits on the sending side. Sheps Center rural closure tracker, accessed 10 July 2026 (S52), and the measured transport-time mechanism (S53).

Measure	Value	Detail
Rural hospital closures and conversions since January 2005	197	109 complete closures plus 88 converted closures (inpatient services ended, some services remain)
Closures and conversions since 2010	154	86 complete plus 68 converted. Rural Emergency Hospital conversions are excluded and tracked separately

Convention warning

The Chartis Center for Rural Health counts 182 since 2010 under a broader convention that treats REH conversion differently. The two counts are not interchangeable; cite one tracker and its convention.

Mechanism, measured. Rural hospital closures raised mean EMS transport time by 2.6 minutes (P=.09) and total activation time by 7.2 minutes (P=.02), with no change in response time: the ambulance reaches the patient as fast, then drives farther. Miller et al., Health Services Research 2020;55(2):288-300, doi:10.1111/1475-6773.13254, PMID 31989591, NEMSIS 2010-2016 (S53). Retrieved from PubMed 10 July 2026.

Panel D. Coordination is a measured lever, not a slide claim.

State medical operations coordination centers (SMOCCs), stood up in eight states during the pandemic, were associated with an IMMEDIATE 35% increase in interhospital transfer rates (rate ratio 1.35, 95% CI 1.05 to 1.74) followed by a long-term rate ratio of 0.94, across 441,709 NEMSIS transfers, June 2020 to December 2022. Transfer throughput responds to coordination infrastructure on a measurable scale. Richert et al., JAMA Network Open 2025;8(12):e2546002, doi:10.1001/jamanetworkopen.2025.46002, PMID 41324958 (S54). Retrieved from PubMed 10 July 2026. Read with Demand_Drivers Panel B: the 287-minute median transfer delay is the friction this lever attacks.

Panel E. The final external anchor. NHAMCS 2022 was the last vintage of the survey.

Measure	Value	Standard error	Source
ED visits, 2022 (thousands)	155,398		NHAMCS 2022 ED Summary Tables, Table 4 (S55)
Ambulance arrival share, all ages	17.0%	0.9%	NHAMCS 2022 ED Summary Tables, Table 4 (S55)
Ambulance arrival share, 65 and over	33.4%	1.6%	NHAMCS 2022 ED Summary Tables, Table 4 (S55)
Ambulance arrival share, 75 and over	39.6%	2.0%	NHAMCS 2022 ED Summary Tables, Table 4 (S55)
Ambulance arrival share, under 15	5.0%	0.8%	NHAMCS 2022 ED Summary Tables, Table 4 (S55)

The 2016 anchor of 16.2% carried on EMS_Transports updates to 17.0% with a standard error of 0.9 points. NEMSIS 2022 coverage of EMS-to-ED transport (17.5% of NEDS visits) sits INSIDE the 95% interval of this final anchor: at full state participation, NEMSIS captures effectively all EMS-to-ED transport. NCHS discontinued NHAMCS after 2022 (successor: National Hospital Care Survey), so this anchor is frozen. Logged as Data_Quality_Register #30.

Table metrics

CAGR of 65+ population per five-year interval (from the projection row above)

1.85%	2.37%	1.27%	0.64%	0.96%
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The subject company, measured: Medicare FFS ambulance book by NPI, 2013 / 2019 / 2024

The question: what does the subject company measurably bill Medicare fee-for-service - volume, realized price, acuity mix, mileage intensity and trajectory - from public provider-grain claims summaries alone? Roll-up of the 23 registered NPIs on MMT_NPI_Estate across MUP_Providers_2013/2019/2024. Medicare FFS is roughly one-sixth of the market by this workbook's own payer arithmetic (GADCS Table S.1: FFS 29% of revenue per NPI on average, and enrollment splits); this tab is the measured PUBLIC FLOOR of the book, not the book.

DATA QUALITY: per-NPI code rows with 10 or fewer beneficiaries are suppressed at source, so every figure is a floor; final-action basis; NPIs enumerated after a vintage year cannot appear in it (Panel F states which and why); billing under a parent or billing-agent NPI is invisible here.

Panel A. Consolidated book by vintage (ground base codes A0426-A0429, A0433, A0434)								
Vintage	NPIs present	Base services (floor)	Beneficiary-code rows (floor)	Allowed \$	Paid \$	Allowed per base service \$	Paid share of allowed	Note
2013		110,162	8,210	\$3,484,246	\$2,736,343	\$342.87	78.5%	beneficiaries summed over code rows: not unique patients
2019		129,999	25,227	\$8,327,999	\$6,594,789	\$277.61	79.2%	
2024		269,684	53,773	\$22,004,764	\$17,292,232	\$315.78	78.6%	

Panel B. Acuity mix by vintage (share of base services) and the volume-weighted RVU								
Level (HCPCS)	2013 services	2019 services	2024 services	2013 share	2019 share	2024 share	AFS RVU	
ALS non-emergency (A0426)	265	4,643	2,927	2.6%	15.5%	4.2%	1.20	RVUs frozen since 2002; same constants as Payment_Rules
ALS emergency (A0427)	5,793	5,929	8,845	57.0%	19.8%	12.7%	1.90	
BLS non-emergency (A0428)	2,865	18,077	52,513	28.2%	60.3%	75.4%	1.00	
BLS emergency (A0429)	1,078	959	4,546	10.6%	3.2%	6.5%	1.60	
ALS level 2 (A0433)	84	177	268	0.8%	0.6%	0.4%	2.75	
SCT (A0434)	77	214	585	0.8%	0.7%	0.8%	3.25	
Volume-weighted RVU (live)				1.61	1.25	1.19		acuity in one number: rises only if the mix shifts up

Panel C. Mileage economics (A0425 ground mileage)						
Vintage	Mileage units (floor)	Base services (link)	Implied avg loaded miles per transport	Mileage allowed \$	Mileage share of total allowed	
2013	266,373	10,162	26.2	\$2,056,338	37.1%	
2019	773,370	29,999	25.8	\$6,024,712	42.0%	
2024	1,014,607	69,684	14.6	\$9,037,972	29.1%	

Implied miles are mileage UNITS divided by base transports: a floor over a floor, so read the level cautiously and the trend directionally. Air codes are excluded from every ground panel; any air rows in the registry are shown in Panel F.

Panel D. Trajectory (live CAGRs over Panel A)			
Metric	2013 to 2019 CAGR	2019 to 2024 CAGR	2013 to 2024 CAGR
Base services	19.8%	18.4%	19.1%
Allowed \$	15.6%	21.4%	18.2%
Allowed per service \$	(3.5%)	2.6%	(0.7%)

Panel E. Per-NPI book, 2024 (the granular floor)								
NPI	Registered name (NPPEs)	State	City	Base services	Allowed \$	Paid \$	Allowed per service \$	Mileage units
1871991125	Midwest Medical Transport Company Llc	NE	Columbus	69,643	\$21,993,137	\$17,282,968	\$315.80	1,013,484
1629831102	Midwest Medical Transport Company Llc	CO	Colorado Springs	41	\$11,627	\$9,264	\$283.59	1,123

Panel F. NPI resolution by vintage (which registered NPIs appear, and why absences are expected)						
NPI	Registered name	In 2013	In 2019	In 2024	Reading	
1003266339	MMT Co LLC	-	-	-	Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI	
1053162420	MMT Co LLC	-	-	-	Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI	

1073296893	MMT Co LLC DBA Southeast Iowa Ambulance	-	-	-	Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI
1134670219	MMT Co LLC - Huron, SD	-	-	-	Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI
1205387545	MMT Co LLC - Aberdeen, SD	-	-	-	Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI
1235681354	MMT Co LLC DBA Fraser Transportation Services	-	-	-	Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI
1235998626	MMT Co LLC	-	-	-	Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI
1346792371	MMT Co LLC DBA Southeast Iowa Ambulance	-	-	-	Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI
1356115562	Midwest Medical Transport Co LLC	-	-	-	Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI
1477490381	MMT Co LLC	-	-	-	Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI
1538612320	MMT Co LLC	-	-	-	Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI
1548034754	MMT Co LLC	-	-	-	Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI
1548047152	MMT Co LLC	-	-	-	Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI
1588115893	MMT Co LLC - Sioux City, IA (subpart)	-	-	-	Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI
1629831102	MMT Co LLC	-	-	Y	Absent early vintages: enumerated later, below the suppression floor, or volume consolidated on another NPI
1689665143	Platte County Ambulance Company DBA Midwest Medical Transport Co	Y	-	-	Resolves across vintages
1740645332	Midwest Medical Transport Co LLC (subpart)	-	-	-	Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI
1760832695	MMT Co LLC	-	-	-	Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI
1871991125	Midwest Medical Transport Co LLC (flagship)	-	Y	Y	Absent early vintages: enumerated later, below the suppression floor, or volume consolidated on another NPI
1922872134	Midwest Medical Transport LLC	-	-	-	Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI
1932650272	MMT Co LLC - Albia, IA	-	-	-	Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI
1982069134	MMT Co LLC	-	-	-	Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI
1992579338	MMT Co LLC	-	-	-	Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI

Read panel

The measured floor and its shape: the registered NPI estate bills Medicare FFS about 69,684 ground base transports in 2024 (allowed about \$22.0M), roughly seven times the 2013 floor, and the mix moved decisively to BLS non-emergency (28.2% of base services in 2013 to 75.4% in 2024) - the scheduled interfacility contract product - while implied loaded miles per transport roughly halved as the book scaled and densified. SCT and ALS2 stay under 1.5% of the MEDICARE-VISIBLE book. What this is NOT: the company's book. Medicare FFS is roughly one-sixth of the market by payer arithmetic, MA and commercial volumes are invisible here, and dedicated-unit facility contracts (Facility_Pay_Layer) never generate these claims at all - which is exactly where high-acuity dedicated work would sit. Direction of bias: every number on this tab understates.

The facility-pay layer: ambulance revenue that never touches a payer claim

The question: how much ambulance revenue is paid by facilities, local governments and contracts rather than through payer claims - the layer every claims-based market size misses? Three primaries, verified against the documents on 11 Jul 2026: the CMS/RAND GADCS Year 1-4 census (the only national revenue-source survey), the DocGo 10-K (the only US-listed pure ambulance P&L), and USAspending PSC V225 (the federal direct-contract channel). This tab bounds incidence and magnitude; the IFT-specific facility-pay share does not exist publicly and stays PENDING.

DATA QUALITY: GADCS values are self-reported survey responses from the Year 1-4 collection cohorts (MedPAC judged the COST side unusable for margins; revenue-source incidence is carried here as the census it is, with that caveat). DocGo figures are audited but one operator. USAspending rows are procurement obligations only; the VA Beneficiary Travel claims channel is separate and larger and must NEVER be summed with these.

Panel A. GADCS: revenue sources of US ambulance organizations (share reporting any; conditional mean; median)				
Revenue source (GADCS Question 5)	Share reporting	Conditional mean \$	Median \$	Locator
Facility contracts (hospitals, SNFs, other facilities)	18.6%	\$922,555	\$48,393	Table 2.32, p.70
Any non-payer revenue (all Question 5 categories)	74.5%	\$2,177,569	\$162,265	Table 2.32, p.70
Local-government contracts	15.4%	\$556,091	\$112,426	Table 2.32, p.70
EMS-earmarked local taxes	23.4%	\$3,135,660	\$352,117	Table 2.32, p.70
Standby event revenue (share reporting)	24.5%			Table 2.32, p.70
Membership programs (share reporting)	7.6%			Table 2.32, p.70

Payer revenue per NPI (GADCS mean)	Mean \$	Locator
All payers	\$3,914,787	Table 2.30, pp.64-65
Medicare FFS	\$1,193,046	Table 2.30, pp.64-65
Medicare Advantage	\$1,238,482	Table 2.30, pp.64-65
Medicaid	\$732,436	Table 2.30, pp.64-65
Commercial	\$1,322,170	Table 2.30, pp.64-65
Medicare FFS share of transport revenue (average per NPI)	29.0%	Table S.1, p.ix
Medicare Advantage additional share	18.0%	Table S.1, p.ix
Organizations that did not always bill in at least one payer category (	20.0%	Table S.1, p.ix; "about one in five"
MA-to-FFS revenue ratio per NPI (the MA book calibrator)	1.04x	DERIVED from Table 2.30 rows above

Interfacility share of transport volume (GADCS)	Share of organizations	Locator
For-profit organizations at 81-100% interfacility	20.2%	Table 2.6, p.33
For-profit organizations at 21-80% interfacility	42.8%	Table 2.6, p.33
All organizations at 81-100% interfacility	7.4%	Table 2.6, p.33

Read: the interfacility-heavy operator is overwhelmingly a FOR-PROFIT phenomenon (20.2% of for-profits vs 7.4% of all organizations run 81-100% interfacility books), which is why claims-only market reads systematically under-see the private IFT segment: it is exactly the segment with facility contracts.

Panel B. DocGo 10-K: the measured share of transportation revenue billed outside per-trip claims				
Metric	FY2025	FY2024	FY2023	Locator
Transportation Services segment revenue \$	\$200,765,608	\$193,429,092	\$181,495,105	Note 2 (F-22); Note 11 (F-37)
Trips (per-trip billed only, by the filing definition)	296,014	283,570	250,114	MD&A pp.62/66
Average trip price \$	\$401	\$402	\$407	MD&A pp.57/62/66; FY23 ATP from Note 2
Implied per-trip billed revenue \$	\$118,701,614	\$113,995,140	\$101,796,398	trips x ATP
Revenue OUTSIDE per-trip billing (dedicated-unit programs billed ho	40.9%	41.1%	43.9%	MD&A p.57 exclusion language; arithmetic live

Guardrail: one operator, and the plus-or-minus 0.1pp rounding from whole-dollar ATP is real. What it establishes is EXISTENCE and scale at a listed operator: roughly 41 cents of every transportation dollar arrives outside a per-trip claim, and the filing itself names healthcare facilities as a billing counterparty class (Note 2, F-22).

Panel C. USAspending PSC V225: the federal direct-contract ambulance channel (obligations, pulled live)				
Fiscal year	VA obligations \$	All-agency obligations \$	VA share	Note
2023	\$173,527,508	\$210,994,517	82.2%	spending_over_time, psc V225

2024	\$190,909,874	\$247,491,935	77.1%
2025	\$246,082,814	\$292,654,678	84.1%
CAGR FY2023-FY2025 (VA)	19.1%	17.8%	two-year CAGR

Top FY2025 V225 recipients	Obligations \$
EXCELSIOR AMBULANCE SERVICE INC	\$19,132,382
RELIANCE MEDICAL TRANSPORT, LLC	\$15,611,296
MEDSHORE AMBULANCE SERVICE, INC.	\$11,461,265
JAN-CARE AMBULANCE OF NORTH CENTRAL W.VA., INC	\$11,443,449
JOURNEY VIA GURNEY, LLC	\$10,256,714
HAWAII LIFE FLIGHT LLC	\$8,614,952
BRADSHER, JAMES E	\$8,485,576
PROFESSIONAL HEALTHCARE ASSOCIATES, INC.	\$8,305,345
MEDEX MEDICAL TRANSPORT SERVICE INC	\$7,980,186
NAVARRE CORPORATION	\$7,737,577

Guardrail: procurement obligations only. The VA Beneficiary Travel program pays ambulance CLAIMS through a separate and larger channel; the two must never be summed. IHS and DoD V225 obligations exist and are additive to VA only within this procurement frame.

Panel D. What this layer does to the sizing math (live links)		
NEMSIS interfacility-scale activations (EMS_Transports)	5,510,664	vehicle legs, all payers
Medicare FFS hospital-to-hospital services 2024 (Medicare_IFT_Serie	871,753	claims-visible floor
The identity this tab prices: activations minus claims-visible volume	4,638,911	wedge, not a market size
IFT-specific facility-pay share of that wedge	PENDING	No public source; would take a claims-vendor panel with facility-remit flags or primary buyer research (bordered slot; see Engagement_Data_Map)

Read panel

What is measured: (1) 18.6 percent of US ambulance organizations report facility-contract revenue, with a conditional mean near \$0.92M and a long right tail (median \$48K); (2) 74.5 percent report some non-payer revenue; (3) a listed pure-play operator books about 41 percent of transportation revenue outside per-trip claims, naming facilities as a payer class in its revenue-recognition note; (4) the federal government alone obligates about a quarter-billion dollars a year buying ambulance service by contract. Together these are the public floor and existence proof for the facility-pay layer that claims-based sizing cannot see. What is NOT measured: the IFT-specific share of that layer; it stays PENDING by design.

Fragmentation, measured: the US Medicare ambulance biller universe across 12 vintages, 2013-2024

The question: how fragmented is US ground ambulance - biller counts, size distribution, top-10/50/100 concentration - and is the long tail exiting? Source: MUP by Provider and Service, ground base codes A0426-A0429/A0433/A0434, all states and territories, 12 manifested vintages 2013-2024; join keys Rndrng_NPI x HCPCS_Cd x vintage. NPI-grain panels first, then a printed, auditable name-pattern roll-up to parents.

DATA QUALITY: NPI grain - one organization can bill under several NPIs, so biller counts OVERSTATE firms and top-N shares UNDERSTATE concentration; the name-pattern parent layer below corrects only what names reveal and is itself a floor. Per-NPI code rows under 11 beneficiaries are suppressed at source: the smallest billers are partly invisible, and an NPI "exiting" can be a dip below the suppression floor rather than a true exit. Final-action claims, Medicare FFS only. Vintages are independent annual snapshots, not a continuous series (until the annual series lands); every consecutive transition below is one year apart.

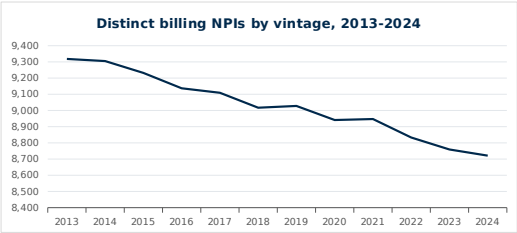
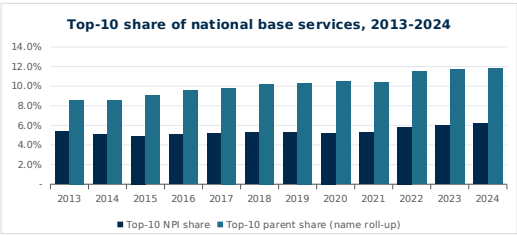
Panel A. Biller count and top-N NPI concentration by vintage (distinct NPIs, ground base codes)									
Vintage	Distinct billing NPIs	National base services (floor)	Top-10 NPI services	Top-10 share	Top-50 NPI services	Top-50 share	Top-100 NPI services	Top-100 share	Note
2013	9,318	14,938,429	801,682	5.4%	2,239,654	15.0%	3,329,992	22.3%	counts overstate firms (NPI grain); services are suppression floors
2014	9,305	15,037,862	766,032	5.1%	2,177,226	14.5%	3,293,063	21.9%	
2015	9,232	14,235,792	700,232	4.9%	2,085,546	14.7%	3,159,170	22.2%	
2016	9,137	13,783,259	700,237	5.1%	2,010,282	14.6%	3,043,316	22.1%	
2017	9,109	13,713,824	710,395	5.2%	1,990,181	14.5%	3,019,137	22.0%	
2018	9,017	13,358,909	701,154	5.2%	1,957,262	14.7%	2,970,069	22.2%	
2019	9,028	12,890,235	678,782	5.3%	1,870,212	14.5%	2,827,987	21.9%	
2020	8,941	11,531,316	601,885	5.2%	1,648,686	14.3%	2,505,563	21.7%	
2021	8,947	10,902,433	573,348	5.3%	1,575,526	14.5%	2,379,182	21.8%	
2022	8,833	10,084,704	586,576	5.8%	1,549,571	15.4%	2,338,034	23.2%	
2023	8,759	9,660,141	579,232	6.0%	1,555,694	16.1%	2,329,728	24.1%	
2024	8,721	9,542,103	588,683	6.2%	1,578,097	16.5%	2,344,426	24.6%	

Panel B. Size distribution by services decile (2024 vs 2013; deciles of each vintage's own biller distribution)								
Decile (by base services)	Billers 2024	Services 2024	Share 2024	Median services 2024	Billers 2013	Services 2013	Share 2013	Note
D1 (smallest tenth)	872	18,546	0.2%	20	931	21,051	0.1%	suppression truncates this decile: the true tail is larger
D2	872	41,255	0.4%	47	932	49,497	0.3%	
D3	872	75,013	0.8%	85	932	89,816	0.6%	
D4	872	123,907	1.3%	141	932	149,667	1.0%	
D5	872	196,112	2.1%	223	932	246,899	1.7%	
D6	872	306,319	3.2%	350	931	397,345	2.7%	
D7	872	464,442	4.9%	529	932	636,388	4.3%	
D8	872	732,547	7.7%	826	932	1,087,182	7.3%	
D9	872	1,343,066	14.1%	1,463	932	2,266,436	15.2%	
D10 (largest tenth)	873	6,240,896	65.4%	4,466	932	9,994,148	66.9%	the whole market story lives here
All billers	8,721	9,542,103			9,318	14,938,429		ties to Panel A rows for the same vintages

Panel C. Name-normalization layer: NPIs rolled to parents by name pattern - the FULL pattern table prints below in blue so the layer is auditable					
Parent family	NPIs matched 2024	States 2024	Base services 2024	Share of national	Match patterns (case-insensitive substring / word)
AMR / American Medical Response	65	17	423,252	4.44%	AMERICAN MEDICAL RESPONSE; word "bAMRb" - mandated pattern; the ground flagship of the GMR family
Global Medical Response / GMR	-	-	-	-	GLOBAL MEDICAL RESPONSE; word "bGMRb" - mandated pattern; no NPI bills under the holding-company name - GMR-family volume rides brand rows (AMR, Lifeguard), so any GMR-level roll-up is a floor
Falck	4	3	72,805	0.76%	FALCK - mandated pattern
Acadian	4	3	151,225	1.58%	ACADIAN - mandated pattern
PatientCare	-	-	-	-	PATIENTCARE; PATIENT CARE EMS - mandated pattern; zero matches - the brand bills under legacy local names, another reason the roll-up is a floor
Priority Ambulance	4	4	6,496	0.07%	PRIORITY AMBULANCE - mandated pattern; subsidiaries billing under local brands stay unrolled
DocGo / Ambulnz	6	5	59,315	0.62%	DOCGO; AMBULNZ - mandated pattern
Superior Air-Ground	5	5	112,351	1.18%	SUPERIOR AIR-GROUND; SUPERIOR AIR GROUND - mandated pattern
Royal Ambulance	1	1	10,909	0.11%	ROYAL AMBULANCE - mandated pattern
Lifeguard Ambulance Service	13	8	36,269	0.38%	LIFEGUARD AMBULANCE - detected same-name multi-state family (8+ states); publicly a GMR brand
Cal-Ore Life Flight	8	3	3,321	0.03%	CAL-ORE; CAL ORE LIFE - detected same-name multi-state family; ground legs of the base codes only

Roll-up rules, printed for audit: an NPI joins a family only when its NPPES organization name matches a pattern above; every unmatched NPI stays its own parent. Municipal and generic same-names (CITY OF X, COUNTY OF X, COMMUNITY AMBULANCE SERVICE) are NOT rolled - they are distinct local entities sharing a name. Direction of bias: this layer can only RAISE measured concentration toward truth, never reach it - brands billing under unrelated legacy names stay unrolled, so top-10 parent share is still a floor.

Vintage	Top-10 NPI share (link)	Top-10 parent services	Top-10 parent share	Parent minus NPI (pp)	Note
2013	5.4%	1,282,256	8.6%	3.2%	top-10 entities after roll-up (families plus unrolled NPIs); denominator is the Panel A national total
2014	5.1%	1,281,223	8.5%	3.4%	
2015	4.9%	1,291,353	9.1%	4.2%	
2016	5.1%	1,318,994	9.6%	4.5%	
2017	5.2%	1,343,974	9.8%	4.6%	
2018	5.2%	1,365,039	10.2%	5.0%	
2019	5.3%	1,332,350	10.3%	5.1%	
2020	5.2%	1,206,349	10.5%	5.2%	
2021	5.3%	1,131,272	10.4%	5.1%	
2022	5.8%	1,157,512	11.5%	5.7%	
2023	6.0%	1,131,747	11.7%	5.7%	
2024	6.2%	1,126,848	11.8%	5.6%	



Panel D. Exit by size decile: NPIs present in one vintage and absent from the next (decile of the START vintage)												
Transition	Exit rate D1 (smallest)	D2	D3	D4	D5	D6	D7	D8	D9	Exit rate D10 (largest)	Exited NPIs	Note
2013 to 2014	20.5%	4.9%	2.4%	2.6%	2.5%	2.7%	2.8%	4.2%	2.9%	1.3%	435	absence includes dips below the suppression floor, so tail exit rates are upper bounds on true exit
2014 to 2015	17.1%	3.2%	1.9%	2.0%	3.0%	3.4%	3.3%	4.3%	6.0%	3.2%	443	
2015 to 2016	22.6%	4.8%	2.5%	2.3%	3.5%	2.0%	2.5%	4.1%	2.8%	1.8%	451	
2016 to 2017	21.9%	4.6%	3.1%	1.8%	1.8%	2.0%	1.9%	3.6%	2.4%	1.5%	406	
2017 to 2018	25.3%	4.0%	3.7%	2.3%	1.9%	2.0%	2.5%	3.0%	1.6%	1.1%	431	
2018 to 2019	19.1%	4.3%	3.4%	3.3%	2.3%	2.3%	2.9%	2.5%	1.8%	1.1%	389	
2019 to 2020	25.5%	3.9%	2.8%	1.9%	2.4%	2.2%	2.7%	2.3%	1.9%	1.1%	421	
2020 to 2021	18.5%	4.6%	2.2%	1.1%	1.5%	1.9%	1.6%	3.7%	2.8%	1.3%	350	
2021 to 2022	23.4%	2.8%	3.6%	3.4%	2.7%	2.3%	2.6%	4.1%	4.5%	1.3%	453	
2022 to 2023	27.4%	6.3%	2.7%	2.5%	3.1%	2.8%	2.1%	3.4%	3.7%	1.0%	487	
2023 to 2024	27.8%	7.0%	3.9%	4.0%	2.3%	2.3%	1.8%	1.4%	2.4%	1.0%	471	

Transition	Entered NPIs (absent start, present end)	Entry rate of end-vintage billers	Note
2013 to 2014	422	4.5%	entry runs below exit in most transitions: the universe shrinks on net
2014 to 2015	370	4.0%	
2015 to 2016	356	3.9%	
2016 to 2017	378	4.1%	
2017 to 2018	339	3.8%	
2018 to 2019	400	4.4%	
2019 to 2020	334	3.7%	
2020 to 2021	356	4.0%	
2021 to 2022	339	3.8%	
2022 to 2023	413	4.7%	
2023 to 2024	433	5.0%	

Read panel

Measured, twice over: the universe is extraordinarily fragmented - 8,721 distinct billing NPIs in 2024, with the top 10 holding 6.2% of national base services, the top 100 holding 24.6%, and even after rolling brands to parents by name the top 10 parents hold only 11.8% (up from 8.6% in 2013: consolidation is real but glacial, and the roll-up is itself a floor). And the long tail is thinning: the smallest decile of billers exits at roughly 23% per year against about 1% for the largest decile, entry runs below exit, and the biller count fell from 9,318 to 8,721 while national Medicare FFS base volume fell about 36%. Read every level as a floor: NPI grain, suppression, and Medicare-FFS-only visibility all push the same direction - the true market is somewhat more concentrated and the true tail somewhat larger than shown.

Medicaid ground ambulance rate card - the footprint states

The question: what does each footprint state Medicaid FFS schedule actually pay for ground ambulance (A0425-A0434), and how much scheduled transport is carved out to NEMT brokers and never priced on these schedules? Sources: nine published state fee schedules named on each panel, each fetched and parsed from the primary file or portal (NEIA/KS/MO/CHW on 11 Jul 2026; MN/IN/KY on 12 Jul 2026); Virginia is PARKED (only 2009-vintage public files). Join key is the HCPCS code. Comparison prices each state against the CY2026 Medicare national unadjusted base on Derived_Rate_Card.

DATA QUALITY: Medicaid FFS schedule rates ONLY - managed-care plans negotiate their own rates and broker-care states route scheduled volume outside these schedules, so the real price of scheduled transport is unobservable here. Effective dates differ by 18 years (WI 7/1/2008 vs MN/IN 1/1/2026); the cross-state comparison mixes vintages by construction. Rates are NOT applies-to-applies product-for-product. MO A0428 is HH+HD-context only; KY has NO ALS1+non-emergency, ALS2, or SCT line and its A0428 is a non-emergency STRETCHER-van line (formerly T2005), not a full BLS ambulance; IN and MO do not cover SCT (A0434). KY is therefore excluded from the A0428 BLS comparison range.

Panel A. Nebraska - Nebraska DHHS fee schedule 471-000-504, Ambulance Services, July 1 2026 (xlsx). Sc					
HCPCS	Service level	Rate \$	Unit	Rate eff.	In-row note
A0428	BLS non-emergency	\$154.89 base		07/01/2026	schedule comment: includes bariatric transfers
A0429	BLS emergency	\$189.93 base		07/01/2026	
A0426	ALS1 non-emergency	\$387.24 base		07/01/2026	
A0427	ALS1 emergency	\$387.24 base		07/01/2026	
A0433	ALS2	\$387.24 base		07/01/2026	
A0434	Specialty Care Transport	\$387.24 base		07/01/2026	flat ALS rate: A0426+A0427+A0433+A0434
A0425	Ground mileage	\$6.35 per mile		07/01/2026	
Source file					per statute mile
PA rule	PA column blank on every ground code (no PA flag); policy 471 NAC 4.				https://dhhs.ne.gov/Medicaid%20Practitioner%20Fee%20Schedules/471-000-504%20Ambulance%20-%20July%20%26%2726.xlsx
Broker carve	YES - Heritage Health MCO brokers (MTM for NE Total Care and Molina; ModivCare for UHC, MTM Health from 1 Jan 2026); DHHS staff arrange residual FFS NET under 471-000-503.				
Dialysis policy	No dialysis restriction printed on the ambulance schedule; scheduled dialysis rides route through the NET benefit.				

Panel B. Iowa - Iowa Medicaid fee schedule portal extract, provider type 11 AMBULANCE (C11.csv), a					
HCPCS	Service level	Rate \$	Unit	Rate eff.	In-row note
A0428	BLS non-emergency	\$84.67 base		07/01/2014	lowest true BLS A0428 in the footprint
A0429	BLS emergency	\$114.30 base		10/01/2015	
A0426	ALS1 non-emergency	\$101.60 base		10/01/2015	
A0427	ALS1 emergency	\$127.01 base		10/01/2015	
A0433	ALS2	\$232.84 base		05/01/2020	
A0434	Specialty Care Transport	\$232.84 base		12/01/2020	same rate as ALS2
A0425	Ground mileage	\$2.61 per mile		07/01/2014	per statute mile
Source file					https://secureapp.dhs.state.ia.us/MedicaidFeeSchedC11.csv
PA rule	No PA; non-emergency requires bed-confinement or equivalent necessity criteria (Ambulance Services Provider Manual Ch. III, 1 Oct 2015).				
Broker carve	YES - NEMT broker Access2Care renamed MTM Health 18 Sep 2025 (IA Health Link MCOs); MTM administers FFS NEMT.				
Dialysis policy	Dialysis NEMT rides schedulable 90 days ahead; ambulance claim destination modifiers G/J are dialysis facilities.				

Panel C. Kansas - KMAP TXIX fee schedule files, 7/1/2026 versions (FeeSchedule_20260701_TXIX_AMB.txt					
HCPCS	Service level	Rate \$	Unit	Rate eff.	In-row note
A0428	BLS non-emergency	\$255.70 base		07/01/2023	SPY2024 rebase
A0429	BLS emergency	\$409.12 base		07/01/2023	
A0426	ALS1 non-emergency	\$306.84 base		07/01/2023	
A0427	ALS1 emergency	\$485.83 base		07/01/2023	
A0433	ALS2	\$703.18 base		07/01/2023	
A0434	Specialty Care Transport	\$831.03 base		07/01/2023	Ambulance rate-type file: legacy \$3.00 row (eff 07/01/2022) persists on the general schedule
A0425	Ground mileage	\$13.20 per mile		07/01/2023	
Source file					https://portal.kmap-state-ks.us/PublicPage/ProviderPricing/FeeSchedules
PA rule	No PA, but ALL nonemergent transports need a Medical Necessity form attached to the claim (KMAP manual).				
Broker carve	YES - KanCare is near-total managed care; NEMT via MCO broker ModivCare (taxi to stretcher van and ambulance).				
Dialysis policy	Dialysis named among critical services eligible for NEMT in the Kansas NEMT provider manual.				

Panel D. Missouri - MHD Ambulance.xlsx, file dated 7 Jul 2026 (provider files updated 07/08/2026)					
HCPCS	Service level	Rate \$	Unit	Rate eff.	In-row note
A0428	BLS non-emergency	\$105.62 base		07/01/2019	HH and HD (hospital-to-diagnostic) contexts ONLY; unmodified code noncovered
A0429	BLS emergency	\$379.69 base		07/01/2024	
A0426	ALS1 non-emergency	\$172.26 base		07/01/2019	
A0427	ALS1 emergency	\$474.75 base		07/01/2024	
A0433	ALS2	\$215.20 base		07/01/2019	
A0434	Specialty Care Transport	NONCOVERED base		07/01/2002	HH (hospital-to-hospital) modifier ONLY; unmodified code noncovered
A0425	Ground mileage	\$7.04 per mile		07/01/2024	
Source file					NONCOVERED under MO HealthNet FFS
PA rule	PA = No on all covered rows; A0428 billable ONLY with HH or HD modifier (MHD hot tip).				per statute mile
Broker carve	YES - statewide NEMT broker MTM (mydss.mo.gov/mhd/nemt).				https://apps.dss.mo.gov/fms/feeschedules/dfiles/Ambulance.xlsx

Dialysis policy

Dialysis runs are NOT billable as FFS A0428 (H+HD only); scheduled dialysis travel is an MTM broker trip.

Panel E. Ohio - Appendix to OAC rule 5160-15-28 (file dated 22 Dec 2023), rule effective 1 Jan 202					
HCPCS	Service level	Rate \$	Unit	Rate eff.	In-row note
A0428	BLS non-emergency	\$203.75 base		01/01/2024	
A0429	BLS emergency	\$244.00 base		01/01/2024	
A0426	ALS1 non-emergency	\$244.50 base		01/01/2024	
A0427	ALS1 emergency	\$289.75 base		01/01/2024	
A0433	ALS2	\$349.50 base		01/01/2024	
A0434	Specialty Care Transport	\$413.00 base		01/01/2024	
A0425	Ground mileage	\$5.05 per mile		01/01/2024	per statute mile
Source file					
PA rule	No PA in rule; necessity per OAC 5160-15-23(B); EMT care, oxygen, or supervised restraint during transport, or qualifying hospital-to-hospital transfer.				
Broker carve	NO statewide broker - county CDJFS NET programs (5160-15-10) plus MCO rides; ambulance stays a direct ODM/MCO claim. GEMT supplemental for public EMS (5160-15-30).				
Dialysis policy	Rules retrieved are silent on dialysis transport.				

https://codes.ohio.gov/assets/laws/administrative-code/pdfs/5160/015/5160-15-28_PH_FF_A_APP1_20231222_0942.pdf

Panel F. Wisconsin - ForwardHealth AMBULANCE MAXIMUM ALLOWABLE FEE SCHEDULE, live portal report accesse					
HCPCS	Service level	Rate \$	Unit	Rate eff.	In-row note
A0428	BLS non-emergency	\$94.90 base		07/01/2008	unchanged for 18 years
A0429	BLS emergency	\$151.84 base		07/01/2008	
A0426	ALS1 non-emergency	\$113.88 base		07/01/2008	
A0427	ALS1 emergency	\$180.31 base		07/01/2008	
A0433	ALS2	\$260.97 base		07/01/2008	
A0434	Specialty Care Transport	\$308.42 base		07/01/2008	
A0425	Ground mileage	\$5.56 per mile		07/01/2008	GM multi-patient half-rate \$2.78
Source file					
PA rule	No code-level PA flag; non-emergency needs ALS/BLS care or supine transport, and manager-coordinated trips need a trip ID within two business days (Update 2014-32).				
Broker carve	YES - statewide NEMT manager (MTM Inc.; Veyo from 11/2021 until MTM acquisition). Manager-coordinated NEMT ambulance claims are paid by the manager, NOT ForwardHealth.				
Dialysis policy	Recurring dialysis rides are standing rides with the NEMT manager; stretcher vans sit on the separate SMV schedule.				

https://www.forwardhealth.wi.gov/WiPortal/Tab/42/csscontent/provider/max/fee/pdf/maxfee06_ambulance.pdf.spaga

Panel G. Minnesota - Minnesota MHCP Fee Schedule.xlsx (mn.gov/dhs), downloaded 12 Jul 2026					
HCPCS	Service level	Rate \$	Unit	Rate eff.	In-row note
A0428	BLS non-emergency	\$284.56 base		01/01/2026	highest A0428 in the footprint - exactly the C-2026 Medicare conversion factor (\$284.56), i.e. Medicare parity
A0429	BLS emergency	\$455.29 base		01/01/2026	
A0426	ALS1 non-emergency	\$341.47 base		01/01/2026	
A0427	ALS1 emergency	\$540.66 base		01/01/2026	
A0433	ALS2	\$782.54 base		01/01/2026	
A0434	Specialty Care Transport	\$924.82 base		01/01/2026	highest SCT in the footprint
A0425	Ground mileage	\$9.51 per mile		04/01/2026	per statute mile (ground codes eff 01/01/2026; mileage 04/01/2026)
Source file					
PA rule	No PA flag on the ambulance rows; MHCP covers emergency and non-emergency ground/air ambulance billed directly to MHCP by enrolled providers (ambulance is not routed through the administered NEMT system).				
Broker carve	MIXED / STATE-ADMINSTERED (no single MCO broker) - split between local county/tribal agency-administered NEMT and State-Administered NEMT (Acentra Health / Kepro); ambulance itself stays FFS. NEMT providers need MnDOT STS certification.				
Dialysis policy	Explicit RETURN-LEG rule: for dialysis members eligible for local NEMT, State-Administered NEMT covers the RETURN trip - Kepro enters a single-day certification for dialysis return trips only.				

https://mn.gov/dhs/assets/mhcp-fee-schedule_tcm1053-293968.xlsx

Panel H. Indiana - Indiana IHCP Professional Fee Schedule.xlsx (in-session portal download), last upd					
HCPCS	Service level	Rate \$	Unit	Rate eff.	In-row note
A0428	BLS non-emergency	\$269.02 base		01/01/2026	100% of CY2025 Medicare (BT2025156)
A0429	BLS emergency	\$430.44 base		01/01/2026	
A0426	ALS1 non-emergency	\$322.83 base		01/01/2026	
A0427	ALS1 emergency	\$511.14 base		01/01/2026	
A0433	ALS2	\$739.81 base		01/01/2026	
A0434	Specialty Care Transport	NONCOVERED base		04/01/2004	NONCOVERED (NOCOV) under IHCP
A0425	Ground mileage	\$9.15 per mile		01/01/2026	per statute mile; reduced-modifier variants (U3/U5, MRT/SE) pay \$1.67
Source file					
PA rule	No PA flag on covered ambulance rows; A0434 SCT is Non-Covered (PA=Y, \$0.00). ALS/BLS ambulance is EXEMPT from the FFS NEMT brokerage and billed FFS directly.				
Broker carve	PARTIAL - non-ambulance FFS NEMT brokered statewide through Verida (formerly Southeastrans); managed care via the MCE. ALS/BLS ambulance is carved OUT of the broker and billed FFS.				
Dialysis policy	Recurring dialysis trips are standing orders through the Verida FFS broker (or the MCE); the ambulance schedule is silent at code level.				

<https://provider.indianamedicaid.com/ihcp/Publications/MaxFee/Professional%20Fee%20Schedule.xlsx>

Panel I. Kentucky - KY Medicaid Transportation Fee Schedule 2026, revised 1/9/2026 (no changes vs prio

	HCPCS	Service level	Rate \$	Unit	Rate eff.	In-row note	
A0428		BLS non-emergency	\$55.00 base		01/09/2026	NON-EMERGENCY STRETCHER line (PT56 Specialty 16), flat \$55 - a stretcher-van rate, NOT a full BLS ambulance; formerly T2005, renamed A0428 eff 1/1/2025 Other (independent) \$60.00 / Hospital-based \$82.50; A0429 GM add-patient \$20 NOT LISTED - no ALS1 non-emergency line on the KY transportation schedule Other (independent) \$60.00 / Hospital-based \$110.00; A0427 GM add-patient \$25 NOT LISTED - no ALS2 line NOT LISTED - no SCT line BLS mileage (A0425 UB) Other \$2.50 / Hospital \$3.00; ALS mileage \$2.50/\$4.00; stretcher and return-trip mileage \$2.00 https://www.chfs.ky.gov/agencies/dms/DMSFeeRateSchedules/2026TransportationRates.pdf	
A0429		BLS emergency	\$60.00 base		01/09/2026		
A0426		ALS1 non-emergency	NOT LISTED	base	01/09/2026		
A0427		ALS1 emergency	\$60.00 base		01/09/2026		
A0433		ALS2	NOT LISTED	base	01/09/2026		
A0434		Specialty Care Transport	NOT LISTED	base	01/09/2026		
A0425		Ground mileage	\$2.50 per mile		01/09/2026		
Source file							
PA rule							
No code-level PA on the schedule; emergency ambulance (PT55) billed FFS. Only non-emergency ambulance line is the PT56 stretcher; broader NEMT goes through the HSTD regional broker, off-schedule.							

Broker carve YES - Human Service Transportation Delivery (HSTD): DMS contracts with the KY Transportation Cabinet, which assigns a regional broker per county; providers contract with the broker, trips booked 72h ahead. Five MCOs administer NEMT for their members.

Dialysis policy Recurring dialysis trips route through the HSTD regional broker (or the MCO); the ambulance/stretcherschedule is silent on dialysis.

Panel J. Virginia (extension state) - rates PARKED, carve verified					Why parked / what fills it
HCPCS	Service level	Rate \$			
A0427/A0429	Emergency codes	PENDING			The only public DMAS emergency ground file is the 2009 sliding scale (VERIFIED "DOS October 31 2009 and Before"): 1-5 mi \$75, 6-10 mi \$150, then +\$2.50/mi. No current-effective FFS rate file is published; the DMAS procedure fee file search portal (interactive per-code lookup) would fill it.
A0428	BLS non-emergency	PENDING			Not payable on a public FFS schedule: DMAS Transportation manual Ch.5 (rev 12/5/2025) requires A0428 to be preauthorized and billed to the FFS NEMT broker.
A0425/A0426/A0433/A0434	Mileage / ALS / SCT	PENDING			Appear only on the 2009-2012 crossover rate table "for crossover calculation only"; no current standalone FFS rate is published.
Source (verified)	https://www.dmas.virginia.gov/media/1569/fee-schedules-for-emergency-ground-ambulance-a0427-and-a0429.pdf				

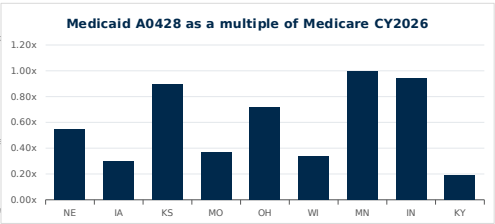
Panel K. Comparison - each state as a multiple of the CY2026 Medicare national unadjusted base (live formulas)											
Medicare column is a GREEN link to Derived_Rate_Card (located by scanning that tab for the code labels; base codes column D, mileage cell B23).											

HCPCS (service)	Medicare CY26 \$	NE	IA	KS	MO	OH	WI	MN	IN	KY	Confound printed with the ratio
A0428 (BLS non-emergency)	\$284.56	0.54x	0.30x	0.90x	0.37x	0.72x	0.33x	1.00x	0.95x	0.19x	IA lowest true-BLS floor, MN at Medicare parity (1.00x); KY excluded (its A0428 is a stretcher-van line, not a full BLS ambulance)
A0429 (BLS emergency)	\$455.30	0.42x	0.25x	0.90x	0.83x	0.54x	0.33x	1.00x	0.95x	0.13x	Same-code comparison; state schedules do not adjust for geography the way Medicare does
A0426 (ALS1 non-emergency)	\$341.47	1.13x	0.30x	0.90x	0.50x	0.72x	0.33x	1.00x	0.95x	n/a	MO A0426 is HH-modifier context; KY has no ALS1-nonE line
A0427 (ALS1 emergency)	\$540.66	0.72x	0.23x	0.90x	0.88x	0.54x	0.33x	1.00x	0.95x	0.11x	Same-code comparison; state schedules do not adjust for geography the way Medicare does
A0433 (ALS2)	\$782.54	0.49x	0.30x	0.90x	0.28x	0.45x	0.33x	1.00x	0.95x	n/a	KY has no ALS2 line (n/a)
A0434 (Specialty Care Transport)	\$924.82	0.42x	0.25x	0.90x	n/a	0.45x	0.33x	1.00x	n/a	n/a	MO and IN NONCOVERED - no ratio; KY has no SCT line
A0425 (Ground mileage)	\$9.15	0.69x	0.29x	1.44x	0.77x	0.55x	0.61x	1.04x	1.00x	0.27x	KS mileage from the ambulance rate-type file, not the legacy \$3.00 row; KY mileage is the BLS/Other rate
Footprint A0428 range - LOWEST multiple (Iowa)			0.30x								min over the eight true-BLS A0428 ratios (KY stretcher excluded)
Footprint A0428 range - HIGHEST multiple (Minnesota)			1.00x								max over the eight true-BLS A0428 ratios; MN = Medicare parity
Dispersion: highest / lowest A0428 rate			3.36x								MN pays about 3.4x what IA pays for the same code
Memo - KY non-emergency STRETCHER (not BLS ambulance)		0.19x									A0428 stretcher-van line (formerly T2005), \$55 flat - a different product; shown for context, excluded from FFS

Panel L. Broker carve-out summary - who actually buys scheduled transport					What is carved
State	Carve?	Broker / manager			
Nebraska	YES	MTM / ModivCare (MTM Health from 1/1/26)			NEMT via Heritage Health MCO brokers; FFS NET by DHHS staff
Iowa	YES	MTM Health (was Access2Care to 9/18/25)			NEMT for MCO and FFS members; ambulance stays FFS-billable
Kansas	YES	ModivCare (KanCare MCOs)			NEMT incl. stretcher van and NEMT ambulance coordination
Missouri	YES	MTM (statewide)			All scheduled NEMT; FFS ambulance only in HH/HD contexts
Ohio	NO	County CDJFS NET + MCO vendors			No statewide broker; ambulance is a direct ODM/MCO claim
Wisconsin	YES	MTM Inc. (NEMT manager; Veyo 2021-22)			Manager-coordinated NEMT ambulance is paid by the MANAGER, not ForwardHealth
Minnesota	YES	Acentra/Kepro (State-Admin) + county/tribal			State-administered + local-agency NEMT (no single MCO broker); ambulance stays FFS
Indiana	YES	Veridia (formerly Southeastrans)			Non-ambulance FFS NEMT brokered; ALS/BLS ambulance EXEMPT and billed FFS
Kentucky	YES	HSTD regional brokers (via KY Transp. Cabinet)			NEMT via county regional brokers; emergency ambulance + PT56 stretcher billed FFS
Broker/administered-carve states of the nine footprint states		8			live count over the column above (OH is the only NO)
Memo - Virginia (extension state)	YES	DMAS FFS NEMT broker			Broker pays FFS NON-EMERGENCY AMBULANCE itself; no public FFS price for scheduled ambulance exists

Read panel

What is measured: nine published state Medicaid FFS ambulance schedules, fetched and parsed from the primary file or portal (NE/IA/KS/MO/OH/WI on 11 Jul 2026; MN/IN/KY on 12 Jul 2026); Virginia is parked because its only public rate files are the 2009 / 2009-2012 vintage. The BLS non-emergency base (A0428), the workforce code of scheduled interfacility transport, ranges from \$84.67 (IA, unchanged since 2014) to \$284.56 (MN, exactly the Medicare conversion factor - full parity) - roughly 0.30x to 1.00x the CY2026 Medicare national base, a 3.4x spread, with WI frozen at \$94.90 since 2008 and IN pinned to 100% of CY2025 Medicare. Kentucky is the outlier: it has NO ALS1-nonemergency, ALS2, or SCT line at all, and its only non-emergency ground line is a \$55 stretcher-van rate (formerly T2005), so it is excluded from the BLS comparison. Eight of the nine states route scheduled NEMT through a broker, transportation manager, or a state-administered system (only OH runs county NET instead), and Missouri will not pay A0428 outside HH/HD contexts nor SCT at all (IN also does not cover SCT). What is NOT measured: managed-care negotiated rates, broker-paid trip prices, and GEMT supplemental payments - the real price of most scheduled Medicaid transport is set inside those channels and is not public.



State TAM assembly: the same row math at state grain, reconciled to the carried top-down model

The question: run the metro row math at state grain for the ten footprint states plus a national roll-up - does the bottom-up assembly reconcile to TAM_Model_National, and where exactly do the two differ? Sources: CMS HSA 2025 (cases by CCN, state = SSA CCN prefix), Acute_IFT_Series rate, Derived_Rate_Card CY2026 base, Insourcing_Bounds Panel C state shares. Join key: CCN prefix -> state. The bridge panel lists every difference with green links to both sides; the SAM filter row states its grouping rule.

DATA QUALITY: state volume is the same Medicare FFS HSA proxy as the metro panels (floor of all-payer admissions; VA/psych/children's hospitals absent). The transfer rate is national, not state-specific - state case mix and rurality shift the true rate and that confound is printed on every state row. Insourcing shares are name-rule bounds, not a census. Nothing on this tab is a TAM; the national roll-up is labeled the proxy floor it is.

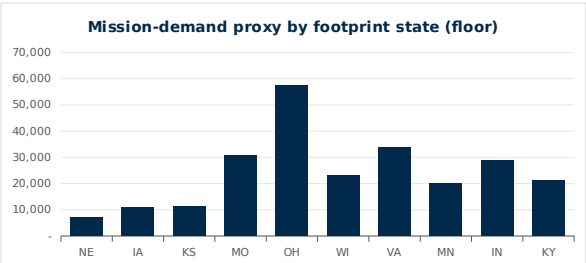
Panel A. The shared spine (green anchors used by every state row)		
Measured transfer rate, mean IFT % of admissions 2019-2023	9.49%	Denominator: AHA all-payer community-hospital admissions (HCUP NEDS+NIS numerator) - disclosed here and applied to Medicare-only cases below
Medicare-anchored \$ per transport (A0428 base, CY2026)	\$285	Base only; mileage excluded (43-45% of Medicare IFT allowed \$)

Panel B. State-grain assembly (footprint states + national roll-up)						
State	HSA cases (Medicare FFS)	Mission-demand proxy	Medicare-anchored \$ floor	Insourcing floor share	Insourcing ceiling share	Note
NE	74,208	7,039	\$2,003,018	0.0%	18.5% CCN prefix "28"; national rate applied - state case mix confound	
IA	113,036	10,722	\$3,051,060	1.5%	30.2% CCN prefix "16"; national rate applied - state case mix confound	
KS	120,270	11,408	\$3,246,320	0.6%	20.3% CCN prefix "17"; national rate applied - state case mix confound	
MO	321,719	30,517	\$8,683,818	-	14.4% CCN prefix "26"; national rate applied - state case mix confound	
OH	606,331	57,514	\$16,366,047	0.1%	20.6% CCN prefix "36"; national rate applied - state case mix confound	
WI	242,525	23,005	\$6,546,219	3.3%	27.3% CCN prefix "52"; national rate applied - state case mix confound	
VA	356,673	33,832	\$9,627,295	2.9%	30.1% CCN prefix "49"; national rate applied - state case mix confound	
MN	212,018	20,111	\$5,722,776	26.6%	67.7% CCN prefix "24"; national rate applied - state case mix confound	
IN	301,874	28,634	\$8,148,163	1.2%	37.7% CCN prefix "15"; national rate applied - state case mix confound	
KY	223,647	21,214	\$6,036,665	2.5%	35.0% CCN prefix "18"; national rate applied - state case mix confound	
FOOTPRINT (10 states)	2,572,301	243,996	\$69,431,382		provisional footprint; matches Insourcing_Bounds Panel C	
NATIONAL roll-up (all CCNs in the HSA file)	14,144,025	1,341,631	\$381,774,607	1.4%	27.0% the bottom-up side of the bridge below	

Panel C. Bridge: this bottom-up assembly vs the carried top-down scenario model (row per difference, green links both sides)				
Difference	Bottom-up side (this tab)	Top-down side (TAM_Model_National)	Direction	Why (stated, public)
Segment scope	\$381,774,607	\$20,595,460,675	top-down higher	The model prices ALL reimbursed ground ambulance (911 + IFT, five payer books); this assembly prices acute-to-acute IFT mission demand only
The model's own scheduled/IFT segment	\$381,774,607	\$12,973,695,487	top-down higher	Model segment = Medicare CODE mix (non-emergency + SCT + ALS2) applied across payers; this tab = HCUP transfer EPISODE definition - different segment definitions, never mixed
Volume basis and payer scope	14,144,025	41.3	assembly lower	Bottom-up counts Medicare FFS inpatient CASES (HSA); top-down builds all-payer TRANSPORTS from lives (top-down cell is in millions) - payer scope and unit both differ
Price basis	\$285	\$475	assembly lower	A0428 base only vs measured allowed per transport incl mileage (mileage share green: see Medicare_IFT_Series)
Episode vs leg vs billed-transport counting	3,061,156	5,510,664	differs by unit	HCUP counts sending-side episodes; NEMSIS counts vehicle legs; claims count billed transports - one transfer can appear once, twice or zero times depending on the counter
Gross-up layers (unbilled; facility direct-pay)	PENDING	not in the model (reimbursed both understate spend only)		No public IFT-specific volume share exists for either layer; anchors live on Facility_Pay_Layer; PENDING GADCS microdata / claims-vendor panel (Claims_Vendor_Recv)

Panel D. SAM filter: multi-hospital-system share of the footprint			
Grouping rule (printed): a system family = hospitals sharing a health-system crosswalk ID (AHRQ Compendium of US Health Systems); interim name-family rule = shared leading corporate	PENDING	PENDING	PENDING dataset: AHRQ Compendium hospital-to-system crosswalk (built on the AHA Annual Survey license); Hosp_Registry carries name/city/state/ownership but no system field, so the share is not computed from a weaker proxy
Read panel			

What the bridge shows: the bottom-up assembly and the top-down model are different instruments and the panel prints why, difference by difference - segment scope (IFT mission demand vs all ground), payer scope (Medicare FFS proxy vs five payer books), price basis (base-only vs measured allowed including mileage), counting unit (episodes vs legs vs billed transports), and the gross-up layers both sides leave unpriced. The assembly is the auditable floor; the model is the scenario envelope; neither is quoted as a single TAM.



Scenario matrix: three volume bases x three price bases, live where measured, PENDING where not

The question: across volume bases (claims-visible / NEMSIS-anchored / gross-up-full) and price bases (Medicare-anchored / blended-observed / MRF-forward), which sizing cells are live from public evidence today, and what would close each gate? Sources: PSPS-derived Medicare_IFT_Series (claims floor), NEMSIS 2024 (EMS_Transports), Insourcing_Bounds, Facility_Pay_Layer, TAM_Model_National. No cell of this grid is "the TAM"; the grid IS the answer, and its gates are printed below it.

DATA QUALITY: the claims-visible basis is Medicare FFS carrier claims only (hospital institutional billing excluded); the NEMSIS basis counts all-payer vehicle legs from a voluntary-participation registry (54 states/territories, coverage grew over time); gross-up volume shares have no public source and stay PENDING. Blended and MRF price bases await Medicaid_Rate_Card, Contract_Corpus rate extraction and Transparency-in-Coverage files. Mixing any cell across rows or columns re-introduces the exact double-counting this grid exists to prevent.

Panel A. The bases (each a green link to its measured cell, or PENDING naming its dataset)			
Basis	Value	Source / dataset named	Note
V1 claims-visible volume: Medicare FFS hospital-to-hospital transpor	871,753	Medicare_IFT_Series 2024 row (PSPS-derived carrier claims)	the hardest, fully measured volume
V2 NEMSIS-anchored volume: hospital-to-hospital activations, 2024	5,510,664	EMS_Transports v3.5 type-of-service table (NEMSIS 2024 End-of-Year report)	all-payer vehicle LEGS, not billed transports
V3 gross-up-full volume: claims-visible + unbilled + facility-paid	PENDING	PENDING datasets: IFT-specific unbilled volume share (GADCS microdata beyond the published one-in-five incidence) + facility-pay volume share (claims-vendor panel, Claims_Vendor_Recv; NEMSIS_State_IFT for the state wedge)	anchors are live on Facility_Pay_Layer; the SHARES are not public
P1 Medicare-anchored price: measured allowed per transport, 2024 (incl mileage)	\$662	Medicare_IFT_Series 2024 row, allowed per transport	reference only: A0428 base-only anchor used at metro grain sits on Derived_Rate_Card
P2 blended-observed price	PENDING	PENDING datasets: state Medicaid ambulance fee schedules (Medicaid_Rate_Card, B.3) + observed public contract rates	blend weights would also need payer-mix evidence
P3 MRF-forward price	PENDING	PENDING dataset: Transparency-in-Coverage negotiated-rate machine-readable files (landing shape Commercial_Rate_MRF, shipped as an empty schema)	

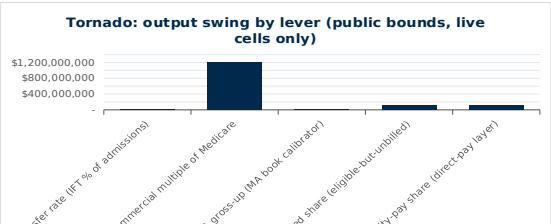
Panel B. The 3x3 grid (LIVE formula where both bases exist; PENDING names the missing input)			
Volume basis \ price basis	Medicare-anchored	Blended-observed	MRF-forward
Claims-visible	\$577,228,322 PENDING	PENDING	PENDING
NEMSIS-anchored	\$3,648,867,667 PENDING	PENDING	PENDING
Gross-up-full	PENDING	PENDING	PENDING
Gate G1: NEMSIS x Medicare cell contains the claims floor	PASS		
Gate G2: NEMSIS legs vs claims transports (volume containment)	PASS		
Honesty metric: LIVE grid cells (of 9)	2		

Panel C. Tornado: five levers, public bounds only, swings live against the grid and assembly cells						
Lever	Base value	Low bound	High bound	Public source of the bounds	Output swing \$	Swings which output / confound
Transfer rate (IFT % of admissions)	9.49%	9.24%	9.85%	HCUP NEDS+NIS / AHA annual ratio band, 2019-2023 (Acute_IFT_Series min/max rows)	\$4,426,562	swings the bottom-up FOOTPRINT assembly (TAM_Assembly_State), not the claims floor – the floor is measured
Commercial multiple of Medicare	1.80x	1.60x	2.00x	Low = pinned cross-year FAIR Health ground multiple (Payer_Rates_Commercial); high = the model's published-bounds high (TAM_Model_National assumption register)	\$1,193,320,088	swings the top-down commercial book at base volume (TAM_Model_National Panel F)
MA gross-up (MA book calibrator)	1.04x	1.00x	1.04x	High = GADCS Table 2.30 MA-to-FFS revenue per NPI, 1.038x (Facility_Pay_Layer calibrator); low = fee-schedule parity (TAM_Model_National MA-multiple low)	\$21,983,181	MA-book adder bound on the claims floor; a per-NPI revenue ratio, not a price index - confound printed
Unbilled share (eligible-but-unbilled)	20.0%	-	20.0%	GADCS Table S.1 "about one in five" organizations did not always bill (Facility_Pay_Layer); low = definitional zero (all bill)	\$115,445,664	incidence of ORGANIZATIONS used as an upper VOLUME bound - stated, not measured
Facility-pay share (direct-pay layer)	18.6%	-	18.6%	GADCS Table 2.32 facility-contract incidence 18.6% (Facility_Pay_Layer Panel A); DocGo books about 41% of transport revenue outside per-trip claims - existence at one operator, not a bound; low = definitional zero	\$107,364,468	incidence used as an upper bound on the claims floor; the IFT-specific share has no public source

Tornado read: swings are LIVE formulas against the grid floor cell (B16), the footprint assembly and the carried model - never against a typed number. A lever without public bounds would stay a PENDING row; the chart renders only when every lever row is live.

Read panel

The grid is the answer. The one fully measured cell - claims-visible volume x the measured Medicare allowed per transport - is the hardest floor, and it equals 2024 Medicare FFS interfacility allowed dollars by construction. The NEMSIS-anchored row shows the all-payer volume scale under gate G2. Every other cell is PENDING and names what closes it: Medicaid_Rate_Card and Contract_Corpus rate extraction (blended-observed), Transparency-in-Coverage files (MRF-forward), and GADCS microdata or a claims-vendor panel (gross-up-full). No cell of this grid is "the TAM".



Investor QA: twelve tearsheets, every number a live cell or a named PENDING slot

The question: the twelve questions an investor asks about this market, each answered in three sentences of plain words, wired to 2-4 live green cells on the home tabs, with the guardrail printed under every block. Sources: the home tabs named per block (this tab wires; it does not create). Join keys: none - green links only.

DATA QUALITY: answers inherit every caveat of their home tabs; where a home tab lands in a parallel module the number row is a bordered PENDING slot that names it, and the link goes live at assembly order. Nothing in this tab is an assertion without a cell behind it.

Tearsheets wired (live count of Q blocks below)	Count	
	12	the twelve banners below

Q1. Is IFT the same business as 911 or NEMT?

THE ANSWER IN THREE SENTENCES: Interfacility transport is scheduled clinical logistics between licensed facilities, paid per transport under the same Medicare fee schedule as 911 but with a different code and acuity mix. The hospital-to-hospital channel runs about 31 percent emergency while residence and scene channels run above 98 percent, and nearly all specialty care transport lives in the facility channels. NEMT is a separate Medicaid benefit of brokered vans and sedans that sometimes shares vehicles but not the ambulance fee schedule.

SCT services in the hospital-to-hospital channel, 2024	57,906	Acuity_by_Channel Panel A
Hospital-to-hospital emergency share, 2024	31.0%	Acuity_by_Channel Panel B
Hospital-to-SNF BLS non-emergency services, 2024	1,221,981	Acuity_by_Channel Panel A
The institutional claims fingerprint (revenue centers 054x)	Ambulance (category name for the full 054x range, verbatim from the	Code_Vocabulary section B

THE GUARDRAIL: Channel shares are Medicare FFS origin-destination coded services only; MA, Medicaid and commercial mixes are not publicly published at this grain.

Q2. Why do claims undercount this market, and by how much?

THE ANSWER IN THREE SENTENCES: Claims-based sizing sees only transports that become payer claims, and a measured share of ambulance revenue never does. Three quarters of ambulance organizations report some non-payer revenue, about one in five did not always bill at least one payer category, and a listed pure-play operator books roughly 41 percent of transport revenue outside per-trip claims. The activation-versus-claims reconciliation prices the wedge; the IFT-specific undercount inside it has no public number and stays PENDING.

Organizations reporting any non-payer revenue (GADCS)	74.5%	Facility_Pay_Layer Panel A
Listed-operator revenue outside per-trip billing, FY2025	40.9%	Facility_Pay_Layer Panel B
Five-universe reconciliation (activations vs claims)	60,298,684	Universe_Reconciliation (title pull)

THE GUARDRAIL: The wedge is a mixed pool of unbilled, facility-paid and non-FFS volume; no public source splits it, so no cell here claims to.

Q3. What does insourced really mean, and how much exists?

THE ANSWER IN THREE SENTENCES: Insourced means the hospital bills the transport itself or carries the ambulance cost center on its own books. Medicare claims bound hospital-billed ground base transports between about 1.4 and 27 percent (a strict-linkage floor against a name-collision ceiling), and the HCRIS panel shows the share of cost-report filers carrying an ambulance cost center. Between the bounds sits the mixed reality of owned services, JVs and contracted units billed under hospital numbers.

Hospital-billed share of ground base transports, 2024 FLOOR	1.4%	Insourcing_Bounds Panel B
Same share, 2024 CEILING	27.0%	Insourcing_Bounds Panel B
Cost-report filers with an ambulance cost center, latest complete FY	14.3%	HCRIS_Ambulance_CostCenters Panel A

THE GUARDRAIL: The floor is a floor twice over (roster gaps understate linkage) and the ceiling is padded by place-name collisions; the AHA ambulance-ownership flag would turn bounds into a measured split (AHA_Recv schema).

Q4. How big is the facility direct-bill shadow?

THE ANSWER IN THREE SENTENCES: Some ambulance revenue is billed to the facility, not a payer, so it never appears in any claims dataset. In the national census 18.6 percent of organizations report facility-contract revenue with a long right tail, and public contracts price the form by the dedicated unit, the hour and the annual subsidy. The corpus proves the channel and its pricing forms; its IFT-specific dollar share is a named PENDING slot.

Organizations reporting facility-contract revenue (GADCS)	18.6%	Facility_Pay_Layer Panel A
Facility-contract revenue, conditional mean per organization	\$922,555	same row: median \$48K prints beside it
Corpus documents with dedicated-unit or unit-hour pricing	2	contract_corpus.json; abstracts on Contract_Corpus

THE GUARDRAIL: GADCS incidence covers ALL ambulance work, not IFT alone; the IFT-specific facility-pay share stays PENDING on Facility_Pay_Layer Panel D by design.

Q5. How much of the market is invisible because of MA?

THE ANSWER IN THREE SENTENCES: Medicare Advantage ambulance claims are not published at provider grain, so the MA half of Medicare is dark to every public claims series. The MA and other share crossed 50 percent of beneficiaries in 2024 and rises every year, mechanically shrinking the FFS window all public series stand on. The GADCS census shows MA revenue per organization at parity or above FFS, so the dark half is not smaller business, just unmeasured.

MA book calibrator	\$1,193,046	MA_Book_Calibrator (title pull)
MA & other share of beneficiaries, 2024	50.1%	Enrollment_ESRD_State Panel A
MA-to-FFS revenue ratio per ambulance NPI (GADCS)	1.04x	Facility_Pay_Layer Panel A

THE GUARDRAIL: MA encounter data exists at ResDAC under DUA (MA_Encounter_Recv schema); until licensed, MA volume is inferred from enrollment and the GADCS revenue census, never from claims.

Q6. What do commercial payers actually pay?

THE ANSWER IN THREE SENTENCES: Commercial allowed amounts run well above Medicare for the same service, and dispersion across states is wide. FAIR Health puts ALS emergency in-network allowed at 758 dollars nationally in 2020, state means span roughly 1.7x, and ground ambulance stayed outside the No Surprises Act. Machine-readable negotiated rates exist in Transparency-in-Coverage MRF files; the receiving schema is built and the pull is the named gap.

ALS emergency average allowed, in-network, 2020	\$758	Payer_Rates_Commercial Panel A (FAIR Health)
State dispersion: high-state over low-state ALS allowed, 2022	1.75x	Payer_Rates_Commercial Panel B
Negotiated-rate slices (5 payers x footprint states)	PENDING	Commercial_Rate_MRF receiving schema lands with B.14; TIC MRF slice pull is the named public gap

THE GUARDRAIL: FAIR Health figures are national/state means from a contributor panel, not a census; negotiated-rate truth arrives only with the MRF slices.

Q7. Where does growth come from without a hockey stick?

THE ANSWER IN THREE SENTENCES: Growth is presented as measured components, never a blended headline: the 65+ base compounds near 3 percent, the current-law price update is 2 percent, and regionalization keeps concentrating sending and receiving. The offsets are equally measured: MA migration shrinks the visible FFS base and the statutory add-ons expire at the end of 2027. Anyone needing one growth number must blend the components and own the blend.

Demographic component (US 65+ CAGR, live)	3.0%	Growth_Outlook_Shell Panel A
Price component (AIF CY2026, live)	2.0%	Growth_Outlook_Shell Panel A
MA migration component (MA share 2024, live)	50.1%	Growth_Outlook_Shell Panel A
Add-on cliff date (per statute)	31 DECEMBER 2027	Growth_Outlook_Shell Panel B

THE GUARDRAIL: No blended growth number exists in this workbook; the conversion-rate slot on Growth_Outlook_Shell is bordered PENDING, not assumed.

Q8. How fragmented is this market, and who do we actually compete with?

THE ANSWER IN THREE SENTENCES: The national biller base is thousands of mostly small organizations, and the top of the distribution holds a modest share, so the real competitive set is local, not national. Concentration only means something per state and per corridor, which is why the workbook prints share panels and HHI at state and NPI grain. Parent roll-ups raise measured concentration, so every printed top share is a floor.

National concentration (top-10 view)	931800.0%	Fragmentation_National top-10 row
State share panels and per-state HHI	Market share, measured: Medicare FFS ambulance billers in the ten footprint	Market_Share_Panels (title pull)

THE GUARDRAIL: NPI-grain shares understate parent concentration; roll-up panels print the correction where public ownership is traceable.

Q9. What makes these relationships sticky?

THE ANSWER IN THREE SENTENCES: Public contracts show multi-year initial terms with renewal options, explicit exclusivity or first-call language in a minority of documents, and enforceable SLAs with liquidated damages in the 911 and NEMT forms. Registry presence in the cohort-corridor states churns slowly, with most 2019 billers still present in 2024. This is an evidence panel, not a stickiness score, and the corpus is a convenience sample.

Longest observed initial term (months)	120	Stickiness_Evidence Panel A summary
Documents with explicit exclusivity language (live count)	3	Stickiness_Evidence Panel A summary
Survivor share of the 2019 org-NPI cohort, cohort states (live)	84.3%	Stickiness_Evidence Panel C

THE GUARDRAIL: Hospital IFT agreements are private and absent from the corpus; churn is registry presence, not contract retention.

Q10. How bad is the labor problem?

THE ANSWER IN THREE SENTENCES: The binding input is EMT and paramedic labor: wages, depth and churn. Over 2019-2024 the compounded payer update roughly matched national EMT median wage growth, so the scissors story is window-dependent, while county-level depth varies by an order of magnitude. Labor risk prices locally: thin-depth counties with hub demand are where service fails first.

Workforce depth panel	\$54,170	Workforce_Depth (title pull)
EMT median wage growth 2019-2024 (live)	16.8%	Workforce_Supply Panel B
Compounded AIF over the same window (live)	18.5%	Workforce_Supply Panel B
Private ambulance employment (QCEW, latest annual)	171,100	QCEW_EMS_Employment Panel A

THE GUARDRAIL: National medians hide the local market; wage-vs-depth tightness is printed state by state on the workforce tabs, and the 2019-2024 scissors read reverses on other windows.

Q11. What is the reimbursement cliff risk?

THE ANSWER IN THREE SENTENCES: Current-law price growth is the AIF, 2.0 percent for CY2026, and the temporary add-ons of 2, 3 and 22.6 percent expire on 31 December 2027 under current statute. A Medicare-heavy ground book therefore carries a dated statutory cliff inside a normal hold period, largest for super-rural mileage. Input costs are tracked separately; the spread between the AIF and measured input inflation is the margin risk to watch.

Latest AIF (CY2026, live)	2.0%	Payment_Rules
Add-on expiry (per statute, carried live)	31 DECEMBER 2027	Growth_Outlook_Shell Panel B
Input cost tracking	PENDING	Input_Cost_Index lands in a parallel module; BLS PPI/ECEC and EIA diesel series already cached

THE GUARDRAIL: The cliff is statutory, not speculative: the add-ons lapsed once already in October 2025 before CAA 2026 s.6203 restored them through 2027.

Q12. What regulatory barriers protect incumbents?

THE ANSWER IN THREE SENTENCES: Entry is regulated locally: state licensure, certificates of need in some states, franchise and exclusive-operating-area statutes, and municipal contracts with performance bonds. Ground ambulance sits outside the federal No Surprises Act, and a growing list of states has enacted its own balance-billing limits, reshaping out-of-network economics state by state. Barriers are real but jurisdictional; there is no single national moat number.

Entry barrier register	5
State balance-billing pegs	22
Ground exclusion from the No Surprises Act (statutory anchor, live pull)	Ground ambulance is EXCLUDED from federal out-of-network balance-

THE GUARDRAIL: Barrier strength varies by state and municipality; the register prints citations, not a barrier score.

Entry_Barrier_Register (title pull)

Balance_Billing_States (title pull)

Payment_Rules statutory anchors

Read panel

Twelve tearsheets, and every number in them is either a green live cell pulled from its home tab or a bordered PENDING slot that names the public dataset or landing schema that would fill it. Nothing on this tab is an assertion: the tab wires evidence that exists elsewhere in this workbook, and where a home tab lands later in assembly order the slot says so and goes live on the next build.

Slide feed: every planned exhibit mapped to live cells, with an honest evidence status

The question: which planned exhibits of the study are backed by live cells today, and where are the named public gaps? Status rule: GREEN = the home tab exists with live cells and no named PENDING input; AMBER = the tab exists but a named input is partial or PENDING; RED = the tab or data is absent from the workbook at this build. Statuses recompute at every assembly because this module scans the workbook it is given. Sources: pointers only - this tab creates no evidence.

OPERATIONAL TAB: this feed powers the deck build and the remaining work plan; it does NOT feed the findings ledger, and honest RED/AMBER statuses are the deliverable. Every non-GREEN row names its gap: either a public dataset/receiving schema or the parallel module that lands the tab in assembly order.

The exhibit register					
Planned exhibit	Home tab (resolved live)	Anchor	Live pull (green = tab title, proves the tab exists)	Status	Named gap (every non-GREEN row)
Market definition: IFT vs 911 vs NEMT	Acuity_by_Channel	Acuity_by_ChannellC21	Each channel has its own clinical character, and the two directions of one interface are opposites.	GREEN	none - live
Code vocabulary and claims fingerprint	Code_Vocabulary	Code_Vocabulary!C11	Code vocabulary - the ambulance/IFT codes beyond HCPCS: place of service, revenue centers, condition/value codes, remittance, modifiers, taxonomies, MACs	GREEN	none - live
Five-universe reconciliation	Universe_Reconciliation	Universe_Reconciliation!A1	Universe reconciliation: why the national ambulance numbers differ, in one ladder	GREEN	none - live
TAM floor grid (Medicare-visible floor)	TAM_Model_National	TAM_Model_National!A1	National TAM model: United States ground ambulance, reimbursed spend	AMBER	MA and Medicaid legs need licensed extracts (MA_Encounter_Recv, TAF_Ambulance_Recv)
Metro TAM panel (20 target metros)	Metro_TAM_Panels	Metro_TAM_PanellA1	Metro TAM panels: the mission-demand line and its priceable floor, one panel per target metro	GREEN	none - live
Gross-up waterfall (floor to all-payer)	Sizing_Playbook	Sizing_Playbook!A1	Sizing playbook: how the team converges on a defensible market number	AMBER	gross-up legs beyond Medicare FFS carry PENDING inputs (MA encounter, TAF, MRF)
Facility-pay triangulation	Facility_Pay_Layer	Facility_Pay_Layer!B8	The facility-pay layer: ambulance revenue that never touches a payer claim	AMBER	IFT-specific facility-pay share PENDING (claims-vendor panel with facility-remit flags)
Insourcing bounds + flow	Insourcing_Bounds	Insourcing_Bounds Panel B	Insourcing bounds: the hospital-billed share of Medicare FFS ground ambulance, floor and ceiling	AMBER	AHA ambulance-ownership flag would turn bounds into a split (AHA_Recv)
HCRIS ambulance cost centers	HCRIS_Ambulance_CostCenters	HCRIS_Ambulance_CostCenters Panel A	Hospital-run ambulance: the Worksheet A line 95 cost center, FY2019 - FY2024	AMBER	FY2024 year-file not yet published (filing lag)
Fragmentation curve	Fragmentation_National	Fragmentation_National!A1	Fragmentation, measured: the US Medicare ambulance biller universe across 12 vintages, 2013-2024	GREEN	none - live
Biller survival curves	Annual_Market_Structure	Annual_Market_Structure!A1	Annual market structure, 2013-2024: census, churn, concentration, survival - twelve vintages, one series	GREEN	none - live
Market share panels (state grain)	Market_Share_Panels	Market_Share_Panels!A1	Market share, measured: Medicare FFS ambulance billers in the ten footprint states, by vintage	GREEN	none - live
HHI / concentration bands	Market_Share_Panels	Market_Share_Panels Panel C	Market share, measured: Medicare FFS ambulance billers in the ten footprint states, by vintage	GREEN	none - live
Cohort corridor map	Cohort_Corridors	Cohort_Corridors!A1	Cohort corridors: where the research-cohort systems draw their inpatients, and who is registered there	GREEN	none - live
Hub counts and hub share	Hub_Spoke_Map	Hub_Spoke_Map Panel B	Hub and spoke: the measured regionalization of hospital corridor volume	GREEN	none - live
Whitespace counties	County_Whitespace_Screens	County_Whitespace_Screens!A1	County whitespace screens: heavy 65+ demand over thin Medicare-billing ambulance supply, ten footprint states	GREEN	none - live
Growth decomposition (price x volume)	Growth_Decomposition	Growth_Decomposition!A1	Growth decomposition: price, volume and mix in the national Medicare ground ambulance book, 2013-2024	GREEN	none - live
Growth outlook components	Growth_Outlook_Shell	Growth_Outlook_Shell Panel A	Growth outlook shell: measured components only, no blended headline	AMBER	forward conversion rate is a bordered PENDING slot (AHA flag or claims-vendor panel)
Denial dollars	Denial_Economics	Denial_Economics!A1	Denial economics: what the Medicare medical-necessity screen filters out, in dollars, 2010-2024	GREEN	none - live

Wage scissors	Workforce_Depth	Workforce_Supply!B17	Workforce depth: the EMS labor base of the footprint, and the wage-vs-AIF scissors at quarterly grain	GREEN	none - live
Boarding-cost shelf	Throughput_Economics_Public	Throughput_Economics_Public!A1	Throughput economics, public spine: what a boarded bed and a delayed transfer cost, and where transport ordering lives	GREEN	none - live
DIDO / transfer delay friction	Transfer_Delay_Burden	Transfer_Delay_Burden!A1	Transfer delay burden: the measured ED-flow environment of the footprint, hospital by hospital	GREEN	none - live
REH conversions and rural closures	REH_Closure_Flow	REH_Closure_Flow Panel A/B	Rural Emergency Hospitals and the closure flow: transfer demand created by statute and by exit	GREEN	none - live
Receiving centers	Receiving_Side	Receiving_Centers!A1	Independent validation from the receiving hospital's record, 2012 to 2023	GREEN	none - live
Medicaid rate floor	Medicaid_Rate_Card	Medicaid fee-schedule tab!A1	Medicaid ground ambulance rate card - the footprint states	GREEN	none - live
Balance-billing pegs	Balance_Billing_States	Balance_Billing_States!A1	State ground-ambulance balance-billing protections: the payment standards, verified statute by statute	GREEN	none - live
Entry barrier register	Entry_Barrier_Register	Entry_Barrier_Register!A1	Entry barrier register: where ambulance market entry is legally gated in the ten footprint states	GREEN	none - live
Input cost index	Input_Cost (planned)	Input_Cost_Index!A1	PENDING	RED	B.11 module lands in parallel; BLS PPI/ECEC + EIA diesel cached
Cohort dossiers	System_Research_Cohort	per-system dossier tabs	System research cohort: what the public record establishes, system by system	GREEN	none - live
Footprint determination	Press_Footprint (planned)	Press_Footprint_Registry!A1	PENDING	RED	X-B/E.5 modules land in parallel; wayback_footprint.json cached
Prospect screen	Prospect_Landscape	Prospect screen tab!A1	Landscape screen: additional multi-hospital systems in the footprint, screened by measured transfer demand	GREEN	none - live
Scenario grid (no tornado without it)	Scenario_Matrix	C.3 scenario tab!A1	Scenario matrix: three volume bases x three price bases, live where measured, PENDING where not	GREEN	none - live
Vendor share stack	Vendor_Share_Stack	Vendor_Share_Stack Panel A	The vendor-share evidence stack: six labeled legs, none blended	AMBER	legs 5-6 PENDING: licensed buyer research; claims-vendor panel (Claims_Vendor_Recv)
Stickiness evidence panel	Stickiness_Evidence	Stickiness_Evidence Panel A	Stickiness evidence: observed terms, exclusivity language, SLA precedents, registry churn - a panel, NOT a score	AMBER	hospital IFT agreements are private; corpus is a convenience sample
Subject-company measured book	MMT_Medicare_Book	MMT_Medicare_Book!A1	The subject company, measured: Medicare FFS ambulance book by NPI, 2013 / 2019 / 2024	GREEN	none - live
Investor QA tearsheets	Investor_QA	Investor_QA (12 blocks)	Investor QA: twelve tearsheets, every number a live cell or a named PENDING slot	GREEN	none - live
MA invisibility calibrator	MA_Book_Calibrator	Imbalance_Ledger!B39	The MA book calibrator: Medicare Advantage revenue per NPI at 1.04x Medicare FFS (GADCS Table 2.30)	AMBER	MA encounter extract under DUA is the named gap (MA_Encounter_Recv)
Commercial rate benchmarks	Payer_Rates_Commercial	Payer_Rates_Commercial!C6	What everyone other than Medicare fee-for-service pays. Published evidence only.	AMBER	TIC MRF negotiated-rate slices PENDING (Commercial_Rate_MRF schema)

Status counts (live)	GREEN	AMBER	RED	Total
Exhibits by evidence status	26	10	2	38

Read panel

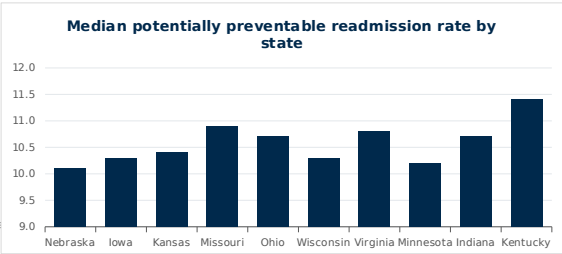
This tab is the deck feed and the remaining work plan in one: every planned exhibit of the study points at its home tab and anchor, GREEN rows are ready to render, AMBER rows render with their named PENDING input printed on the slide, and RED rows are not exhibits yet - they are work items whose gap column names the module or public dataset that unblocks them. The statuses are recomputed live at every assembly, so this page is only ever as optimistic as the workbook it sits in. It feeds the deck, not the findings ledger.

SNF return-leg quality: discharge-to-community vs bounce-back rehospitalization

The question: which footprint SNFs send patients home (discharge to community, less future transport) versus bounce them back to the hospital (potentially preventable readmission, the SNF-to-hospital return leg = transport demand)? Source: CMS SNF Quality Reporting Program provider data (dataset fykj-qjee), risk-standardized measures S_005_02 (DTC) and S_004_01 (PPR-PD). Join key: CMS Certification Number (CCN) and state; both are national, risk-adjusted rates in percent.

DATA QUALITY: these are CLAIMS-BASED, RISK-STANDARDIZED, SNF-REPORTED facility measures - not counts of transports. Transport demand is INFERRED (a readmission implies a SNF-to-hospital transport leg), not measured. CMS suppresses facilities below the minimum eligible-stay count (they arrive as "Not Available" and are dropped here). Risk standardization already adjusts for case mix, so residual variation is facility signal; the worst-decile floor is the NATIONAL 90th-percentile PPR rate. Rates mix a ~2-year claims window (Start/End Date), not a single year.

Panel A. Footprint state distribution - discharge-to-community and rehospitalization (risk-standardized, percent)						
State	SNFs (n)	Median DTC rate (%)	Median PPR rate (%)	SNFs in worst natl PPR decile	Worst-decile share	Note
Nebraska	179	43.9	10.1	2	1.1%	higher PPR / lower DTC = more bounce-back transport
Iowa	389	49.6	10.3	15	3.9%	
Kansas	296	50.1	10.4	13	4.4%	
Missouri	487	48.9	10.9	44	9.0%	
Ohio	921	53.5	10.7	56	6.1%	
Wisconsin	323	51.8	10.3	18	5.6%	
Virginia	289	55.4	10.8	27	9.3%	
Minnesota	338	54.5	10.2	5	1.5%	
Indiana	507	54.2	10.7	58	11.4%	
Kentucky	268	47.2	11.4	48	17.9%	
Footprint (pooled)	3,997	52.0	10.6	286	7.2%	pooled over all footprint SNFs
National (all SNFs)	14,695	51.5	10.7	1,287	8.8%	worst decile = top 10% of PPR-reporting SNFs; as a share of ALL SNFs it is lower bc



Panel B. Footprint vs national gap (live formulas) - is the bounce-back worse here?				
Measure	Footprint median	National median	Gap (fp - natl)	Read
Discharge to Community (higher is better)	52.0	51.5	0.5	negative = footprint sends fewer home
Potentially Preventable Readmission (lower is better)	10.6	10.7	(0.1)	positive = footprint bounces back more
Footprint SNFs in the worst NATIONAL PPR decile	286		22.2%	footprint share of the national worst-decile tail

Panel C. Worst-decile SNFs in footprint states - highest rehospitalization (top 22 of 286 at/above the national 90th-percentile PPR)						
SNF (provider name)	State	City	PPR rate (%)	PPR eligible stays	DTC rate (%)	vs national
Spring Mill Health Campus	IN	Merrillville	19.0	227	42.4	Worse than the National Rate
August Healthcare At Iliff	VA	Dunn Loring	18.3	197	21.4	Worse than the National Rate
August Healthcare At Leewood	VA	Ammandale	17.4	263	55.8	Worse than the National Rate
Eastgate Health Care Center	OH	Cincinnati	16.6	272	50.0	Worse than the National Rate
Austinwoods Rehab Health Care	OH	Austintown	16.4	181	56.9	Worse than the National Rate
Autumn Woods Health Campus	IN	New Albany	16.2	343	59.6	Worse than the National Rate
Lincoln Hills Of New Albany	IN	New Albany	16.1	163	45.2	Worse than the National Rate
Sikeston Convalescent Center	MO	Sikeston	16.1	117	31.5	Worse than the National Rate
Nhc Healthcare, Kennett	MO	Kennett	15.9	162	44.5	Worse than the National Rate
Westminster Village Kentuckiana	IN	Clarksville	15.8	128	54.4	Worse than the National Rate
Carriage Hill Health & Rehab Center	VA	Fredericksburg	15.6	442	49.2	Worse than the National Rate
Gasconade Manor Nursing Home	MO	Owensville	15.5	65	52.0	No Different than the National Rate
Altercare Cambridge Inc.	OH	Cambridge	15.5	134	50.7	Worse than the National Rate
Princeton Nursing & Rehabilitation	KY	Princeton	15.5	114	27.5	Worse than the National Rate
Signature Healthcare Of Bowling Green	KY	Bowling Green	15.5	203	43.9	Worse than the National Rate
Signature Healthcare Of Terre Haute	IN	Terre Haute	15.4	253	44.1	Worse than the National Rate
Charlestown Place At New Albany	IN	New Albany	15.4	197	44.3	Worse than the National Rate
Our Lady Of The Valley	VA	Roanoke	15.1	278	63.2	Worse than the National Rate
Cumberland Trace Health & Living Community	IN	Plainfield	15.1	247	67.9	Worse than the National Rate
Signature Healthcare Of Elizabethtown	KY	Elizabethtown	14.9	318	51.5	Worse than the National Rate
Greendale Park Nursing And Rehab	WI	Greendale	14.9	527	64.2	Worse than the National Rate
Park Terrace Health Campus	KY	Louisville	14.9	478	67.3	Worse than the National Rate

Read panel

What is measured: two risk-standardized SNF QRP claims measures across every reporting footprint-state SNF. Discharge to community runs about 52.0% in the footprint (national 51.5%) and potentially preventable readmission about 10.6% (national 10.7%). The tail is what matters for transport: 286 footprint SNFs sit in the worst NATIONAL decile for rehospitalization - each is a structural bounce-back node that generates SNF-to-hospital transport demand rather than discharging patients home. Panel C names the worst of them by CCN and city so they can be joined to the corridor and hub-spoke maps. What is NOT measured: actual transport counts - a readmission implies a transport leg but the dataset carries quality rates, not trips; and managed-care or non-Medicare stays are outside the claims-based measure.

SNF return-leg structure: who bounces patients back (ownership, scale, star rating)

The question: X-A.4 measured the return-leg rates; this tab asks whether a SNF's ownership, scale and quality rating predict how much return-leg transport it generates. It joins the SNF QRP claims measures (snf_qrp, dataset fykj-qjee) to the CMS Nursing Home provider-info file (dataset 4pq5-n9py) on the CCN - a 1:1 join over all reporting SNFs - and cross-tabs the risk-standardized potentially-preventable readmission (PPR = bounce-back transport) and discharge-to-community (DTC) rates by ownership, certified-bed scale, and 5-star rating. Higher PPR / lower DTC = a stronger structural transport-demand node.

DATA QUALITY: both inputs are national CMS facility files joined on the CMS Certification Number; the join covers 14,695 SNFs that appear in both. Rates are CLAIMS-BASED and RISK-STANDARDIZED (case-mix adjusted), so residual variation across ownership / scale / rating is facility signal, not case mix; transport demand is INFERRED (a readmission implies a SNF-to-hospital transport leg), not counted. Ownership buckets collapse the CMS sub-categories (For profit - LLC / Corporation / Individual / Partnership -> For profit, etc.). The worst-decile floor is the national 90th-percentile PPR.

Panel A. Return-leg rates by ownership bucket (higher PPR = more bounce-back transport)							
Ownership	National SNFs	Natl median PPR (%)	Natl median DTC (%)	Footprint SNFs	Fp median PPR (%)	Fp median DTC (%)	Note
For profit	10,855	10.7	50.8	2,644	10.7	51.1	for-profit tends to bounce back more than non-profit
Non profit	2,897	10.4	55.4	1,060	10.3	54.0	
Government	943	10.6	50.5	293	10.5	53.4	

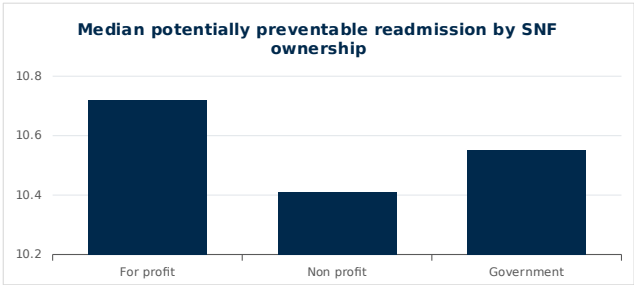
Panel B. Return-leg rates by certified-bed scale							
Bed scale	National SNFs	Natl median PPR (%)	Natl median DTC (%)	Footprint SNFs	Fp median PPR (%)	Fp median DTC (%)	Note
1. Under 60 beds	2,669	10.5	56.5	1,141	10.4	52.9	scale vs bounce-back
2. 60 to 119 beds	6,642	10.7	51.7	2,078	10.6	52.1	
3. 120 to 179 beds	4,041	10.8	50.8	641	10.7	51.0	
4. 180 or more beds	1,335	10.6	48.1	137	10.7	50.2	

Panel C. Return-leg rates by overall 5-star rating (the quality gradient)				
Overall rating	National SNFs	Natl median PPR (%)	Natl median DTC (%)	Note
1	2,873	10.7	46.4	1 star = worst care; note the PPR gradient
2	3,025	10.8	49.9	
3	2,844	10.7	52.0	
4	2,783	10.7	53.1	
5	3,045	10.4	57.6	

Panel D. For-profit vs non-profit gap (live formulas over Panel A national medians)				
Measure	For profit	Non profit	Gap (FP - NP)	Read
Median PPR - bounce-back (lower is better)	10.7	10.4		0.3 positive = for-profit bounces back more
Median DTC - sent home (higher is better)	50.8	55.4	(4.6)	negative = for-profit sends fewer home

Read panel

What is measured: the same two risk-standardized SNF QRP claims measures as X-A.4, cut by facility structure. Ownership sorts the bounce-back: for-profit SNFs run a median PPR of about 10.72% against 10.41% for non-profit - the for-profit facilities return more patients to the hospital, so they generate proportionally more SNF-to-hospital transport. The 5-star rating is a clean gradient: 1-star SNFs sit near 10.74% PPR versus about 10.42% for 5-star, i.e. the worst-rated facilities are the heaviest bounce-back nodes. For an operator these are targeting screens: the return-leg transport demand concentrates in for-profit, lower-rated SNFs, and ownership concentration among them (chains) is a consolidation and partnership signal. What is NOT measured: actual transport counts (a readmission implies a leg but the file carries quality rates, not trips), and non-Medicare / managed-care stays outside the claims measure.



Style standard: the CIM-presentation rules every tab must meet

The question: what does a tab have to satisfy to be pasted into a confidential information memorandum appendix without edits? These rules are the written standard behind the committed format gate (format_gate.py); the gate checks them on every build and no revision ships unless it passes.

The rules	
Dimension	Rule
Title block	Title in A1: Arial 15 bold, navy 00294C. The question line and the data-quality line in grey 9pt beneath it (rows 2-3).
Panel headers	Panel/banner headers Arial 10 bold navy; table header rows white-on-navy fill.
Body	Body Arial 9. Labels left-aligned, numbers right-aligned. Wrapped text with row heights sized so nothing clips.
Source line	A source line under every panel naming the source IDs that power it.
Numbers	Every numeric cell carries an explicit format: thousands separators on counts; dollars with \$ and no decimals above 1,000 (two decimals only for per-unit rates); percentages at one decimal; zeros as dashes; years as bare text; no raw floats, no unformatted integers, no scientific notation.
Layout	Gridlines off; freeze panes below the title block; column widths sized to content so nothing truncates or shows #####; cursor at A1; zoom 100 on save.
Tab colours	Tab colours grouped by section (governance navy, demand teal, supply green, analysis slate, assembly amber, receiving/reference grey) matching the Index groupings; sheet order matches the Index.
Charts	Every chart passes the V9 style gate plus: title present, axis labels present, single value axis, category axis on the bottom (left for horizontal bars) with explicit delete=0, no smoothed lines, no default-grey Excel styling.
The gate	These rules are enforced by format_gate.py, committed beside the verification gates and run on every final build. No revision ships without the format gate passing and a render proof attached.

Read panel

What this tab is: the presentation contract. The evidence tabs answer what the market looks like; this tab answers what a finished exhibit looks like. Every tab in this workbook is built to be screenshot-worthy at 100 percent zoom - a titled, sourced, cleanly formatted exhibit rather than a working file. The format gate turns that intent into a test: gridlines off, a section tab colour, freeze panes, a bold A1 title, and an explicit number format on every numeric cell, on all tabs, or the build fails.